

# Homochiral ideas: symmetry centers

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# Objectives

In the whole infinite spectre tiling:

- Find the location of global 2-fold and 3-fold symmetry centers

Around a global 2-fold or 3-fold symmetry center:

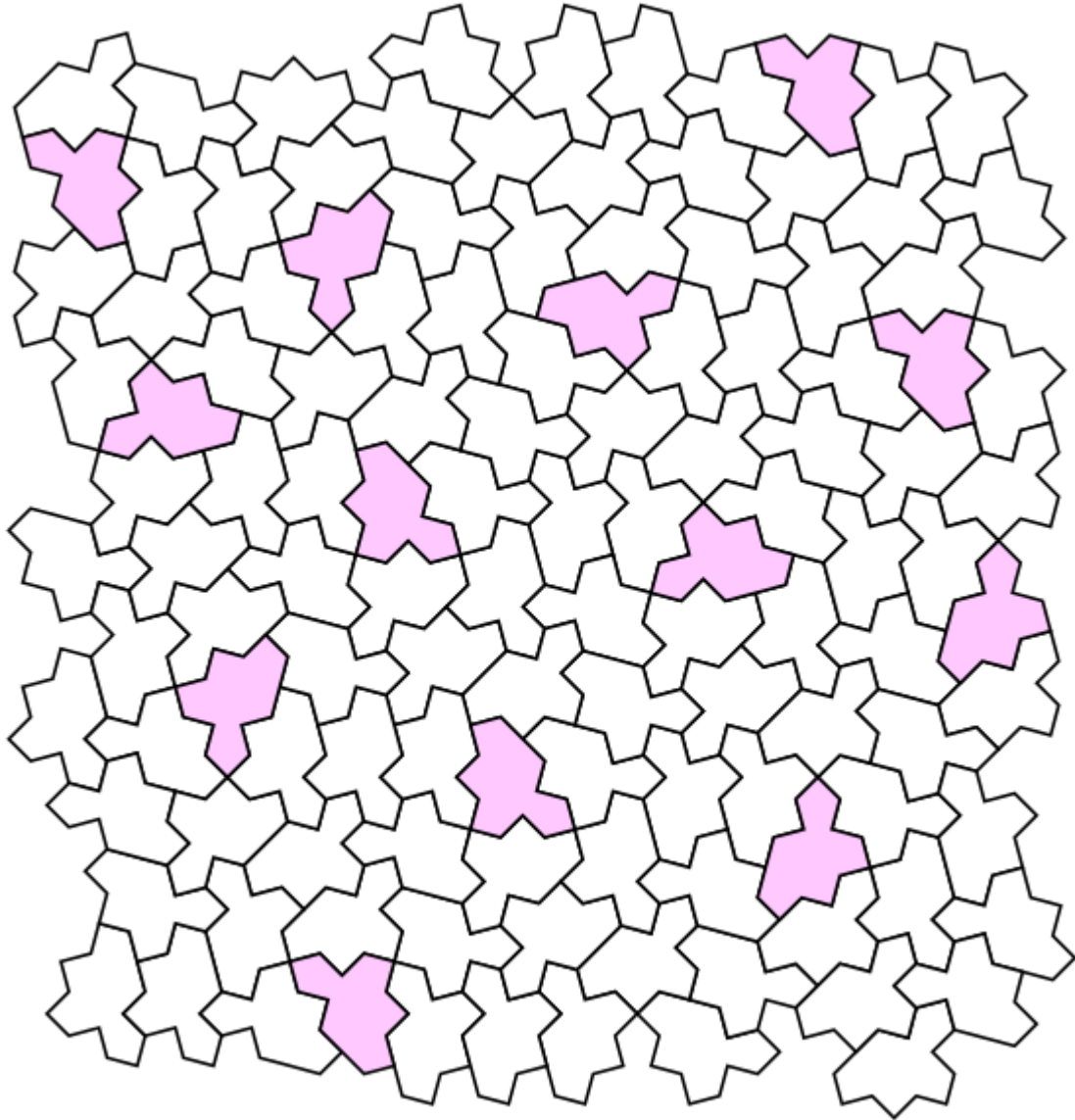
- Build an homochiral inflation construction

# Difference between global and local symmetry

We distinguish between:

- a global symmetry center where rotation operation around the center applies to the whole infinite tiling
- And a local symmetry center, where rotation operation around the center applies locally to a finite subset of tiles

# Spectre tiling (left chirality): ODD and EVEN monotiles



- ODD (pink):  
M1,M3,M5,M7,M9,M11
- EVEN (no color):  
M2,M4,M6,M8,M10,M12



# Six orientations for ODD and EVEN monotiles

$M_1 = M_{O_1}$

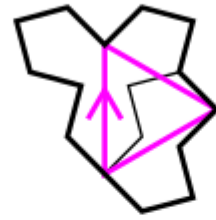
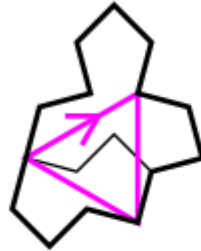
$M_3 = M_{O_2}$

$M_5 = M_{O_3}$

$M_7 = M_{O_4}$

$M_9 = M_{O_5}$

$M_{11} = M_{O_6}$



$M_2 = M_{E_1}$

$M_4 = M_{E_2}$

$M_6 = M_{E_3}$

$M_8 = M_{E_4}$

$M_{10} = M_{E_5}$

$M_{12} = M_{E_6}$

# Identification of the greek letters for tiles

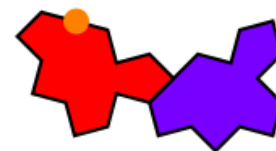
- ODD tiles  
Pink color

$\Gamma$

$\Gamma_2$



$\Theta$



$\Lambda$

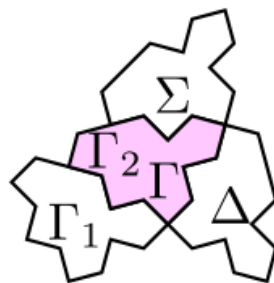
- $\backslash$ theta and  $\backslash$ lambda always in couple

- Small cluster  $\backslash$ sigma,  $\backslash$ delta,  $\backslash$ gamma1  
white color

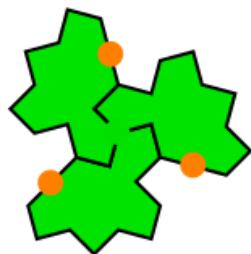
$\Gamma_1$

$\Sigma$

$\Delta$



$\Phi$



- $\backslash$ phi: Small green stars + some isolated EVEN tiles



$\Pi$



$\Psi$

$\Xi$

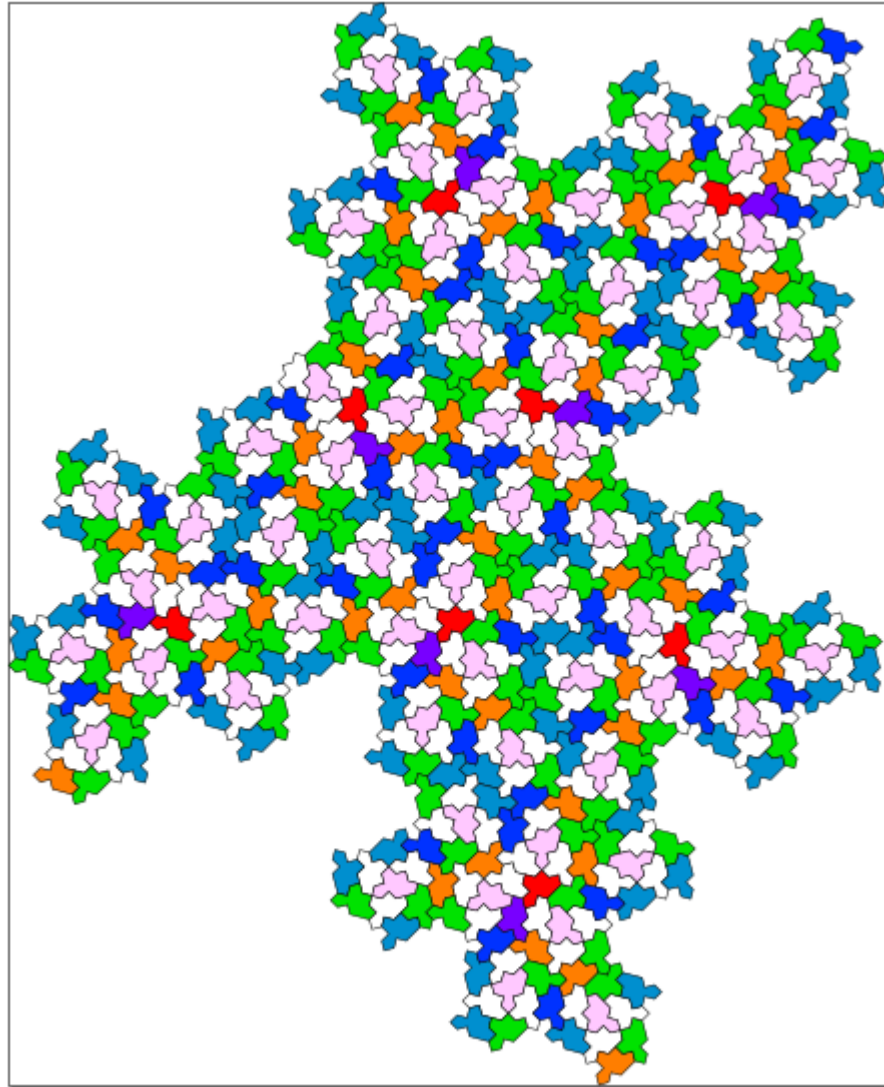
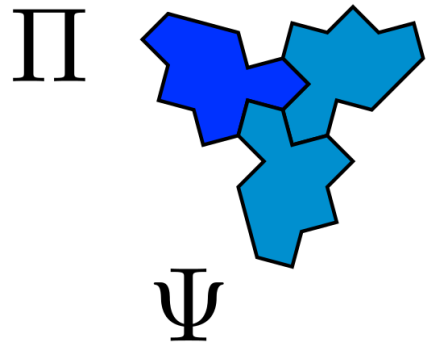


$\Pi$

- All  $\backslash$ chi tiles are linked to a  $\backslash$ pi tile and consequently to a large star.

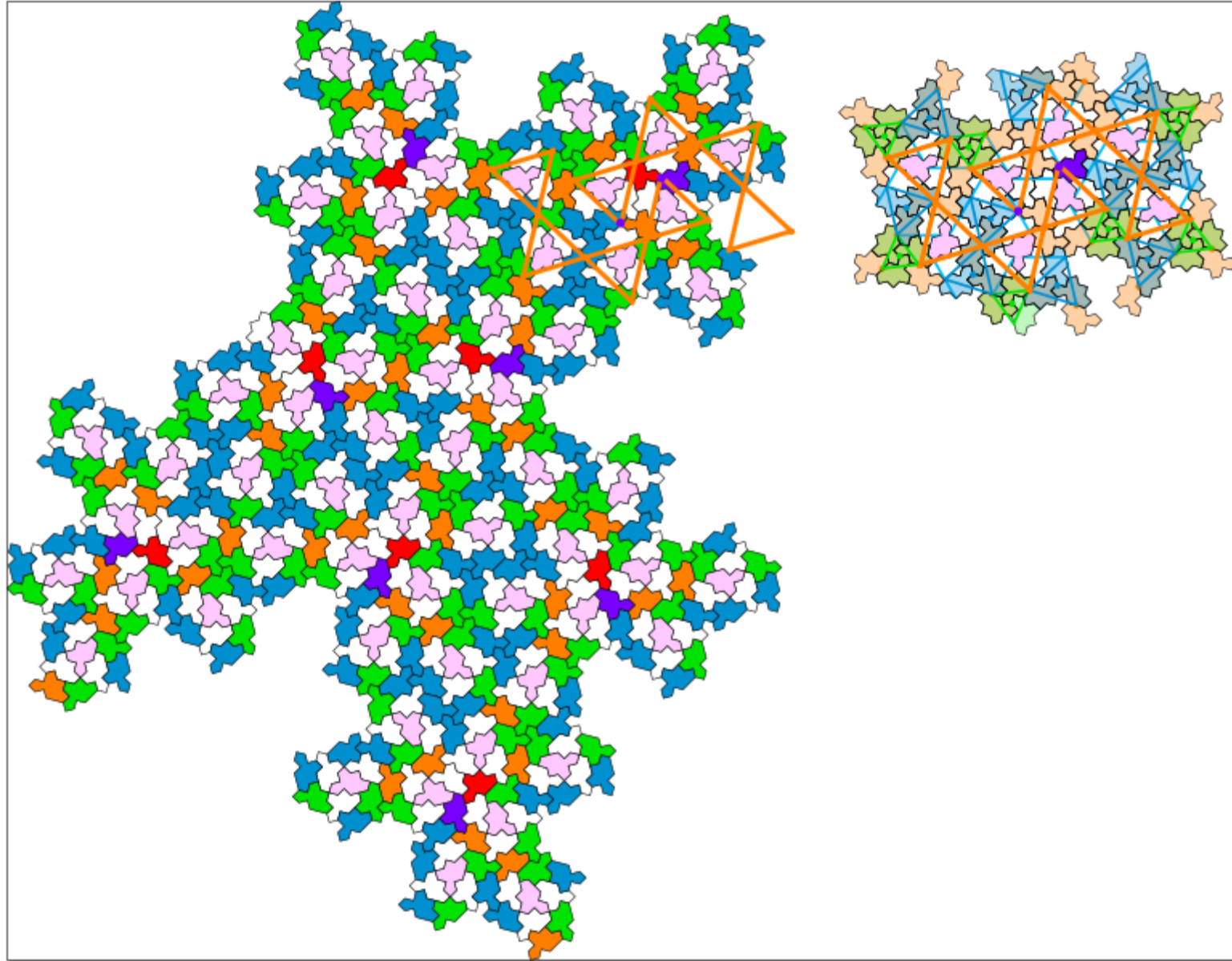
- Blue large stars correspond to all  $\backslash$ Pi and  $\backslash$ Psi tiles with different combinations

# Inflation using python code

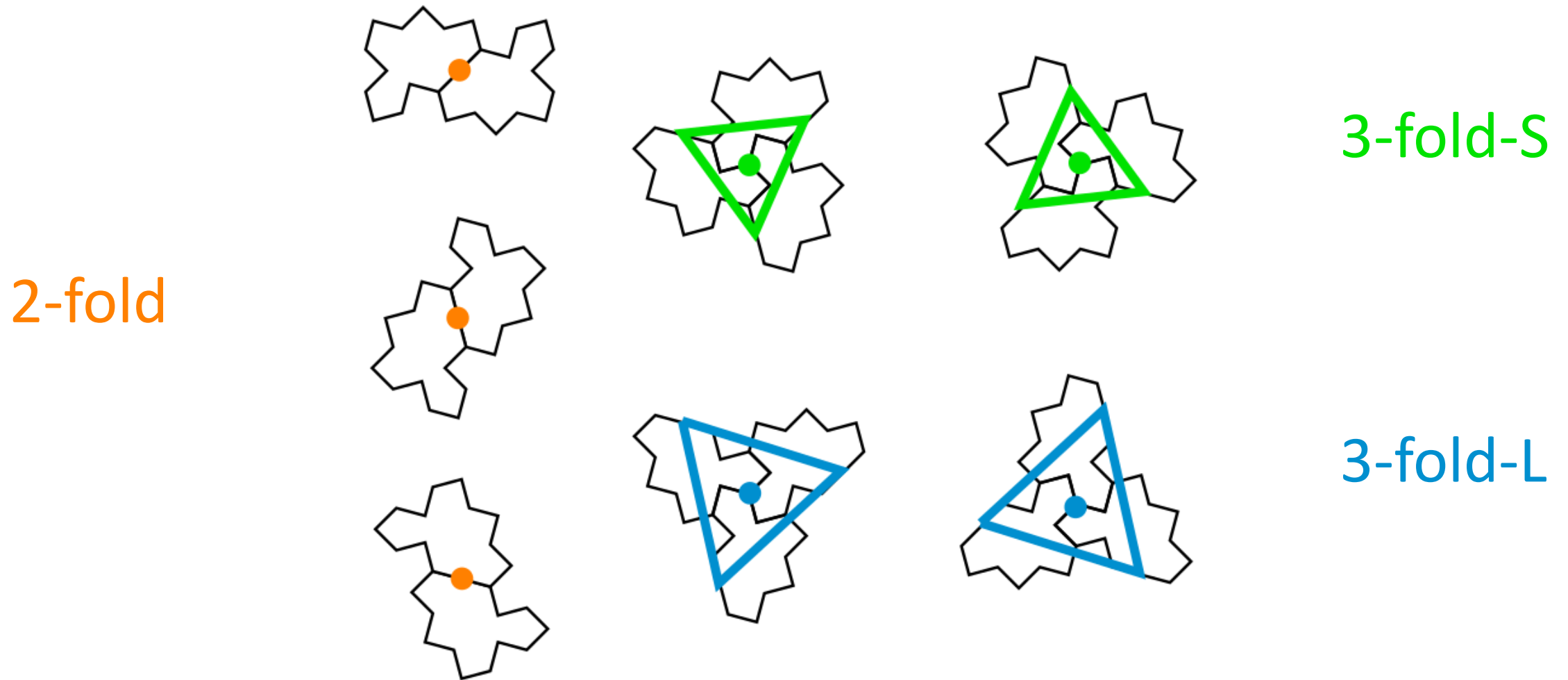


- Blue large stars correspond to all  $\Pi$  and  $\Psi$  tiles with different combinations

# Inflation using python code

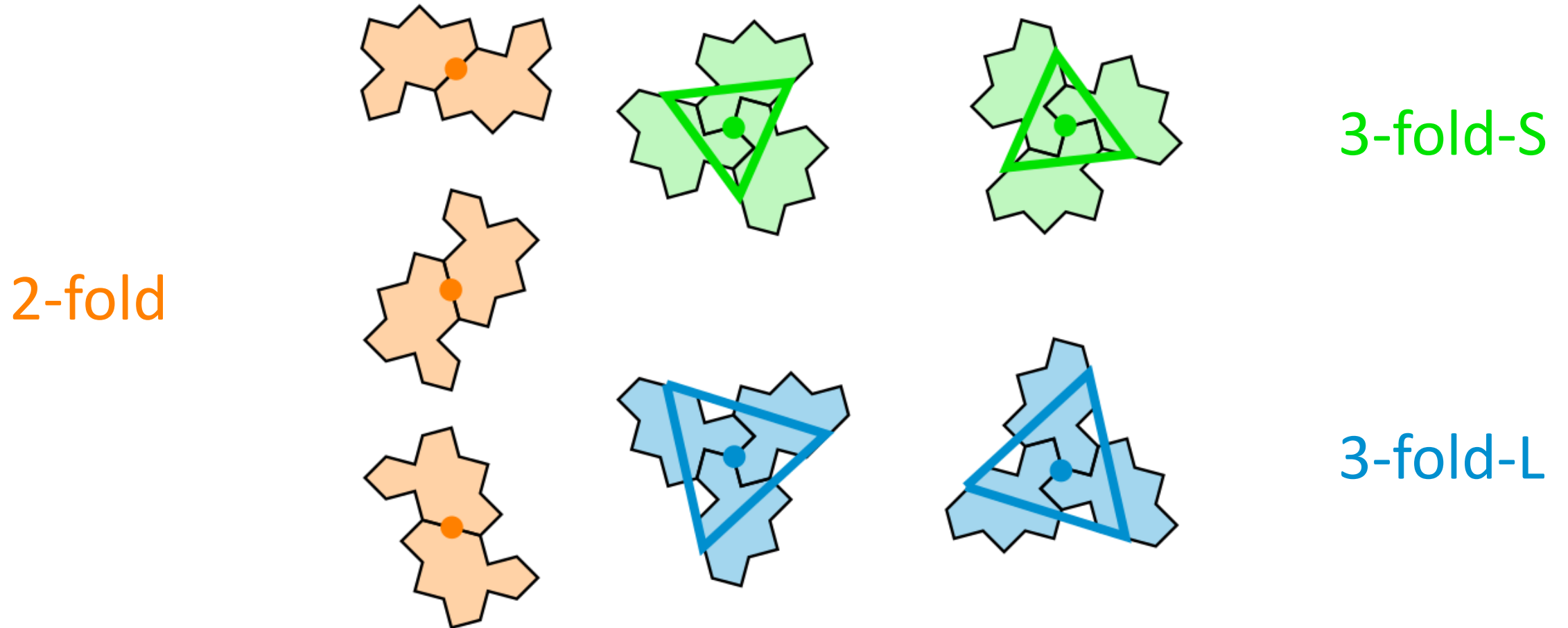


# configuration of monotiles at a symmetry center



- One configuration for 2-fold (orange color) and two configuration for 3-fold (green and blue). Figures show all possible orientations in the tiling.

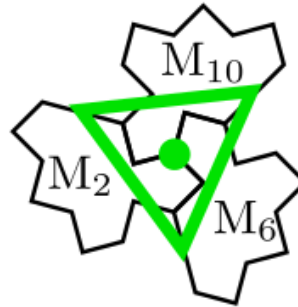
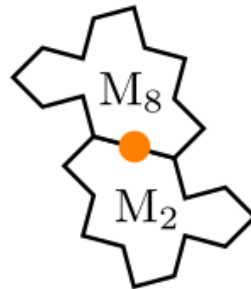
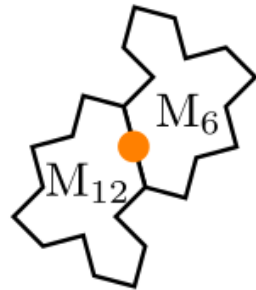
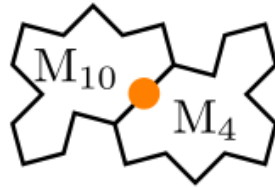
# configuration of monotiles at a symmetry center



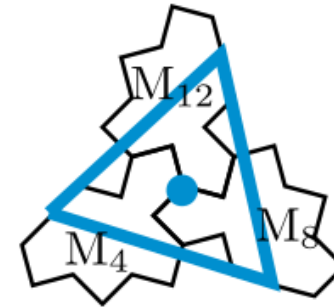
- Color is used when rotation around symmetry center is applied locally.

# Symmetry centers are located on the contour of adjacent EVEN monotiles

2-fold



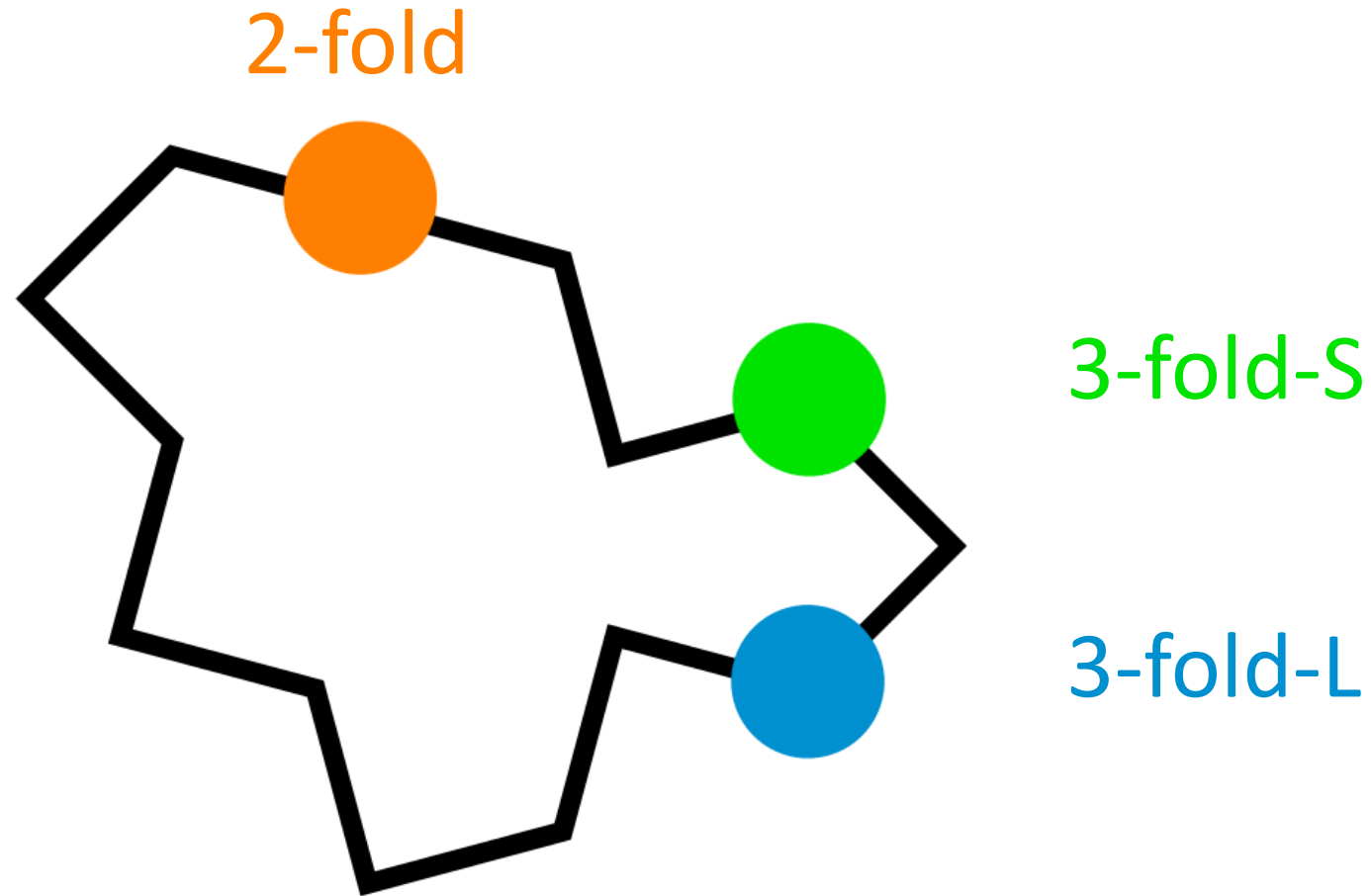
3-fold-S



3-fold-L

- EVEN monotiles are M2, M4, M6, M8, M10 and M12

# Location on the contour of one EVEN monotile

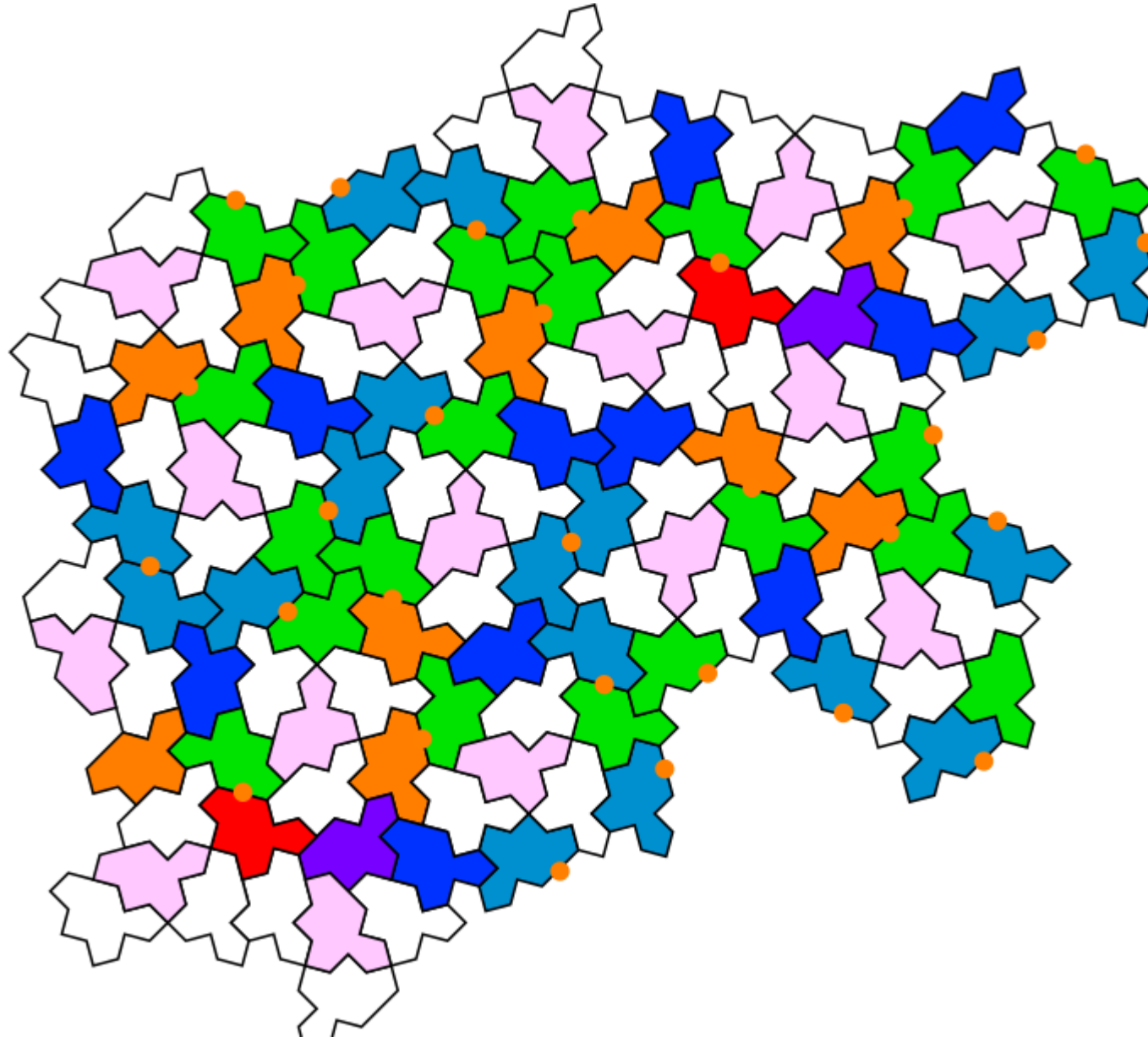


- This figure shows the origin of the two different, S and L, 3-fold centers
- It is linked with the chirality of the spectre tile (here with left handiness)

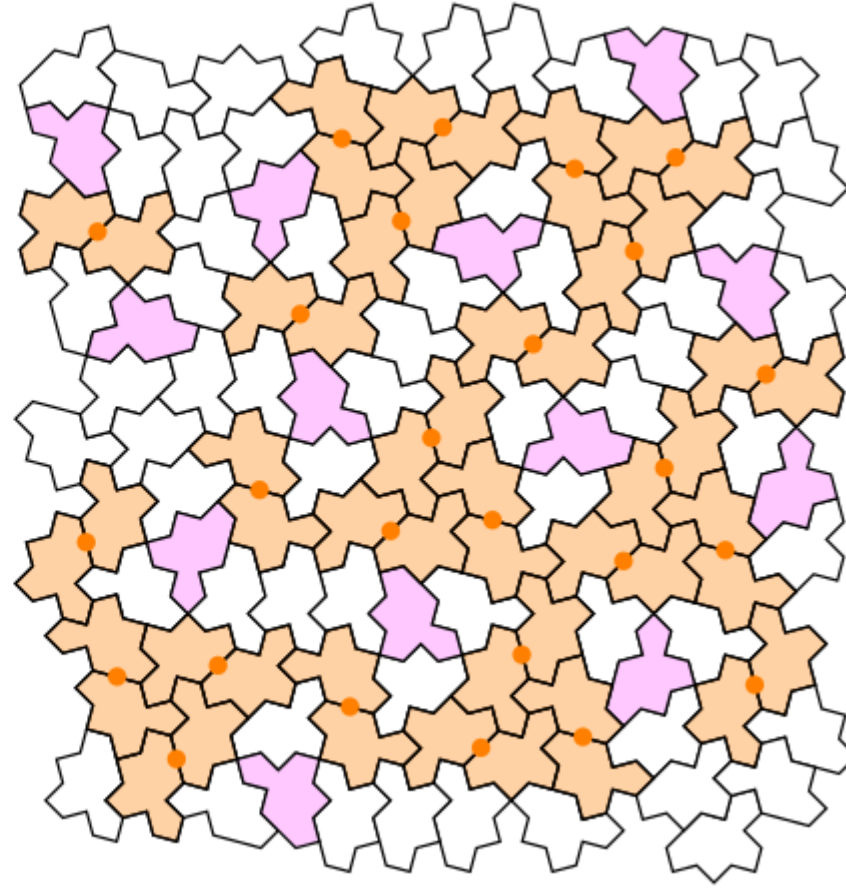
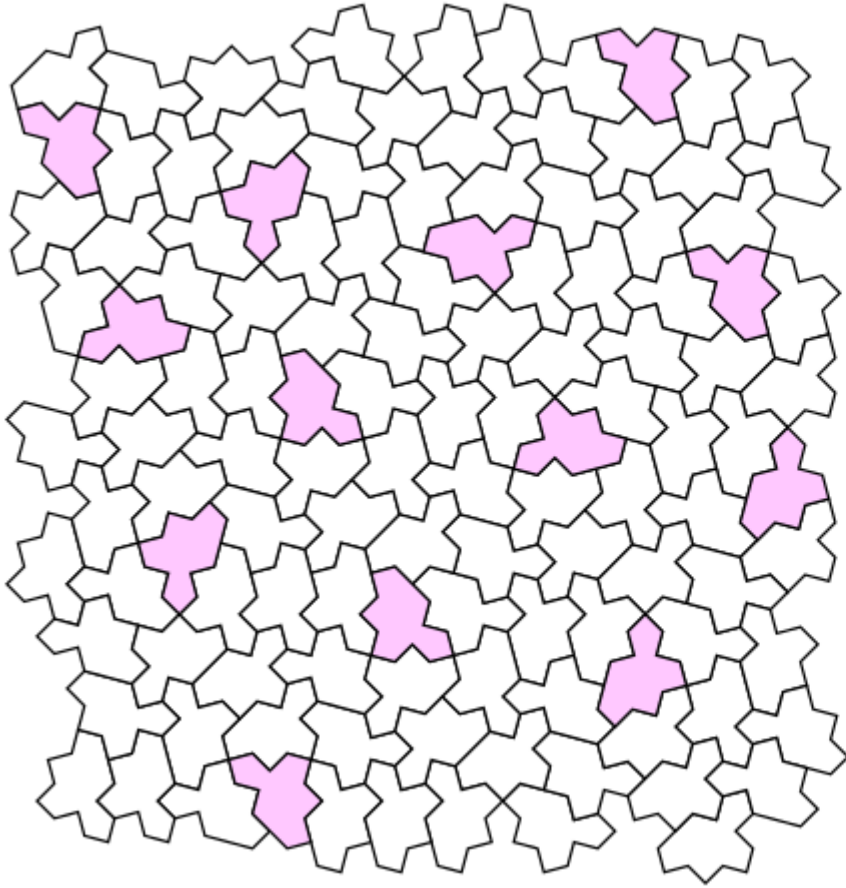


# Local two fold symmetry centers

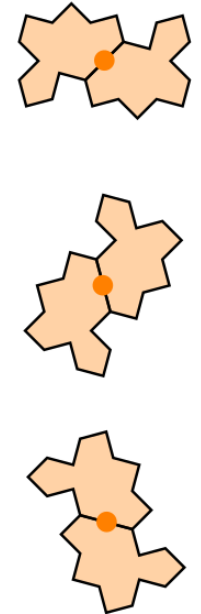
2-fold



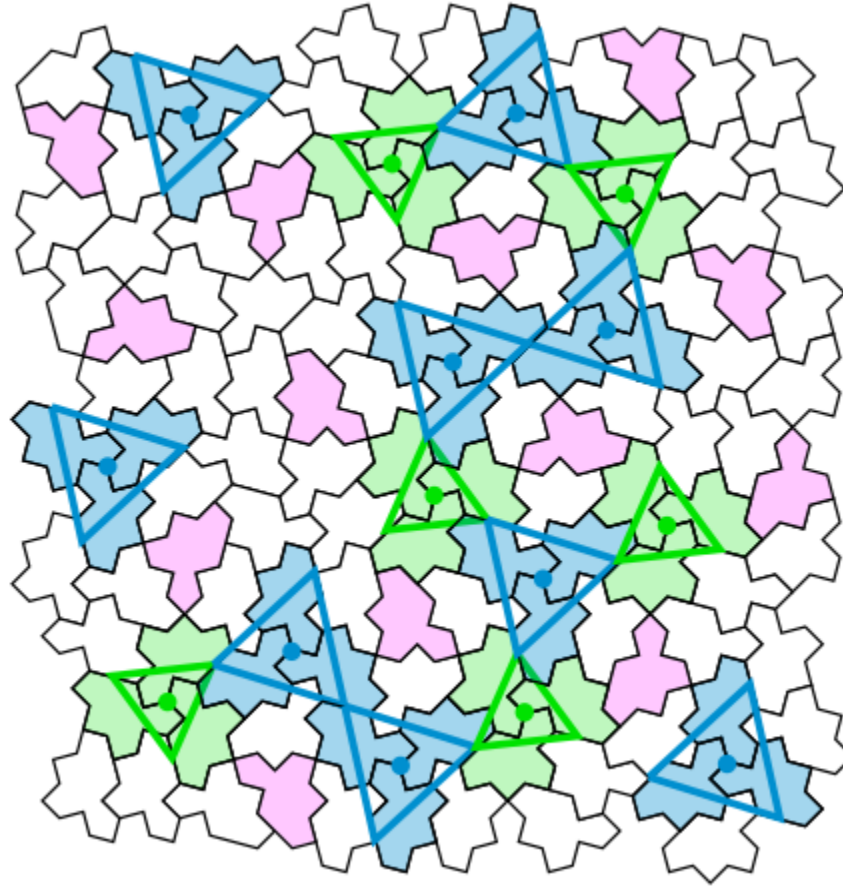
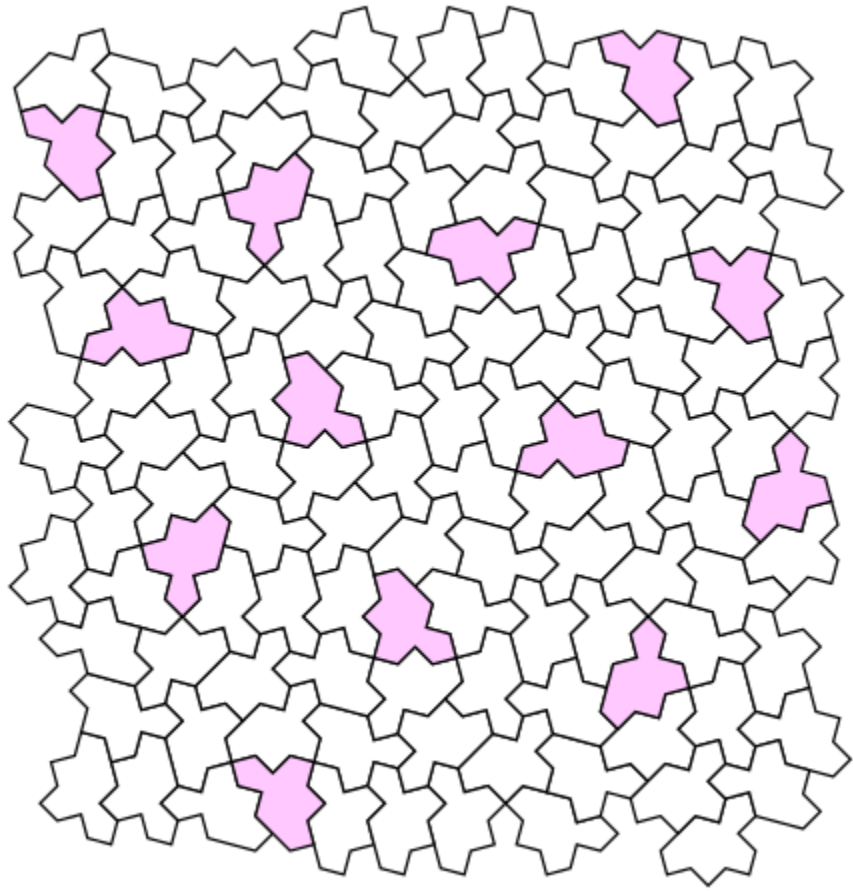
# Example patch with all local 2-fold centers (orange color)



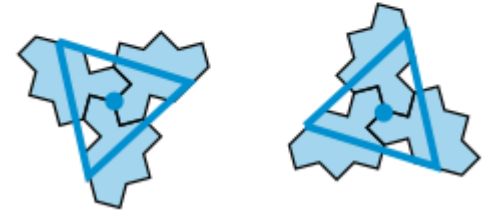
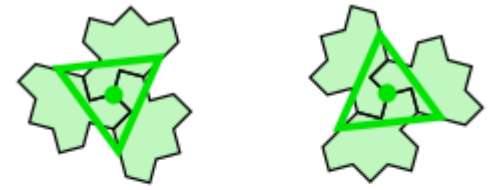
Local 2-fold



# Same patch with all local 3-fold centers (green and blue colors)



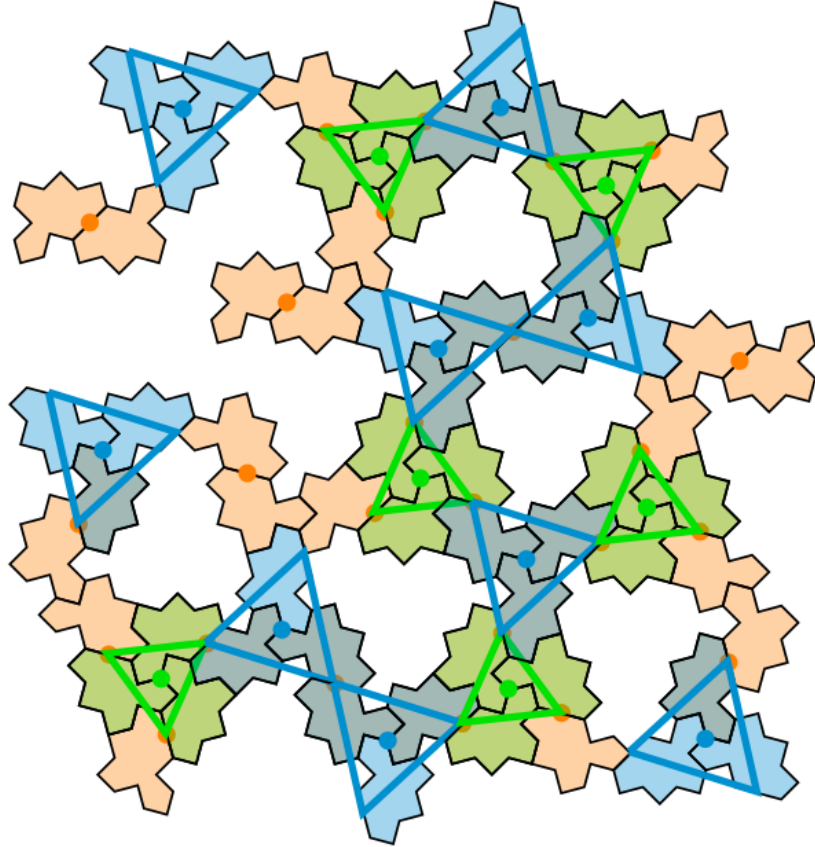
3-fold-S



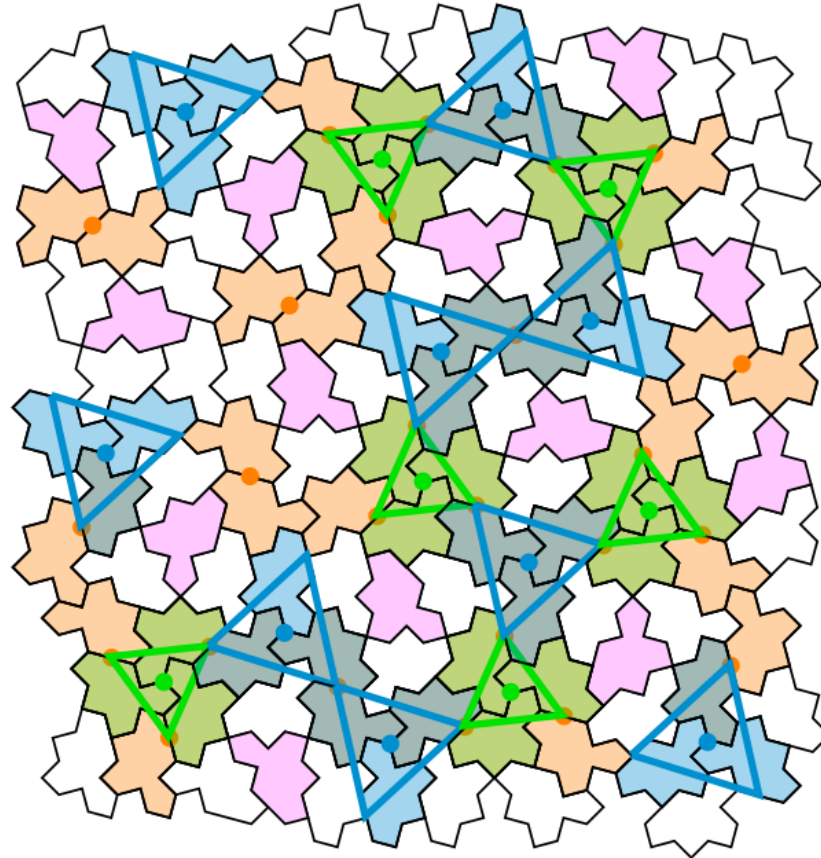
3-fold-L

# Superposition of (orange/blue/green) colors

2-fold



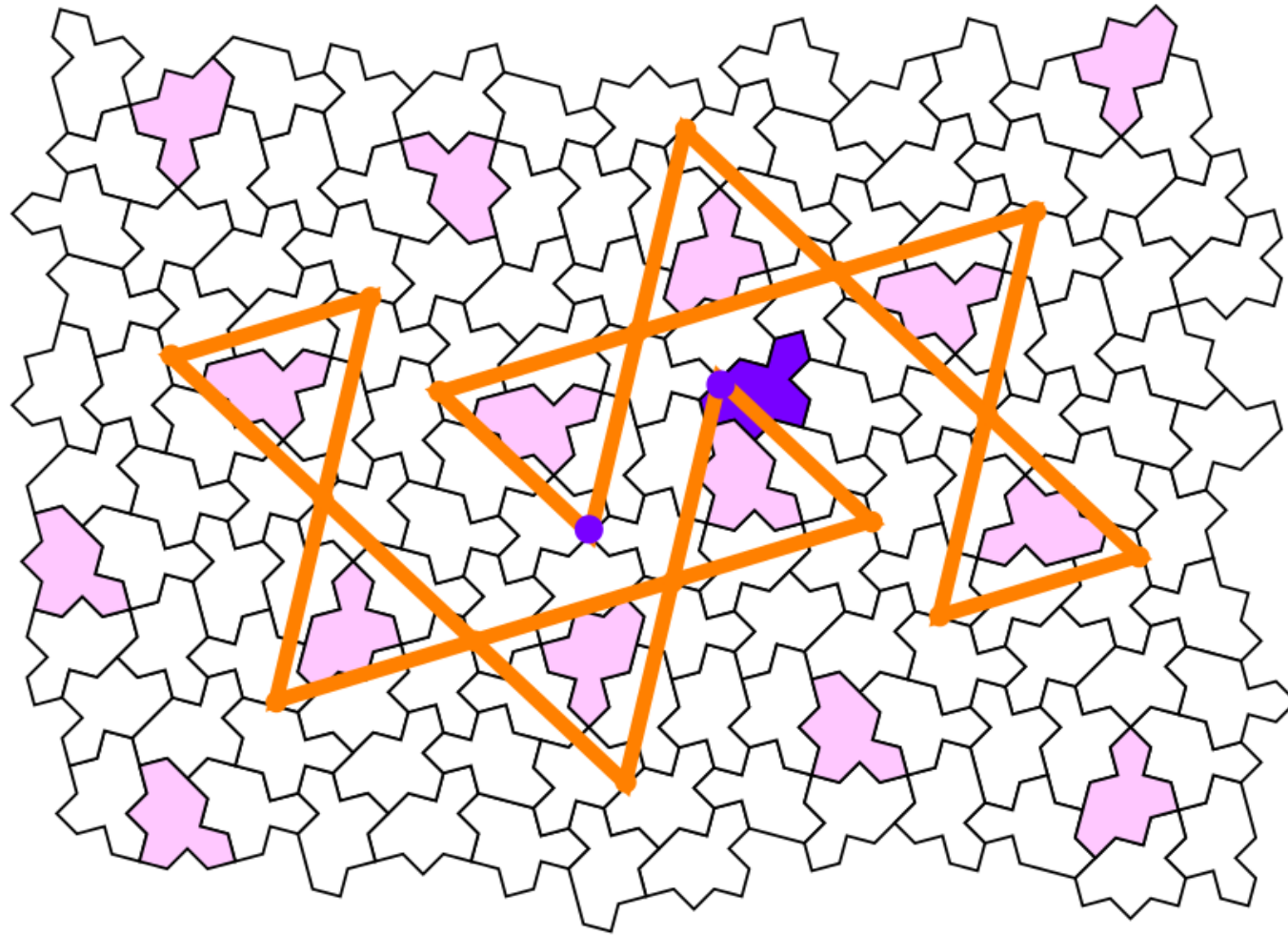
3-fold-S



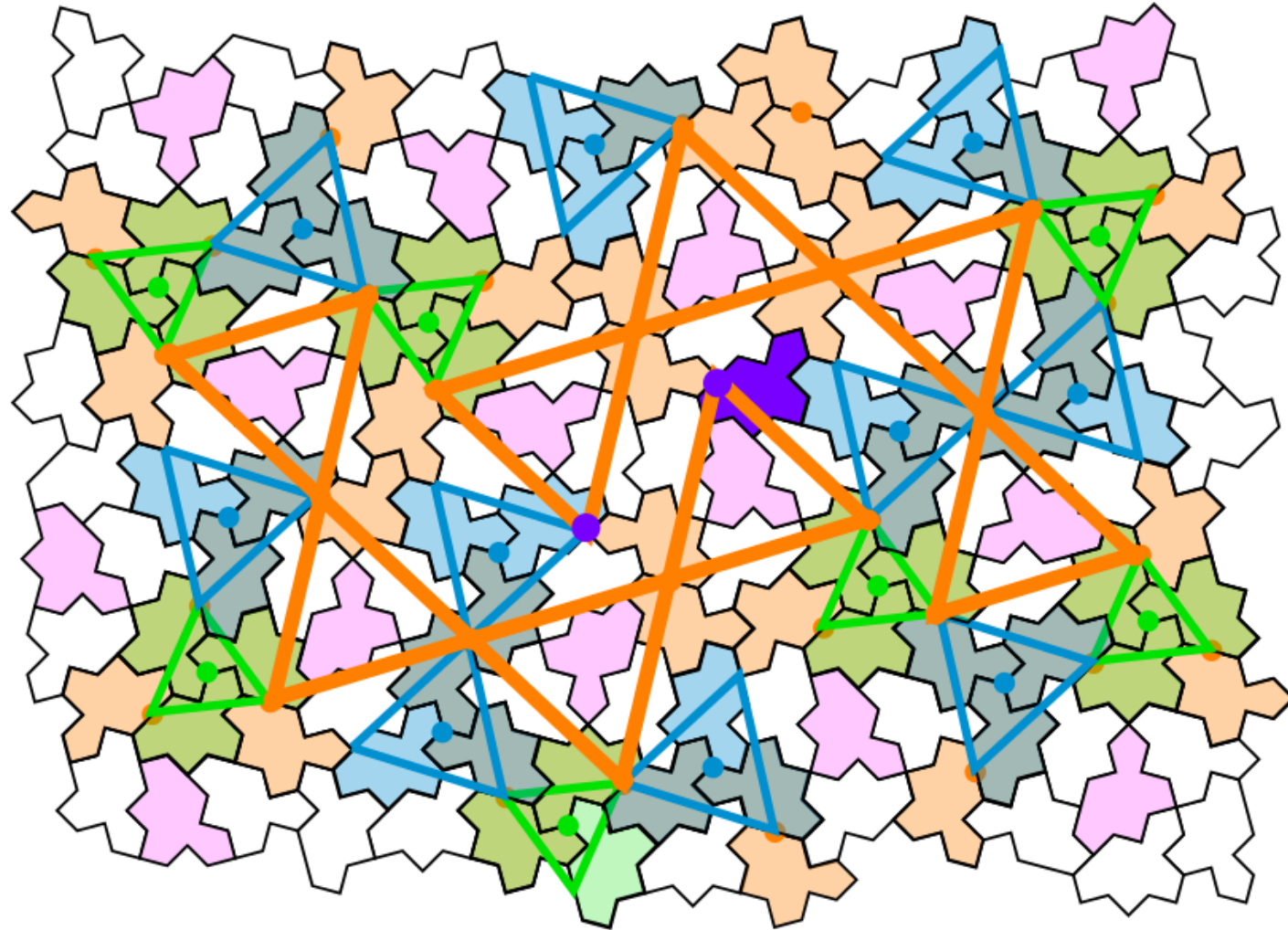
3-fold-L



## 2-arms

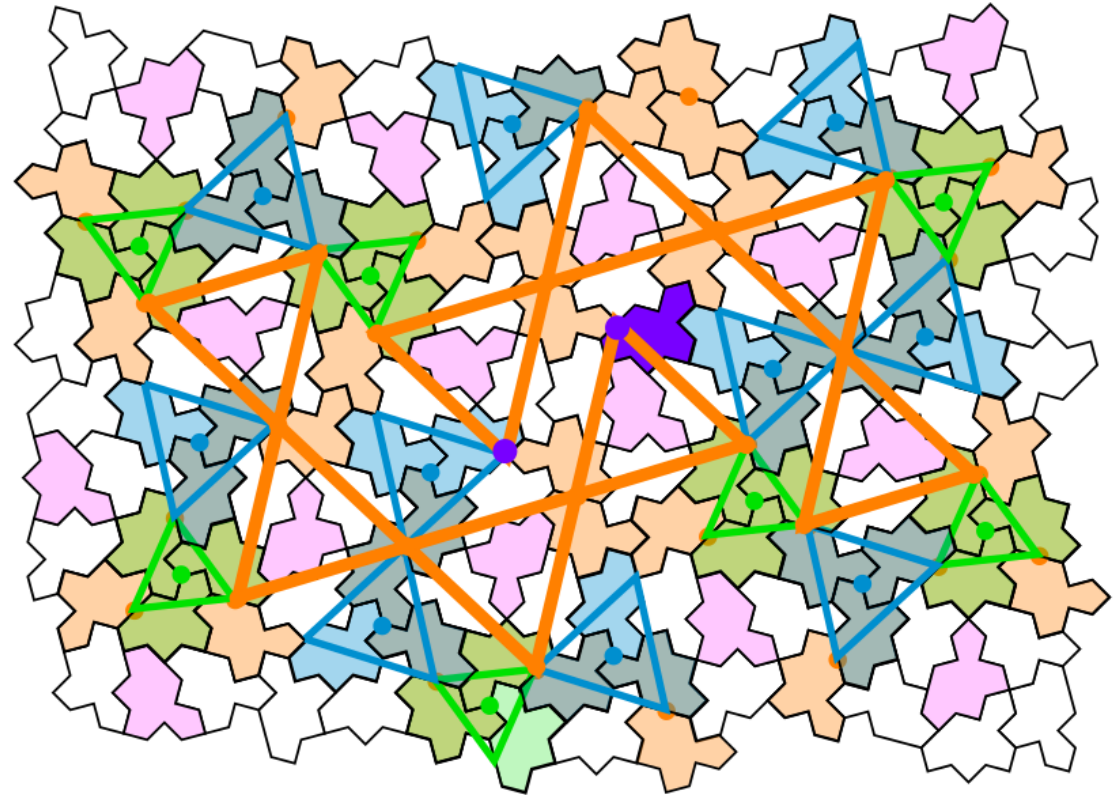
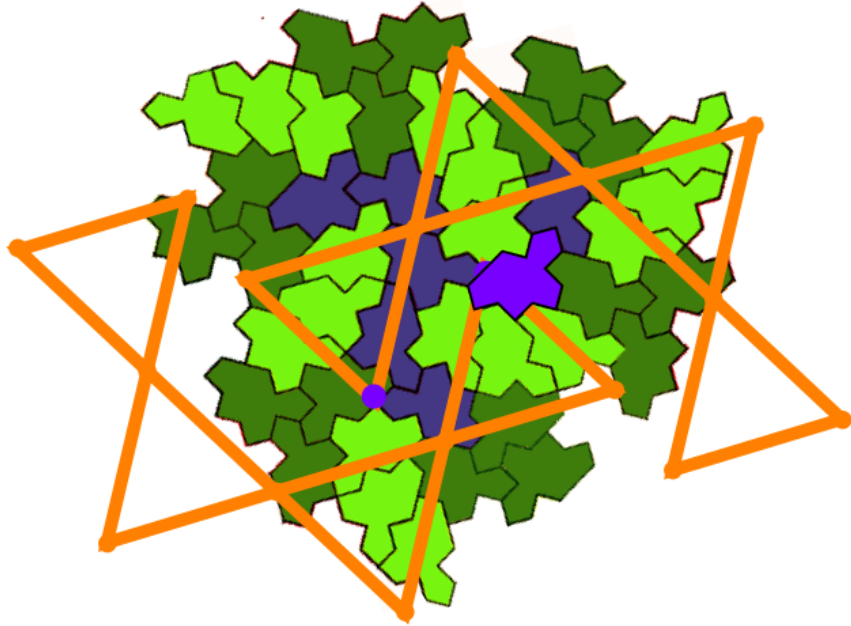


ALL EVEN tiles:  
orange/blue/green colors + one violet isolated tile



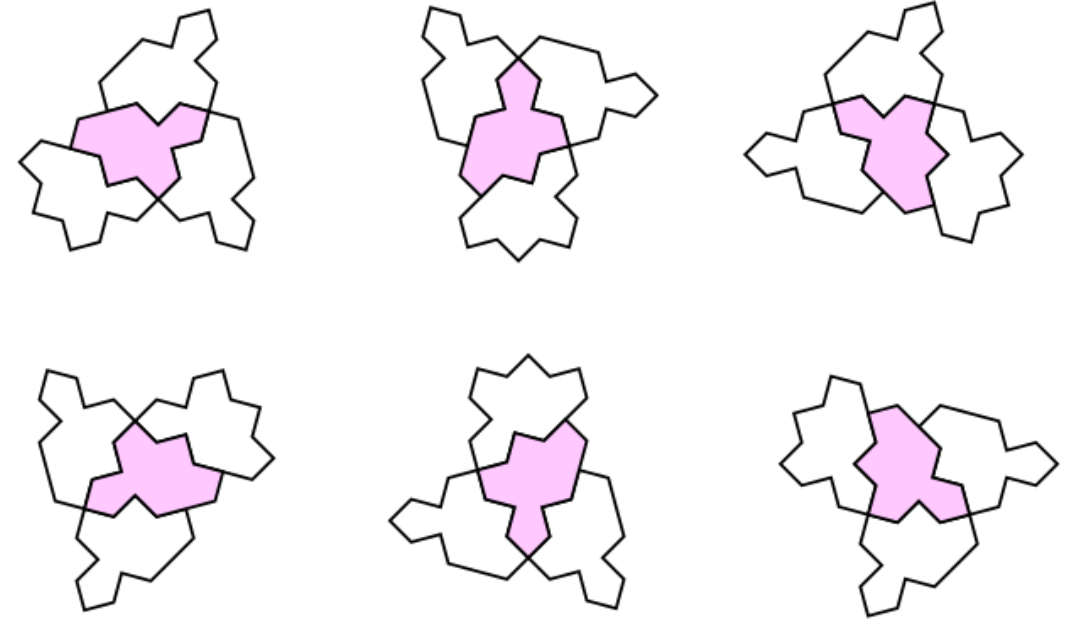
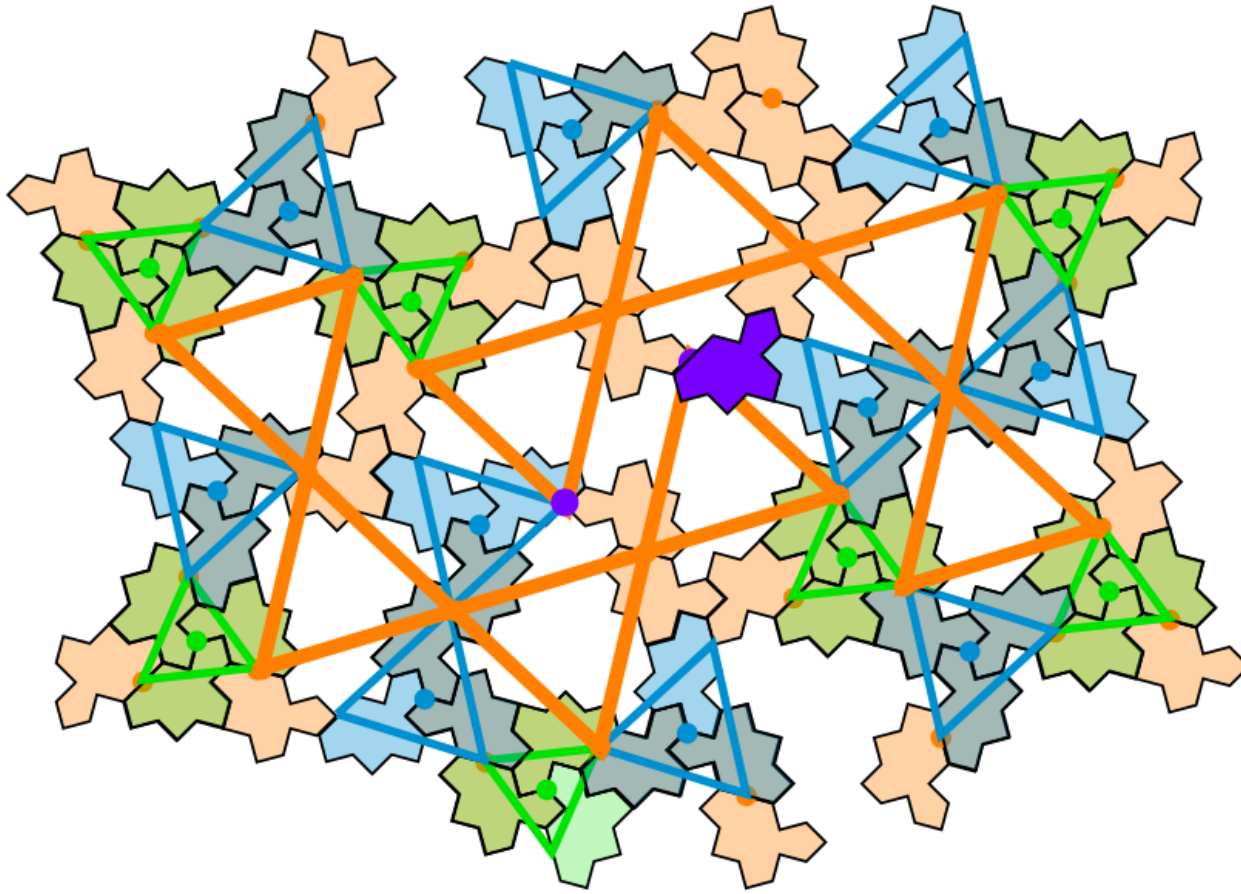
■ 2-arms

ALL EVEN tiles:  
orange/blue/green colors + one violet isolated tile



- 2-arms and superposition with chimera pattern

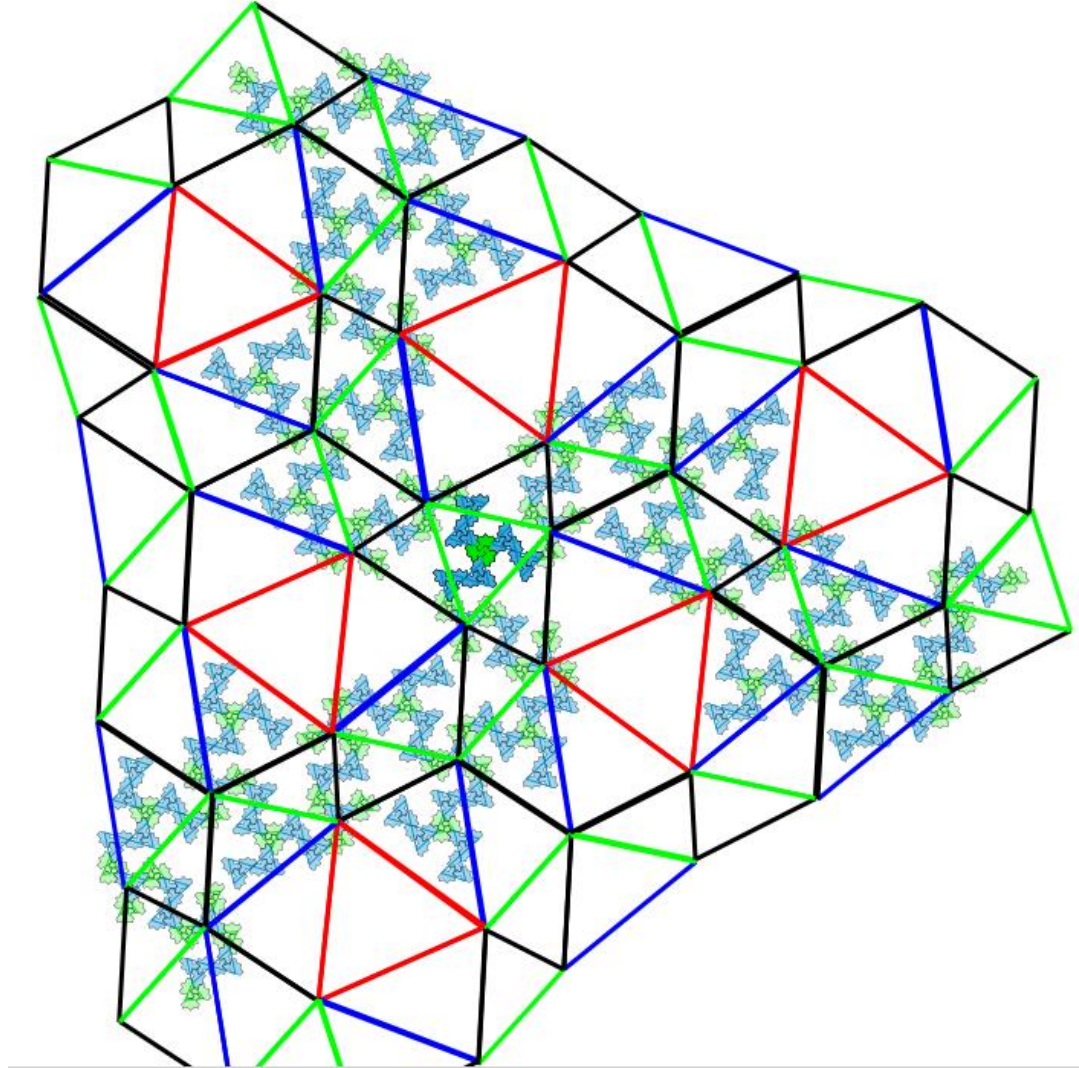
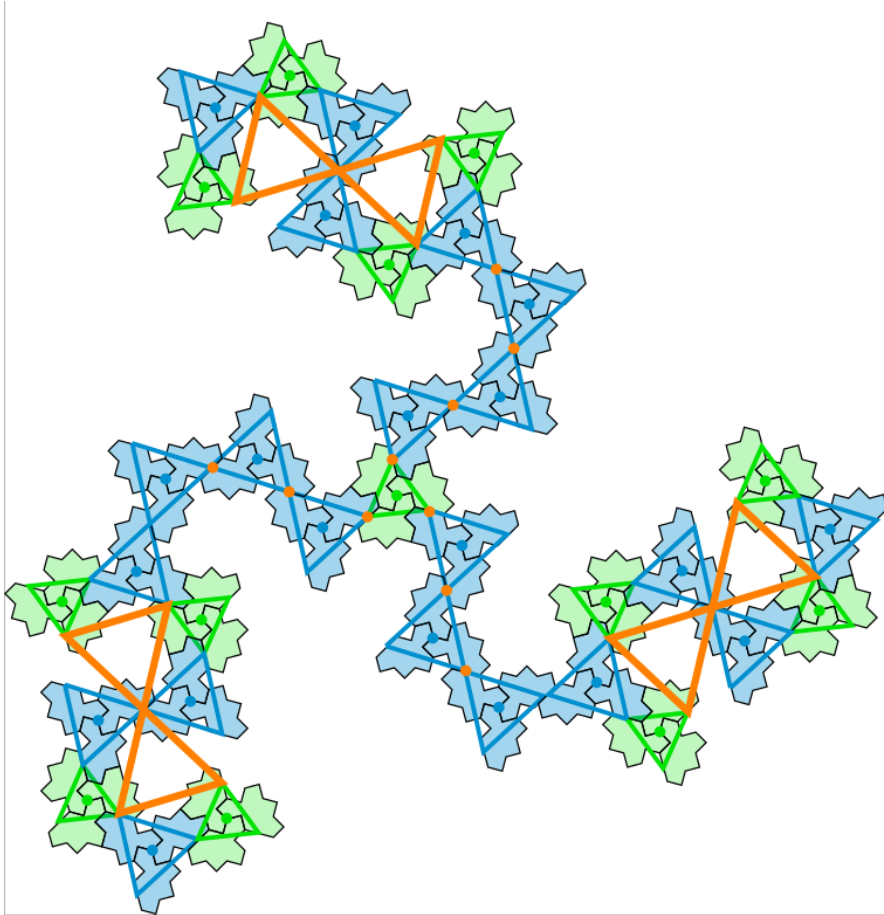
Orange/blue/green + violet: a continuous cluster of ODD tiles around small clusters.



- Small clusters of 4 tiles (one ODD in pink  $\gamma_2$  + three EVEN  $\gamma_1$ ,  $\Delta$  and  $\Sigma$ )

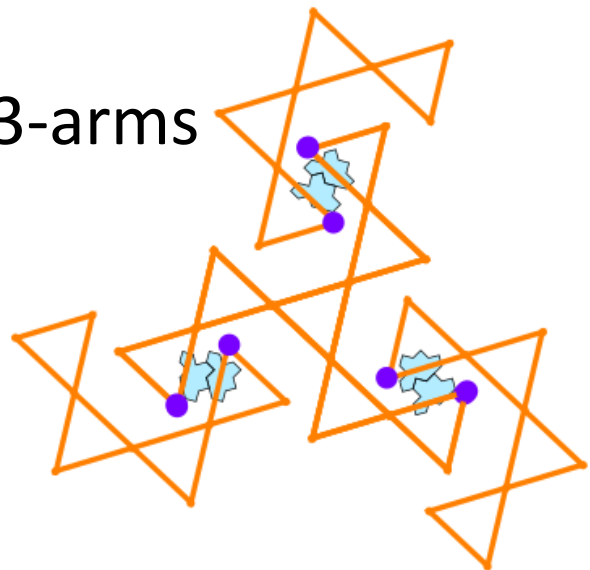


To continue ? generate star clusters by applying locally 2-fold and 3-fold symmetry between EVEN monotiles

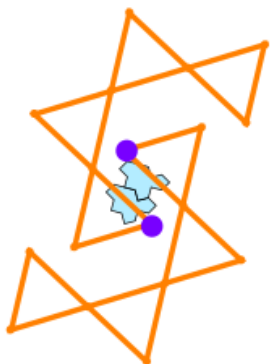


# Three patterns in the whole tiling 😊

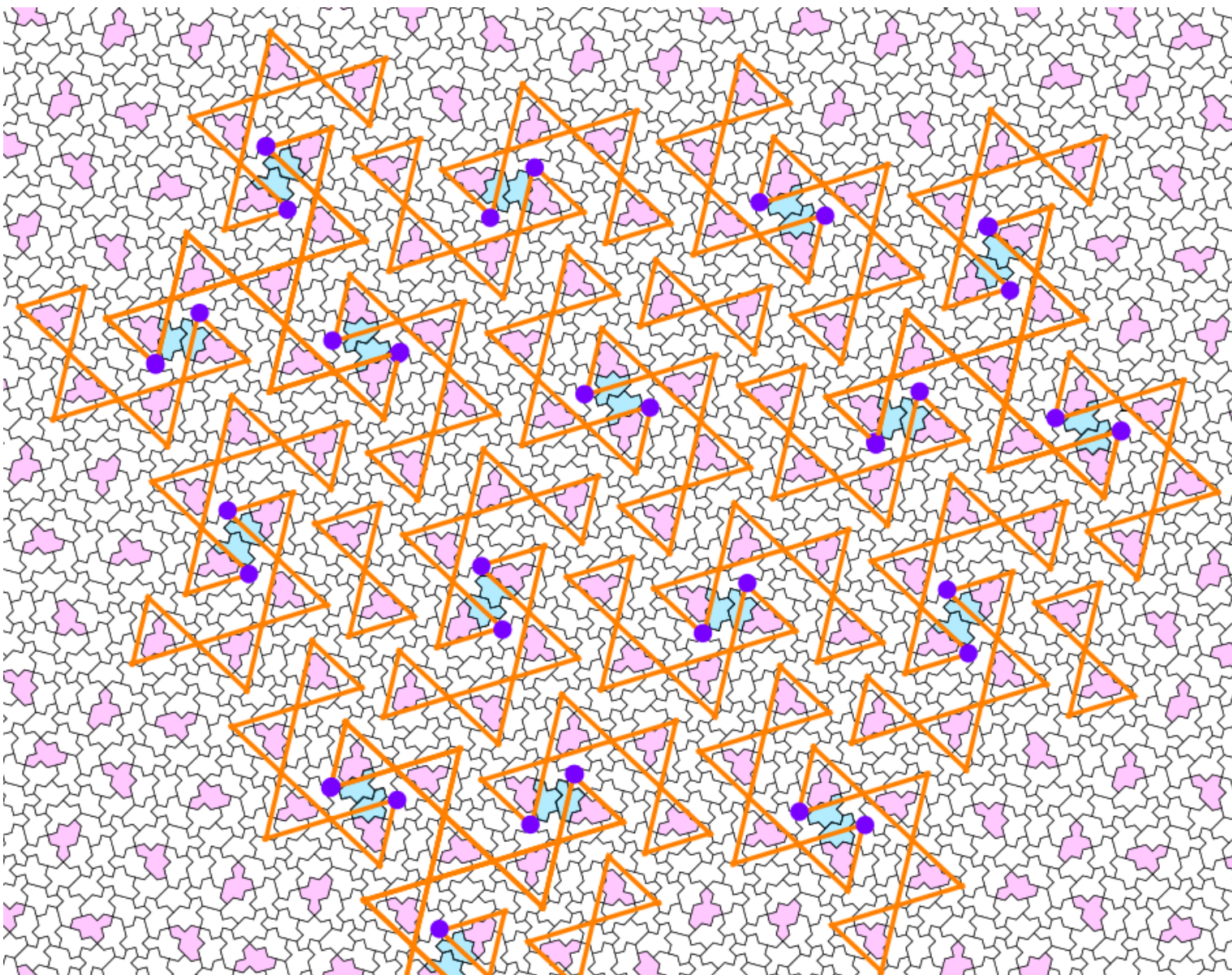
■ 3-arms



■ 2-arms

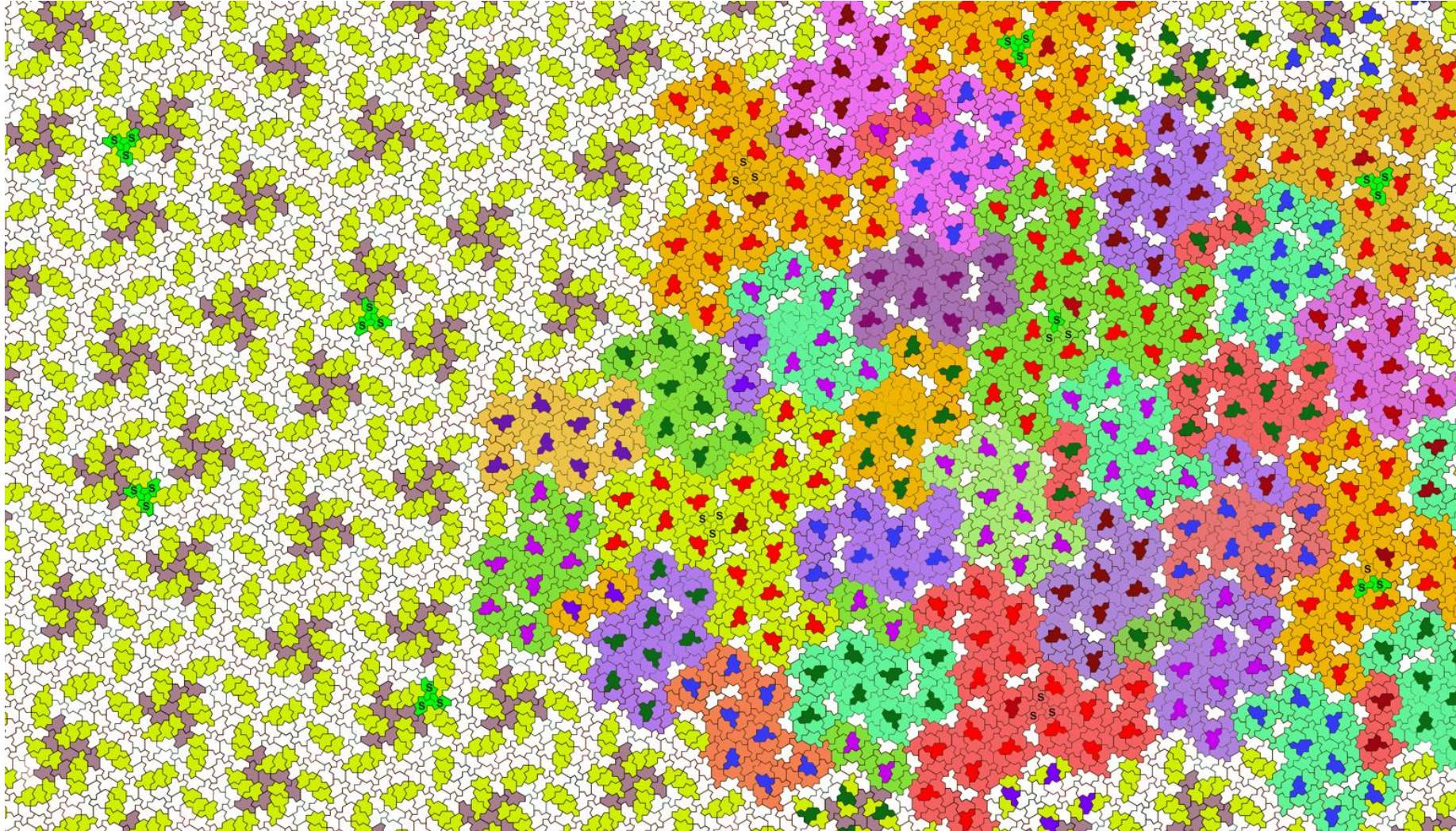


■ diabolos





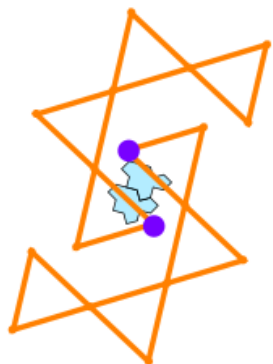
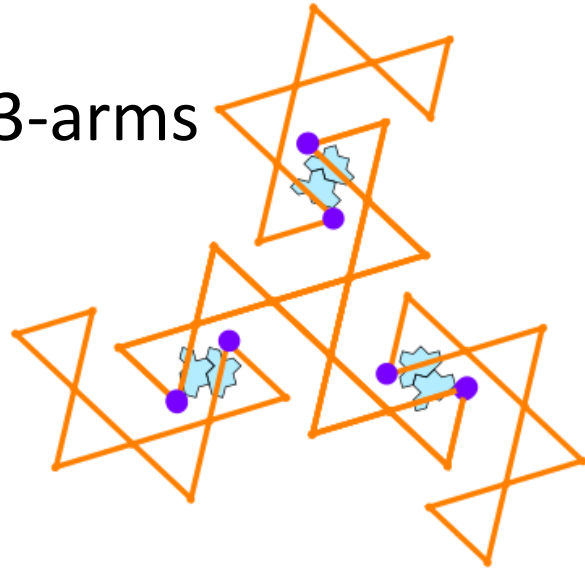
# Diabolo/2-arms/3-arms at level 2 by Boris Horvat





# Frequency of the three patterns (Boris Horvat)

## ■ 3-arms



## ■ 2-arms

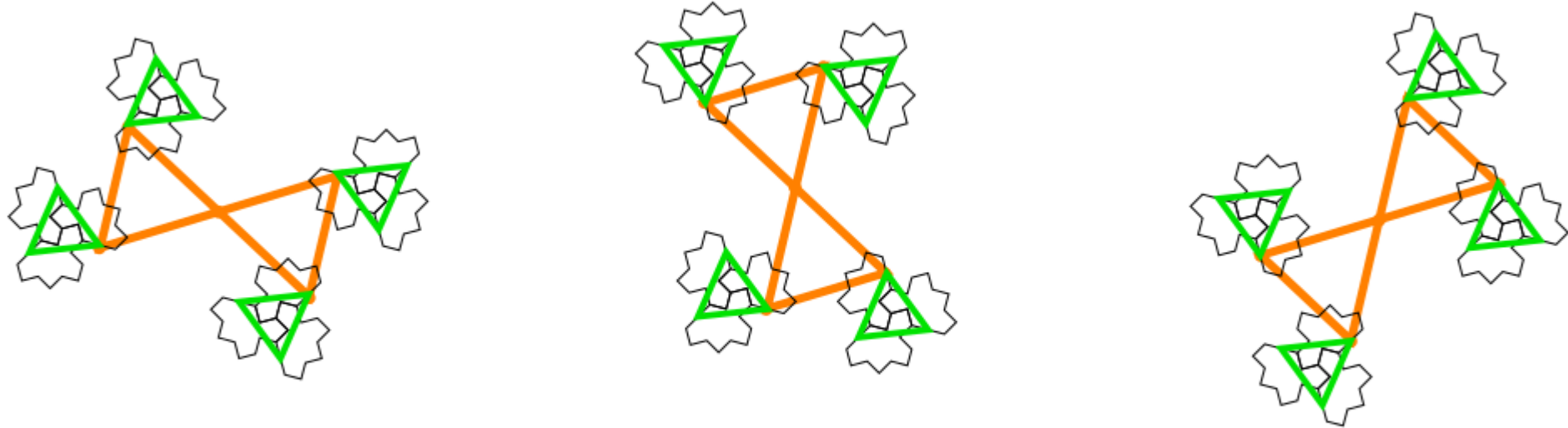
## ■ diablo



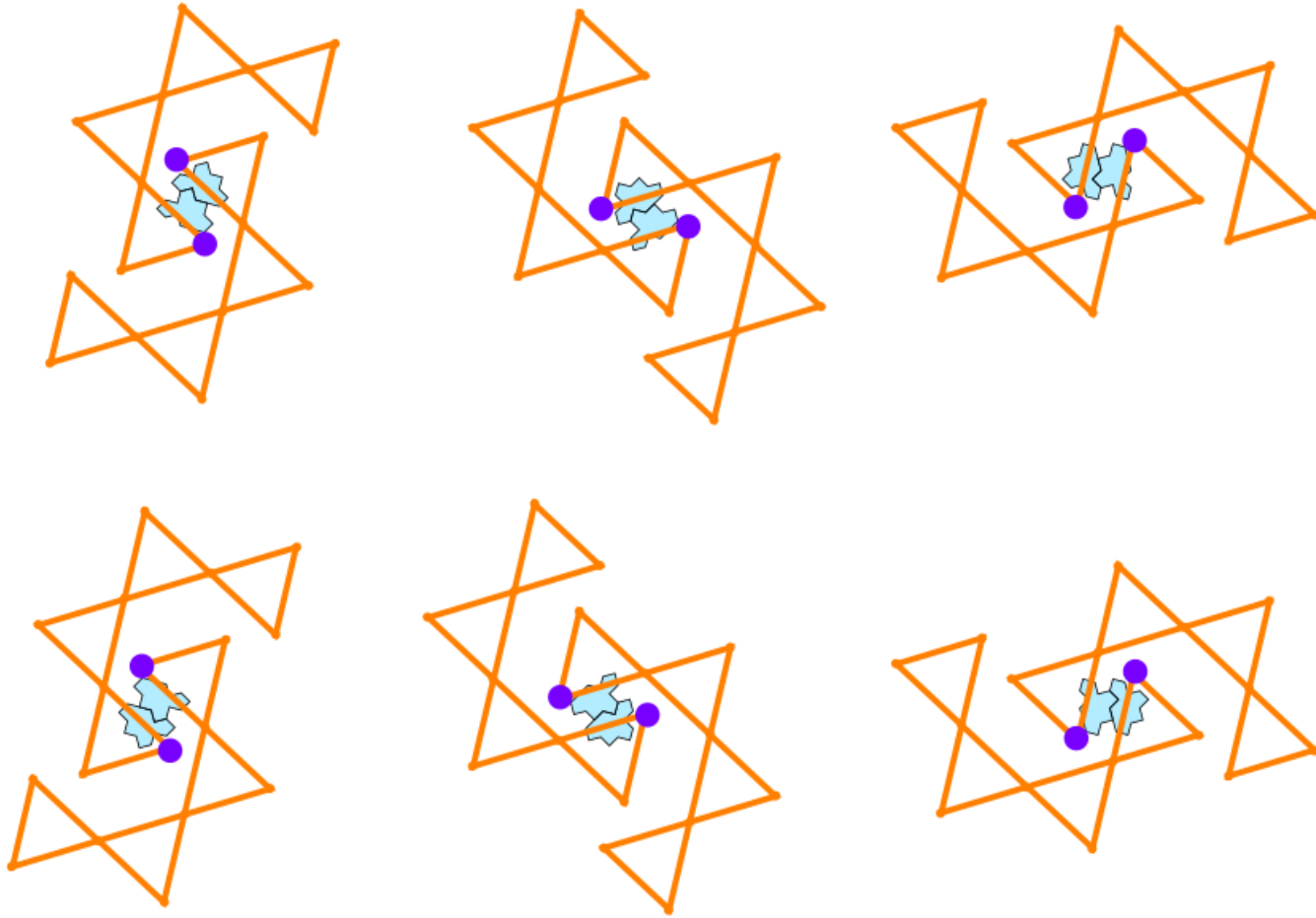
After colouring (manually) the tiling with your objects, I realized that the frequency of diabolos is exactly twice the frequency of 3-way objects. My observations confirm that **every 3-way object (Object-19) has exactly two diabolos attached to two of its arms in a consistent topological orientation**. This allows us to define a new primary super-structure: Object-23 ( $1 \times 3\text{-way} + 2 \times \text{Diablo}$ ), thus partitioning the tiling into two large objects only (Object-23, and Object-8). The 3-way objects (the nodes of the tiling, centred around lone stars) connect to each other via three distinct modes:

- 1) Direct Contact: Arm-to-arm contact between two 3-way objects (Object-19).
- 2) Diablo Link: Connection mediated by a single diablo (Object-2).
- 3) 2-way link: Connection mediated by a 2-way object (Object-8).

# three orientations for diabolos

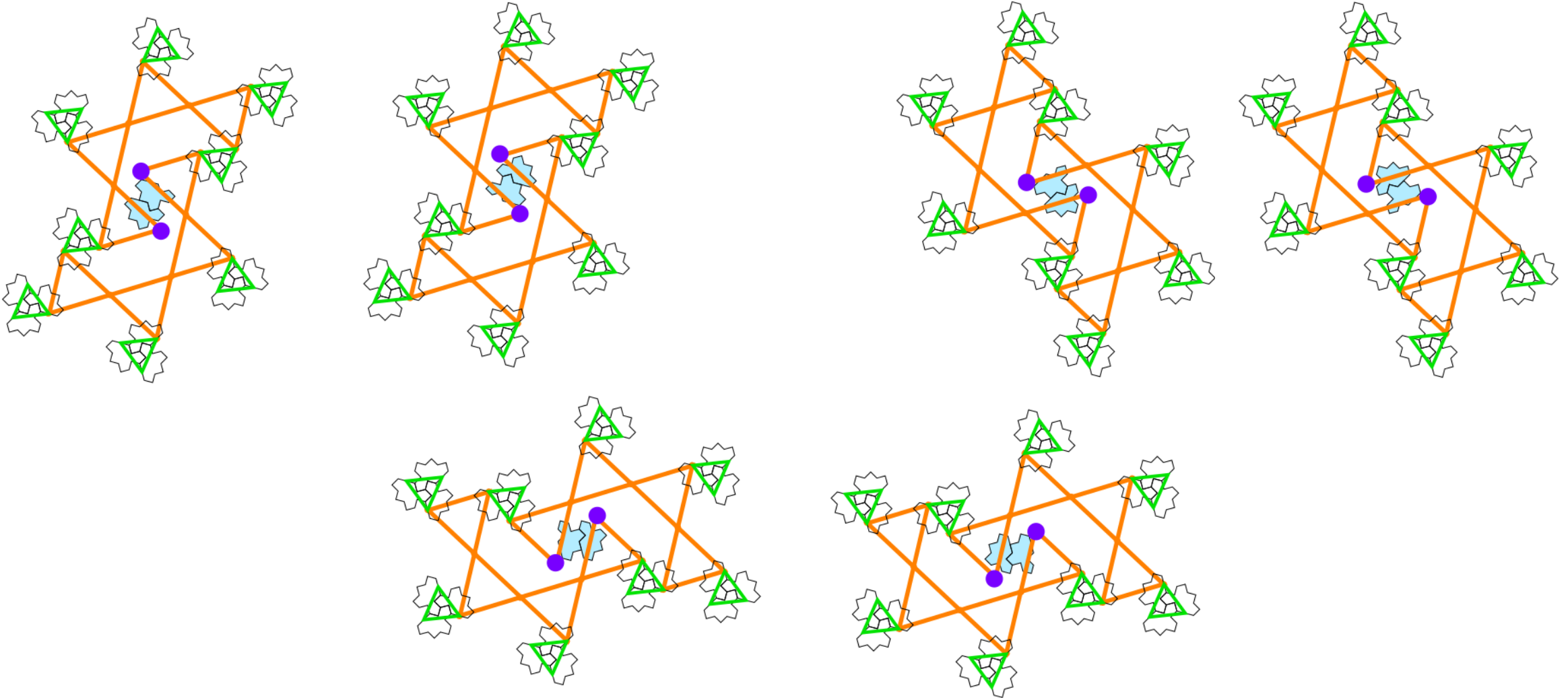


# Six orientations of the 2-arms

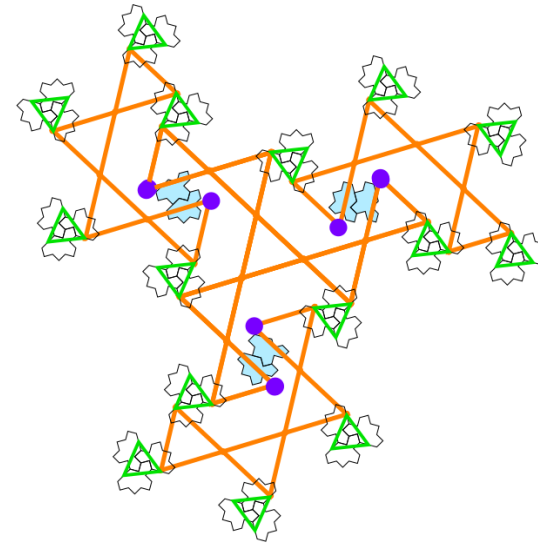
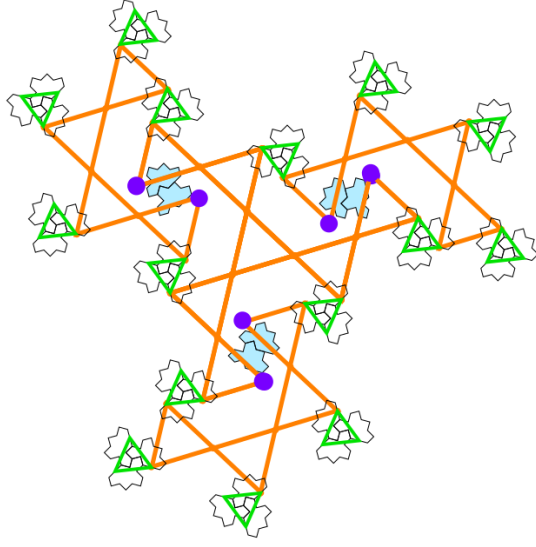
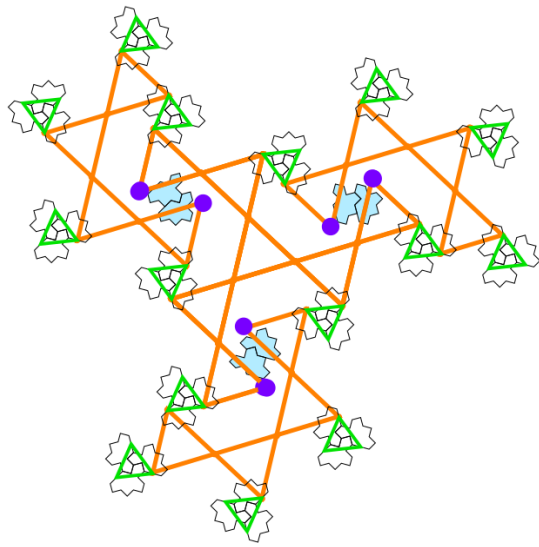
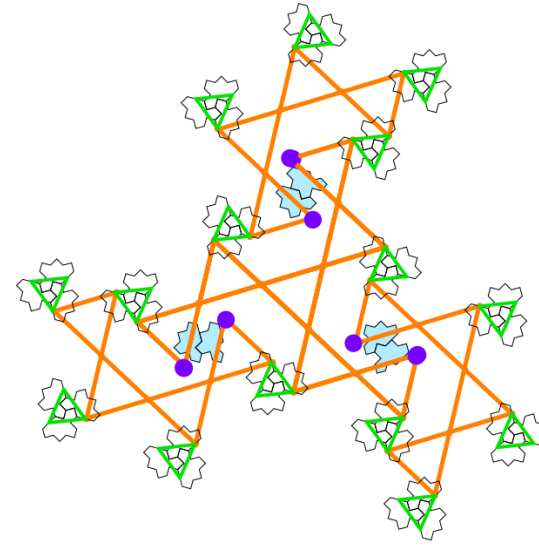
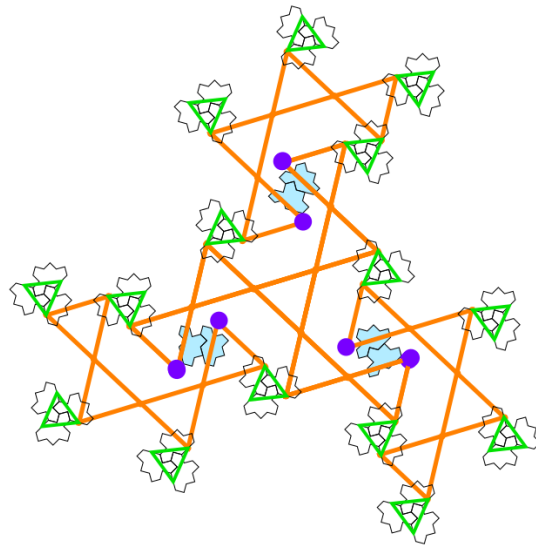
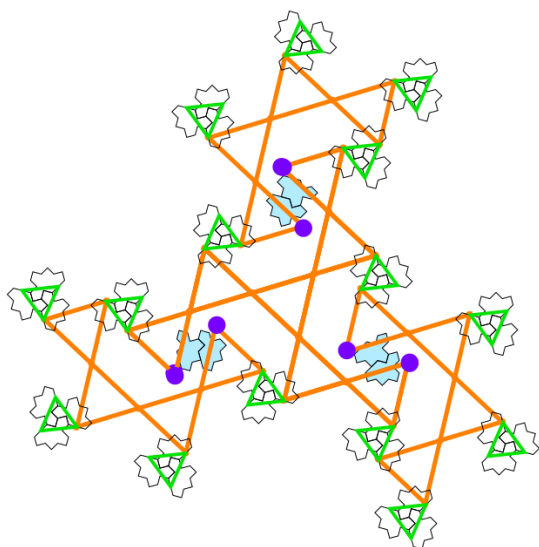




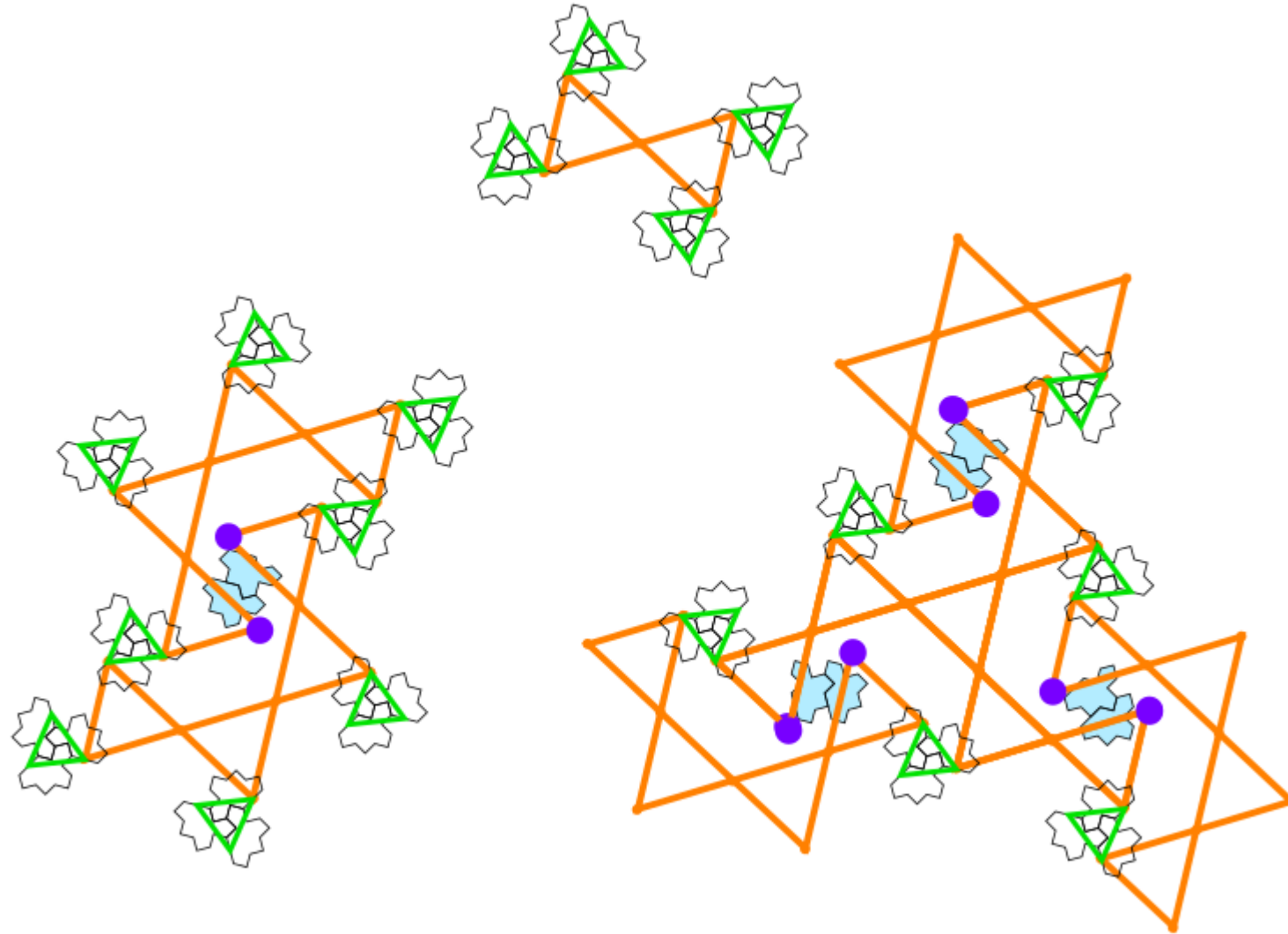
# Six orientations of the 2-arms



# six orientations of the 3-arms

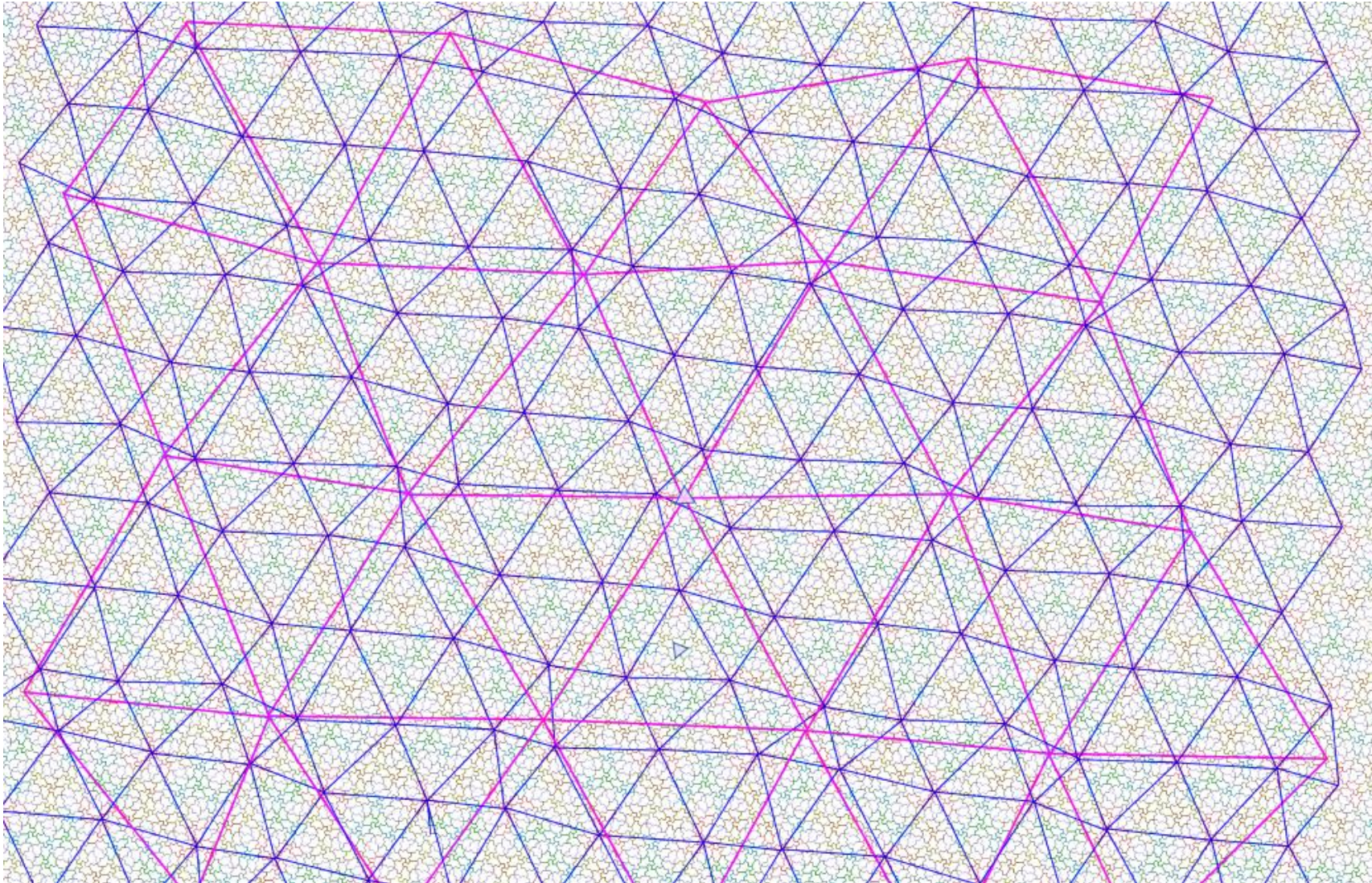


# New puzzle





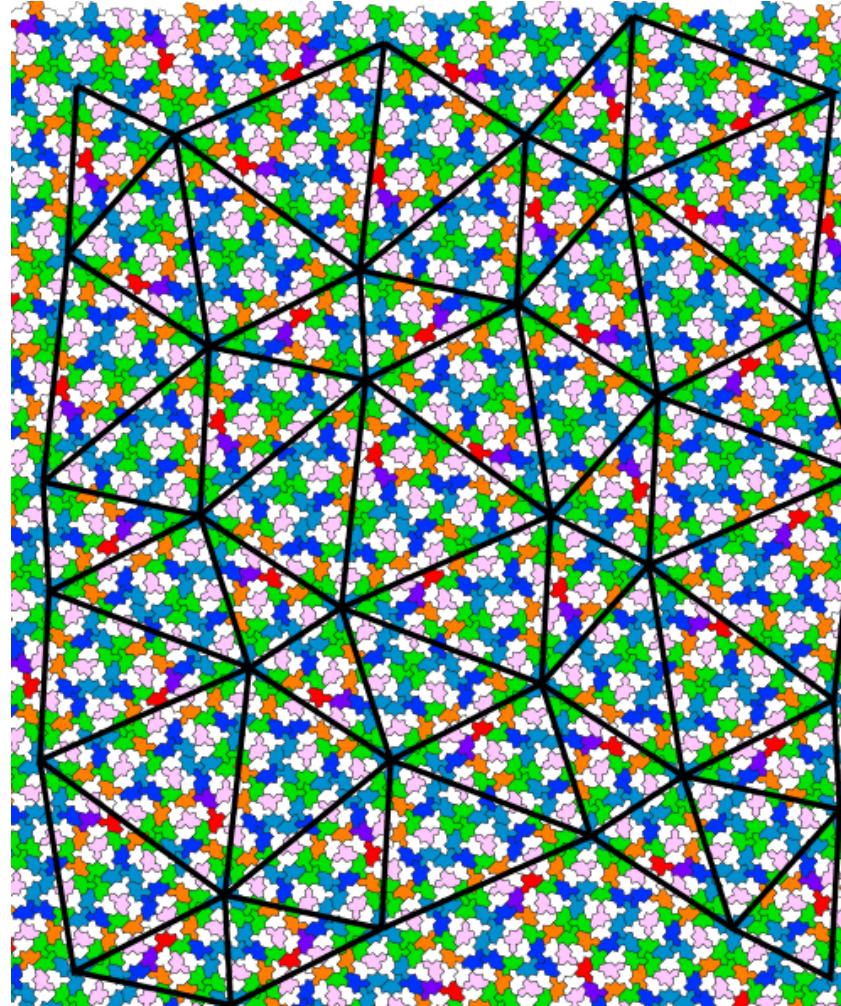
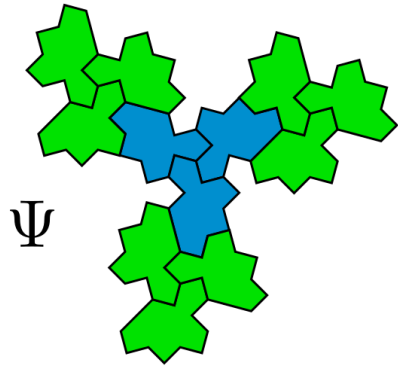
# Triangular array by Boris Horvat



**blue color: triangular array between the “lone stars”**



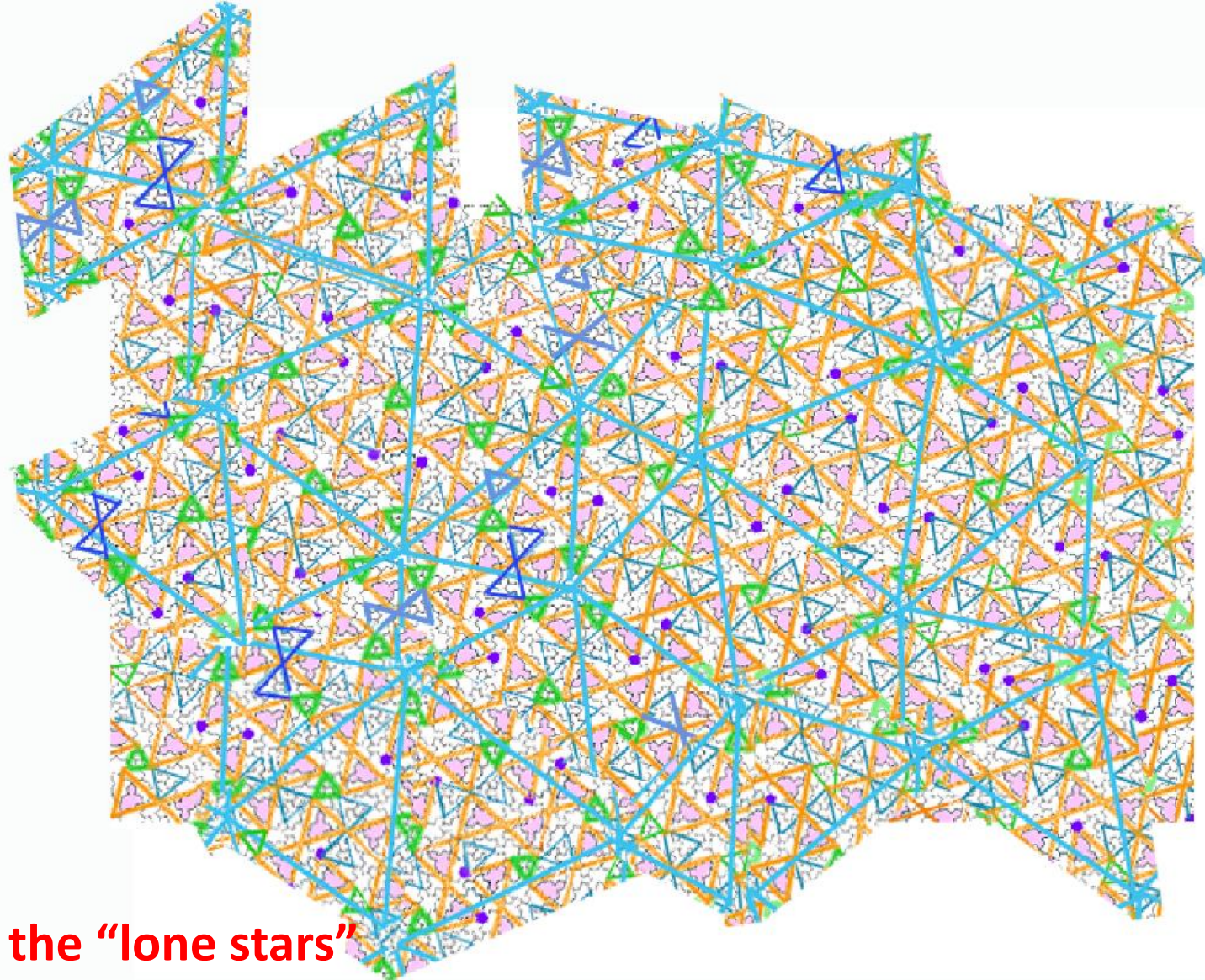
# Inflation using python code



- Triangular array between large blue stars made of three  $\Psi$  tiles surrounded by three S2 green stars

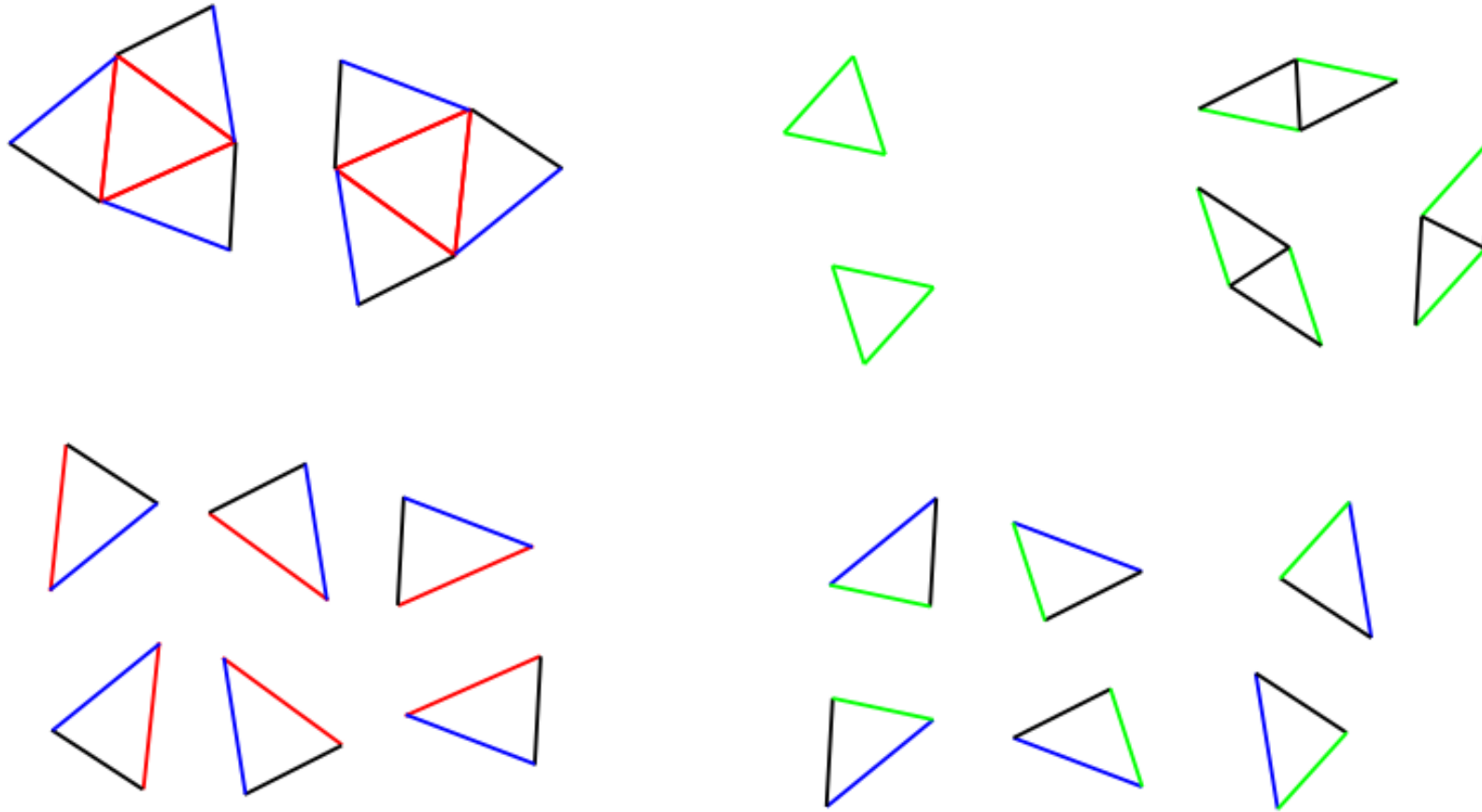


# Triangular networks by Jean-François



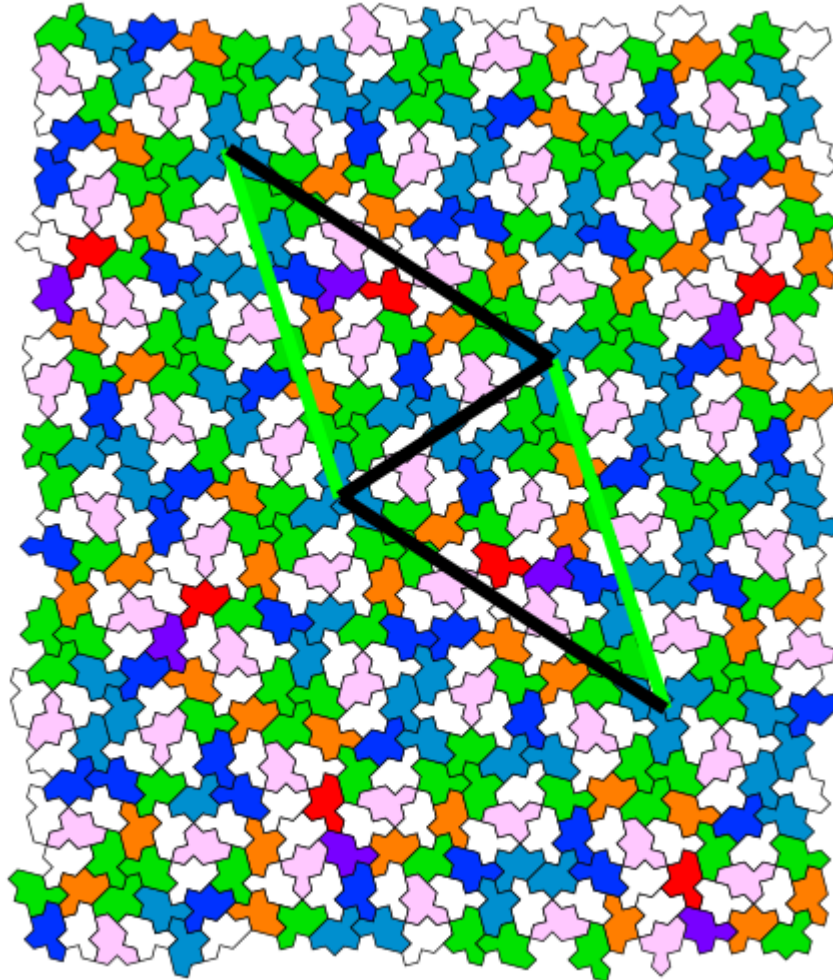
Same network between the “lone stars”

# A finite set of triangles ☺



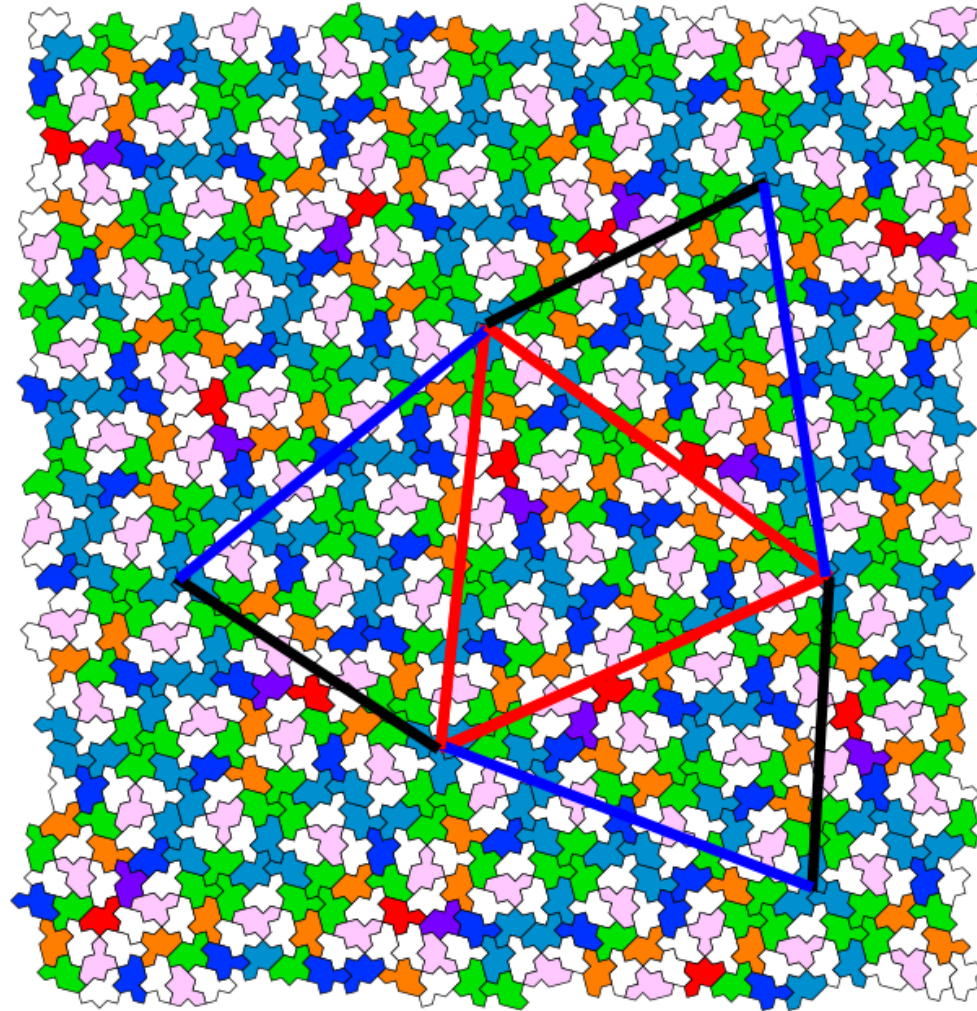
**5 different triangles with two regular ones -> Calculate the edge's lengths ?**

# Set of triangles: diabolos



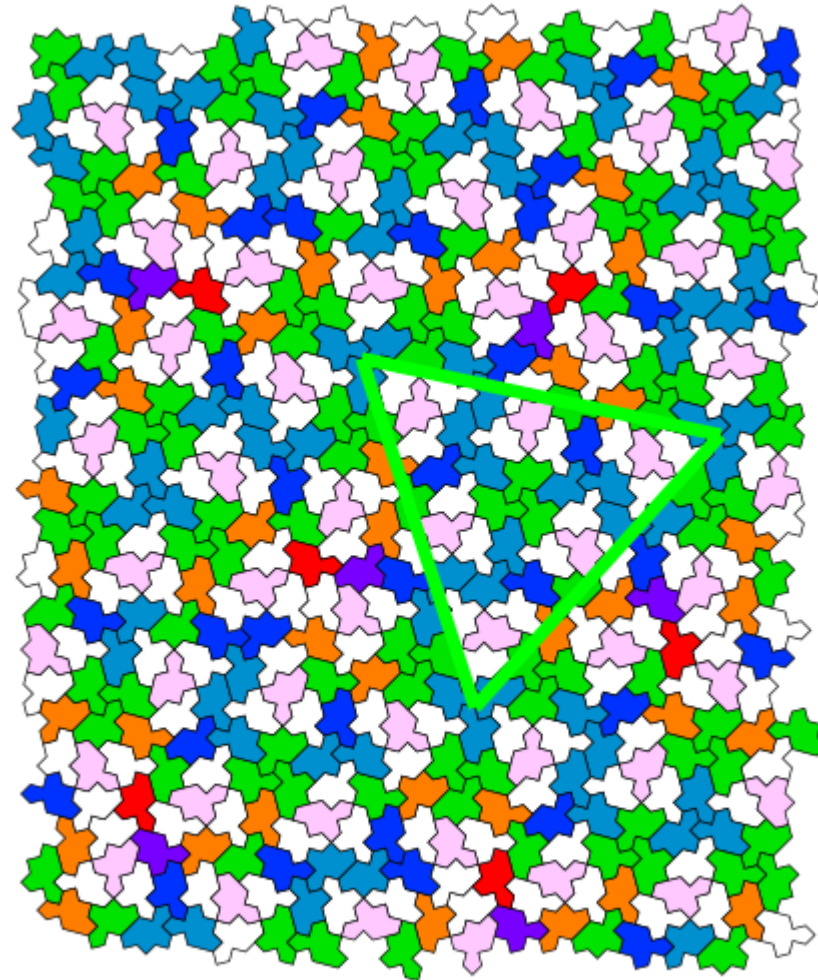


# Set of triangles: three arms

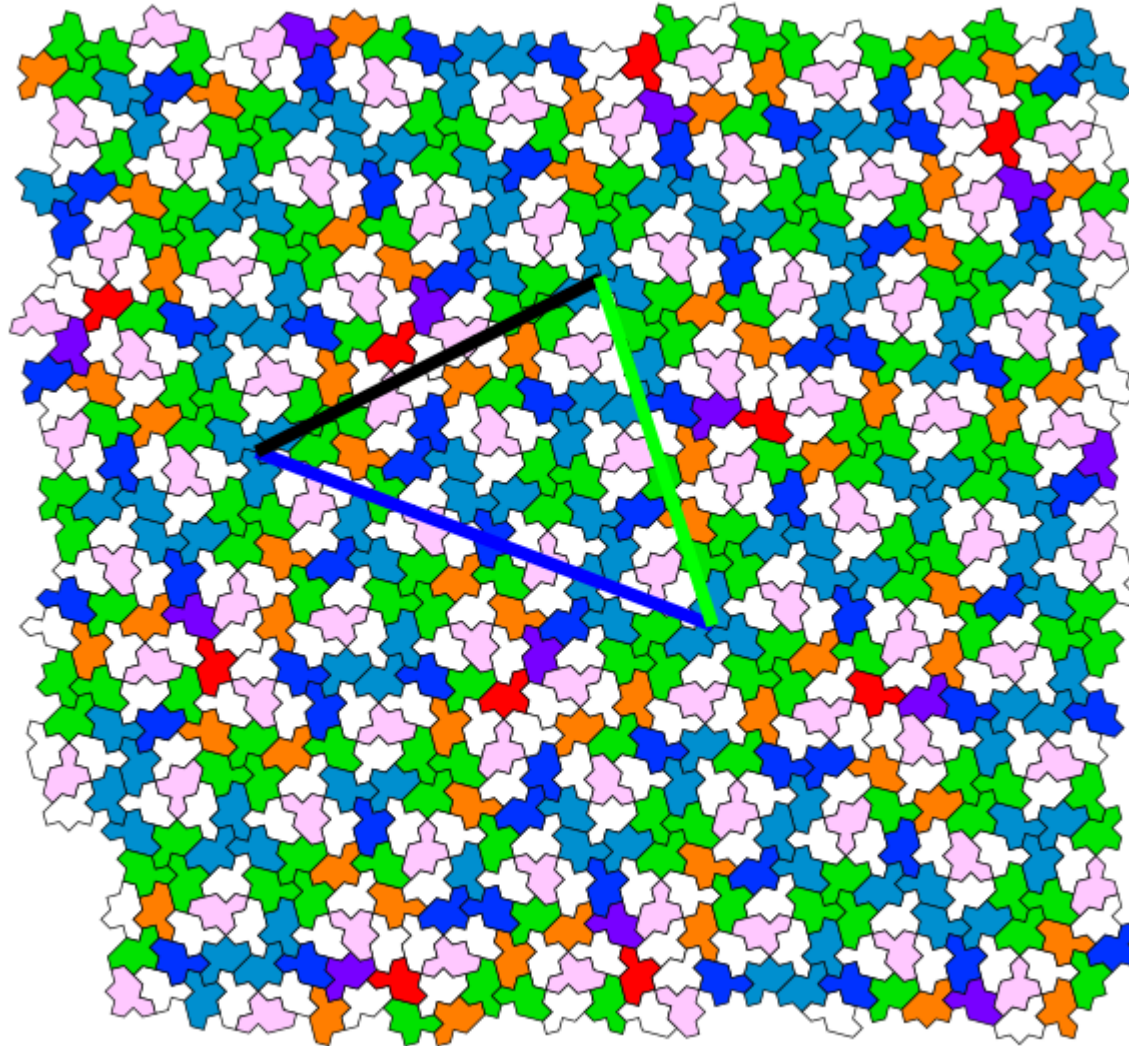




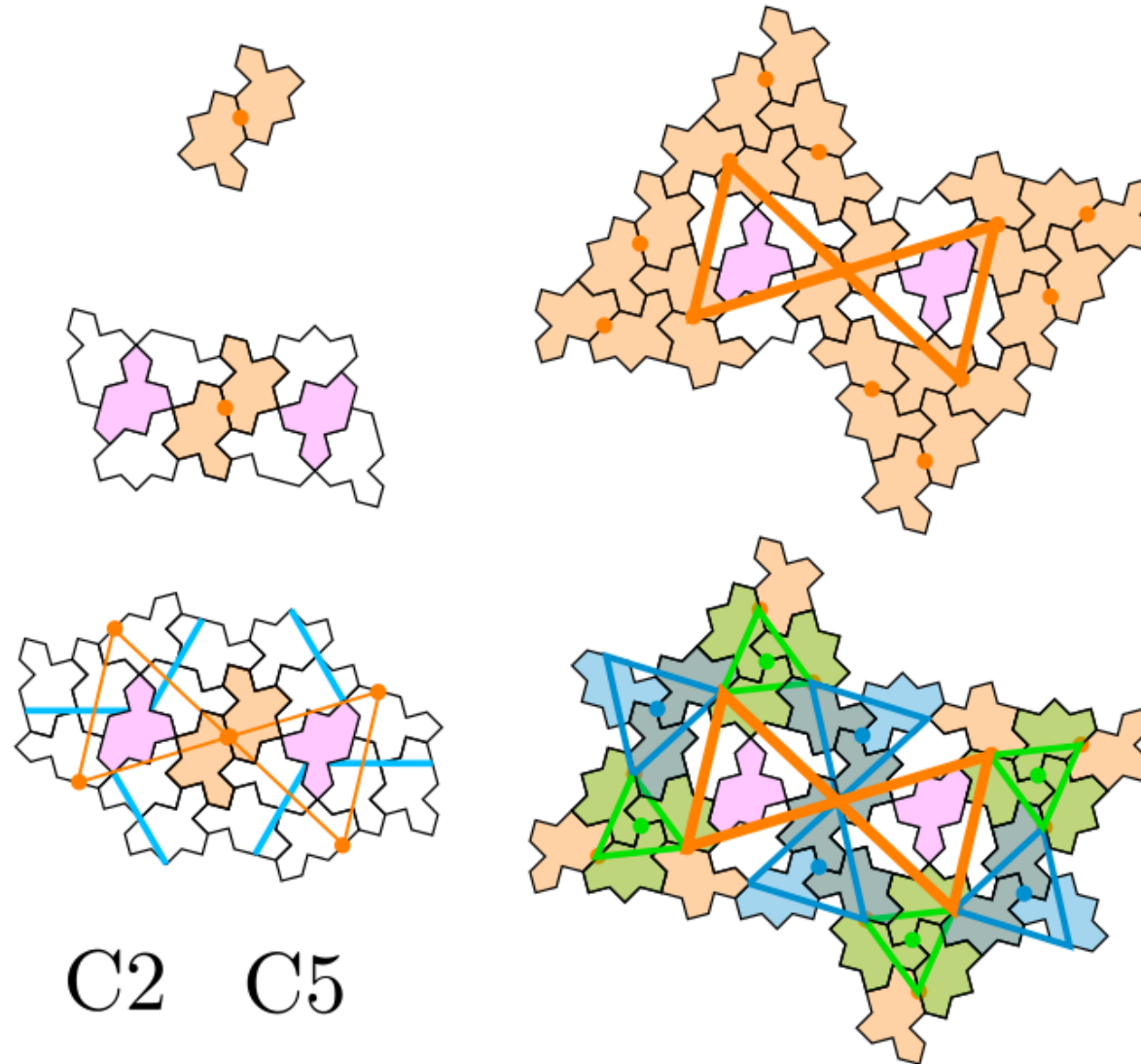
# Set of triangles: green triangle



# Set of triangles: last triangle



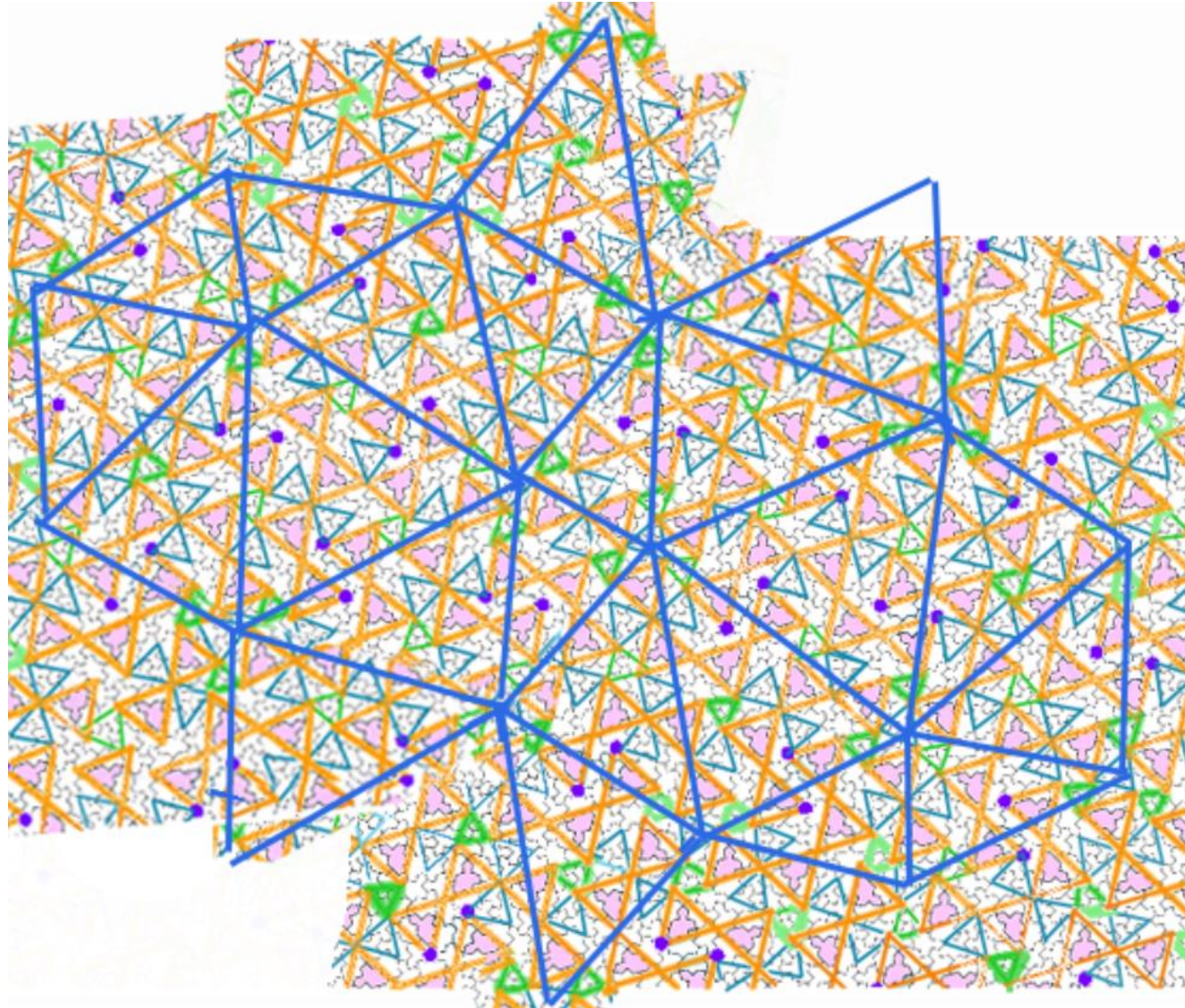
# Global 2-fold symmetry: center of a diablo pattern



- From local to global 2-fold symmetry ???

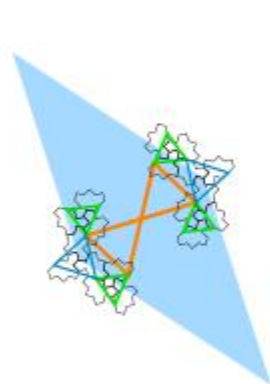
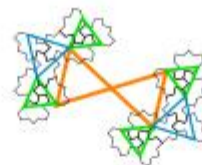


# Global 2-fold symmetry: center of a diabololo pattern





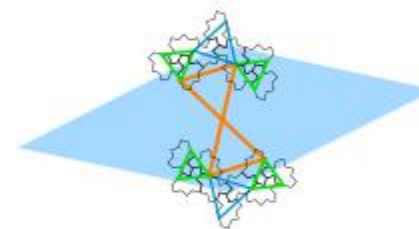
# Rhombi for diablo



D1



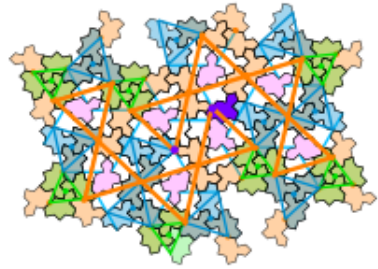
D3



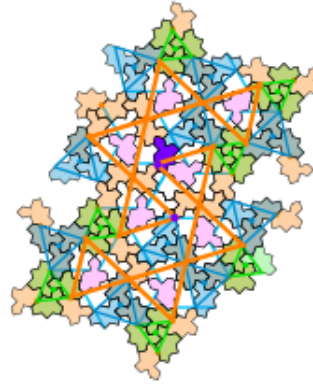
D5

# Six ORIENTATIONS OF 2-arms pattern

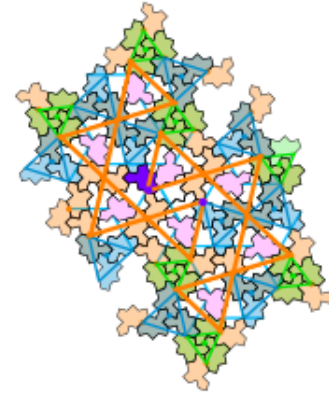
P1



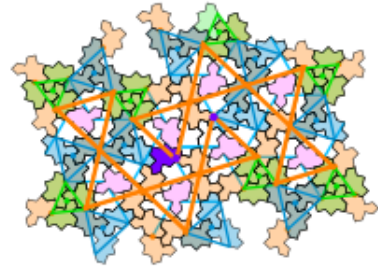
P2



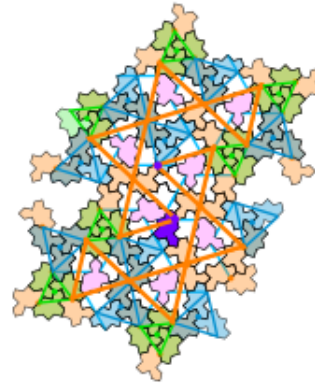
P3



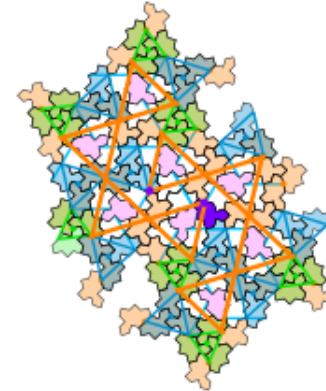
P4



P5



P6



# FOUR possible seeds with 3-fold symmetry

- COMBINE Three 2-arms rotated from each other by  $2\pi/3$

3-fold-S1



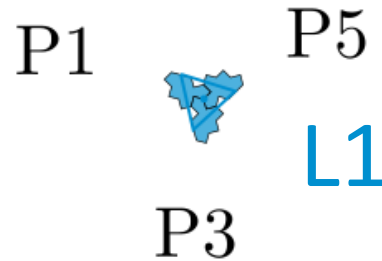
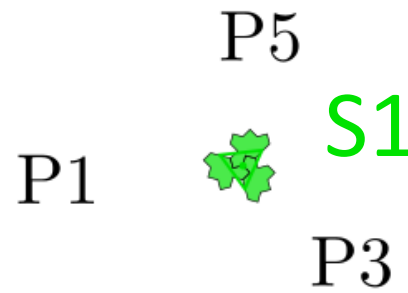
3-fold-S2



3-fold-L1

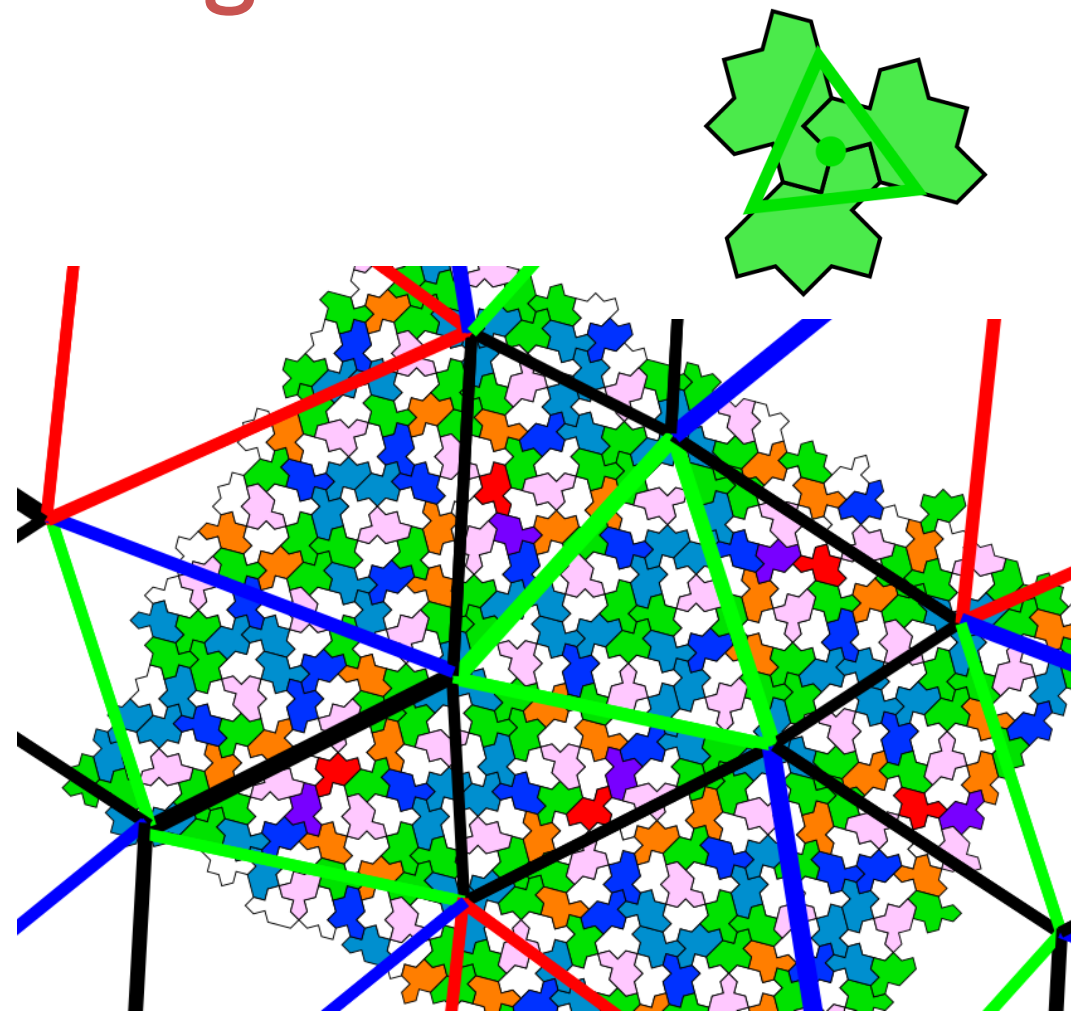
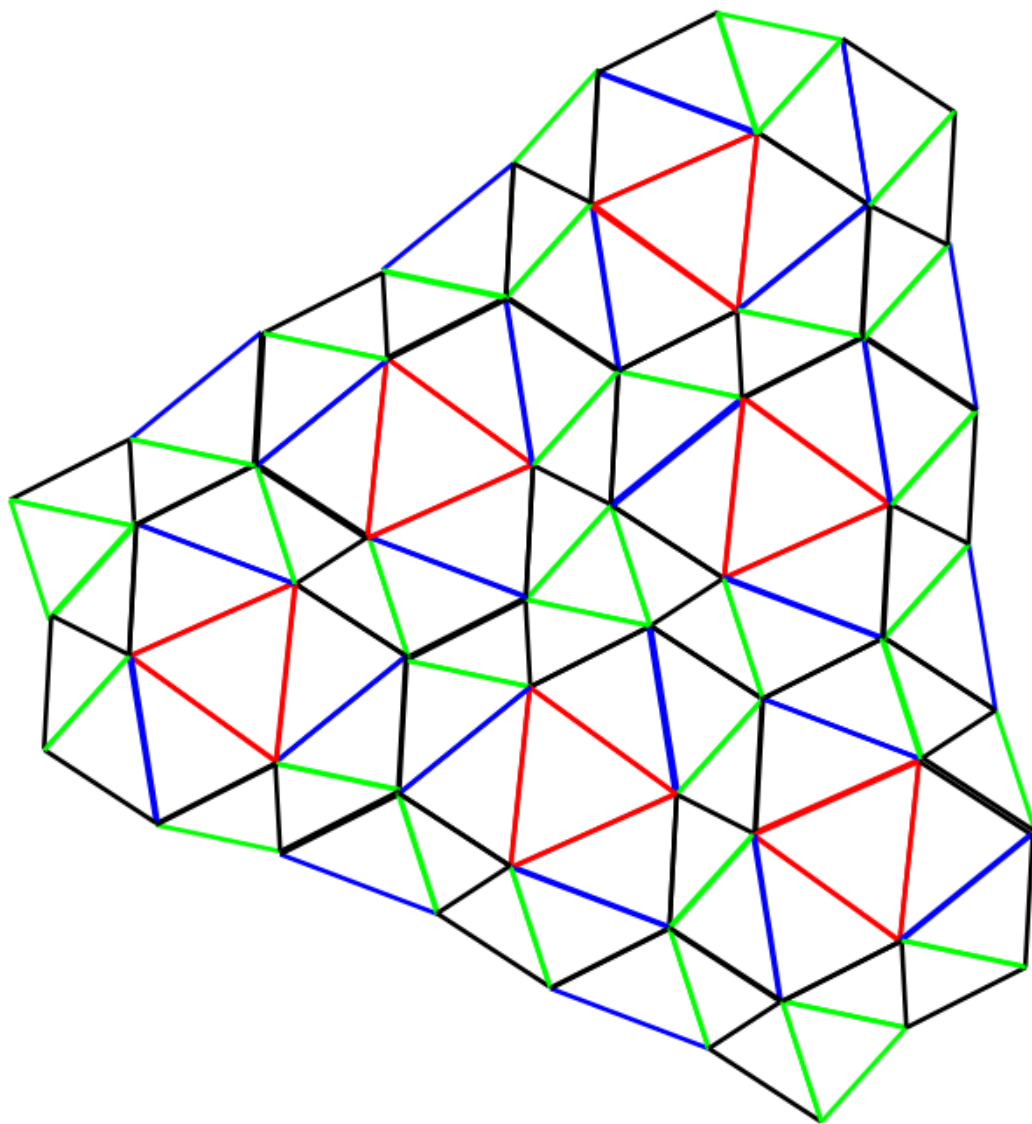


3-fold-L2



- Question: are all of them present in the infinite tiling ???

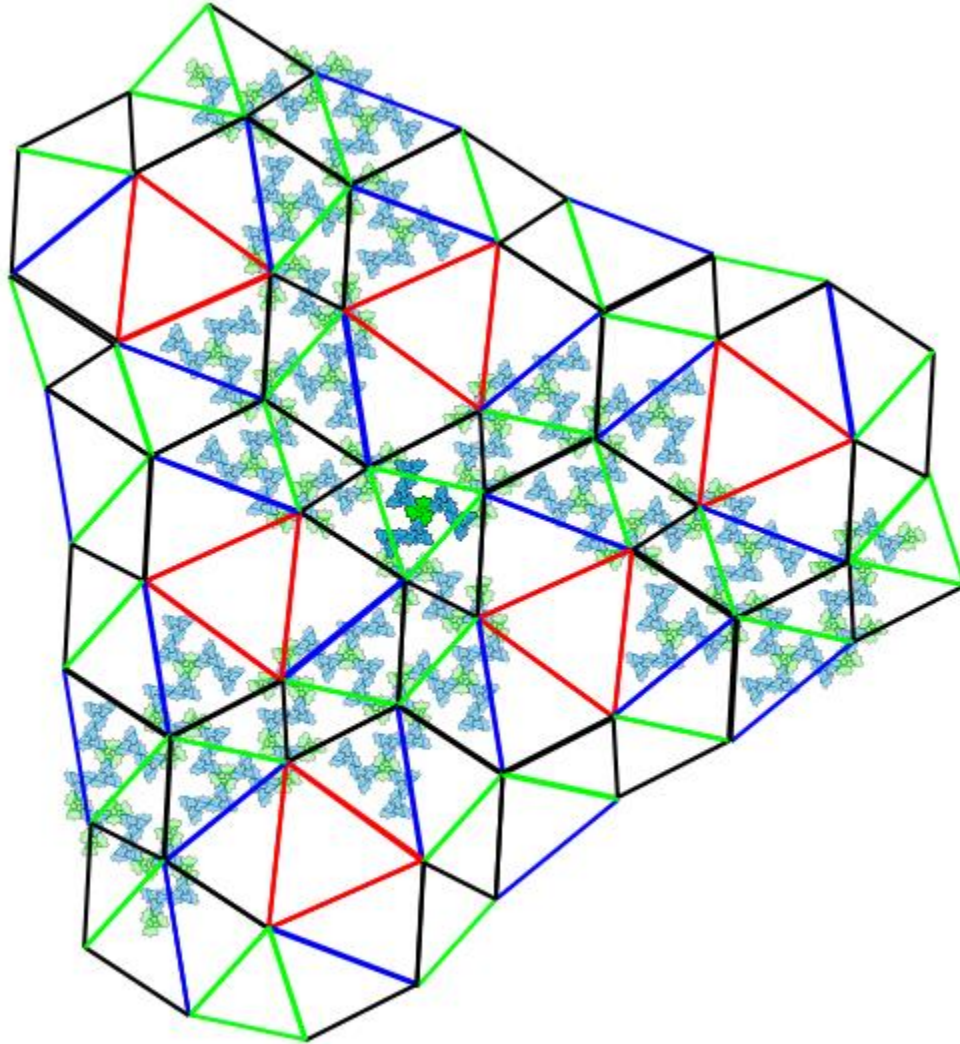
# Using set of triangles



3-fold-S2



# Using set of triangles

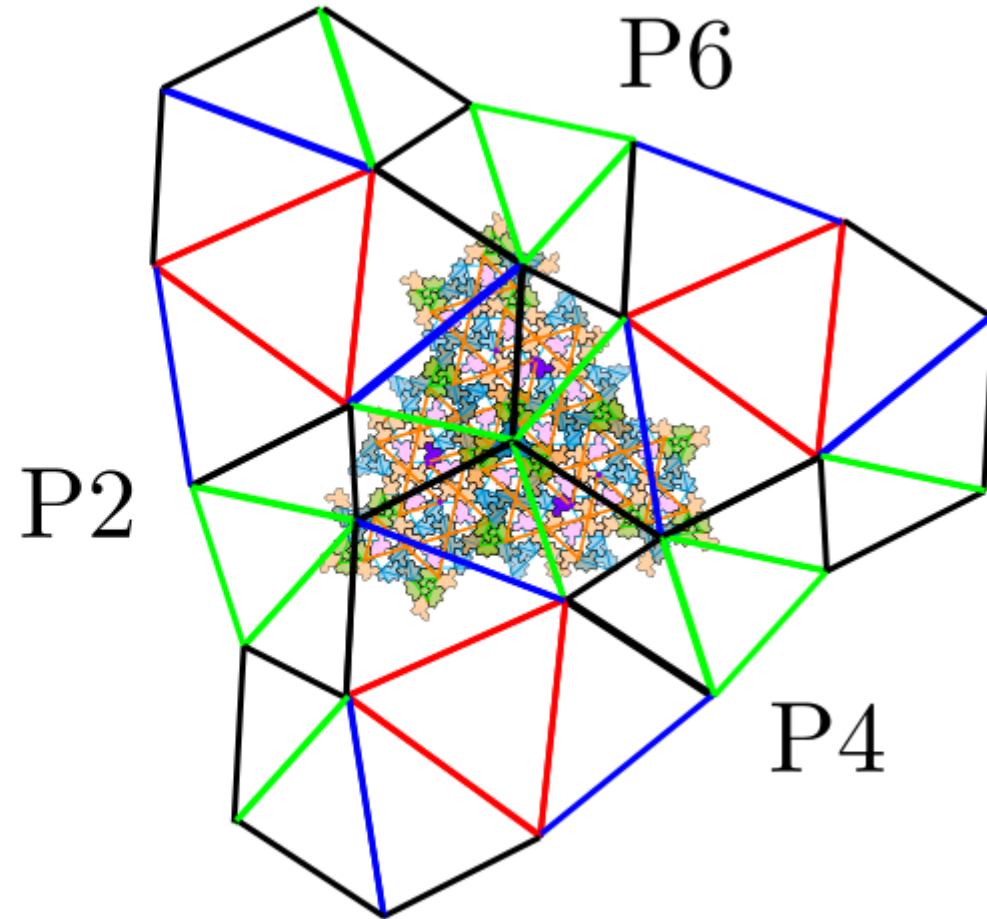


3-fold-S1

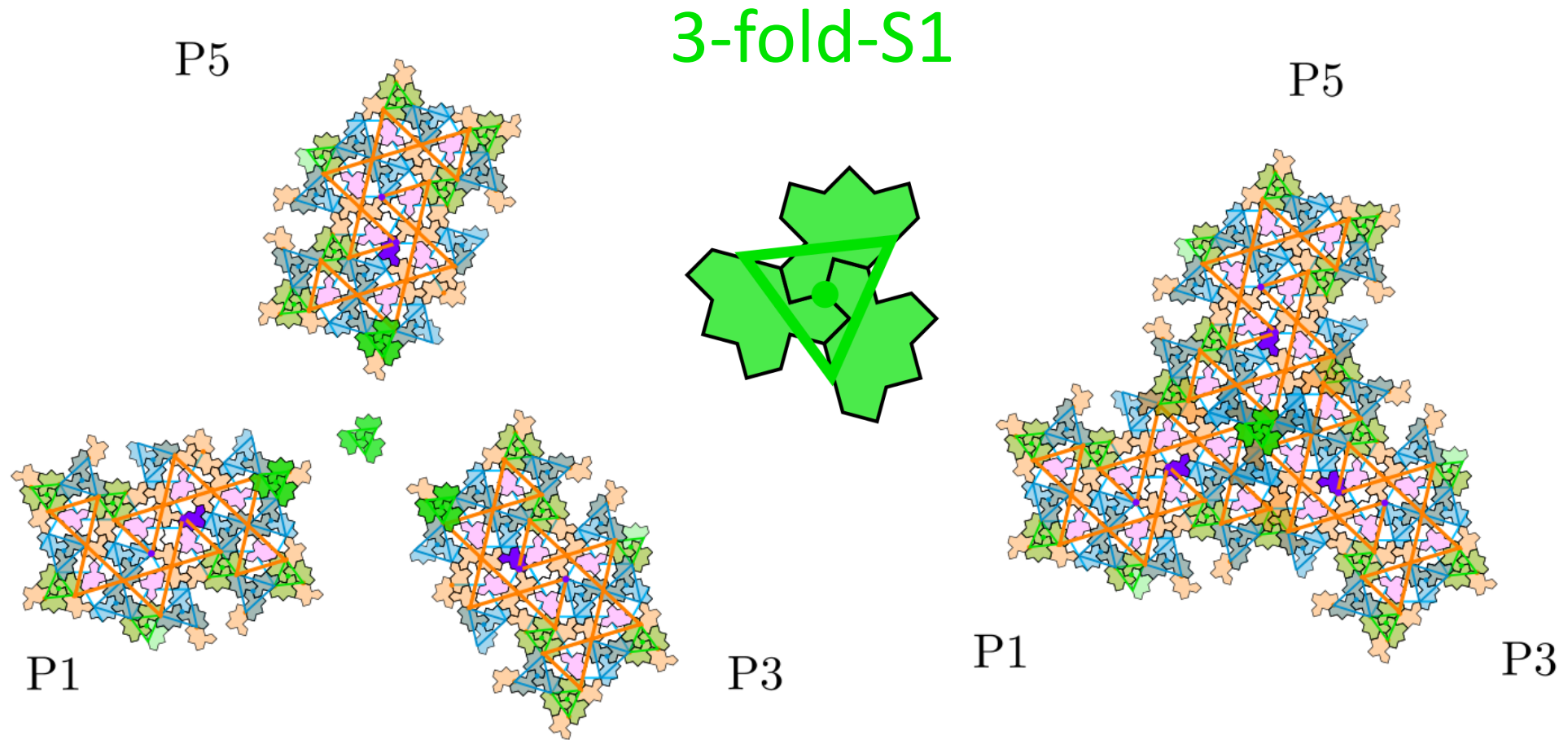
# Using set of triangles



3-fold-L2

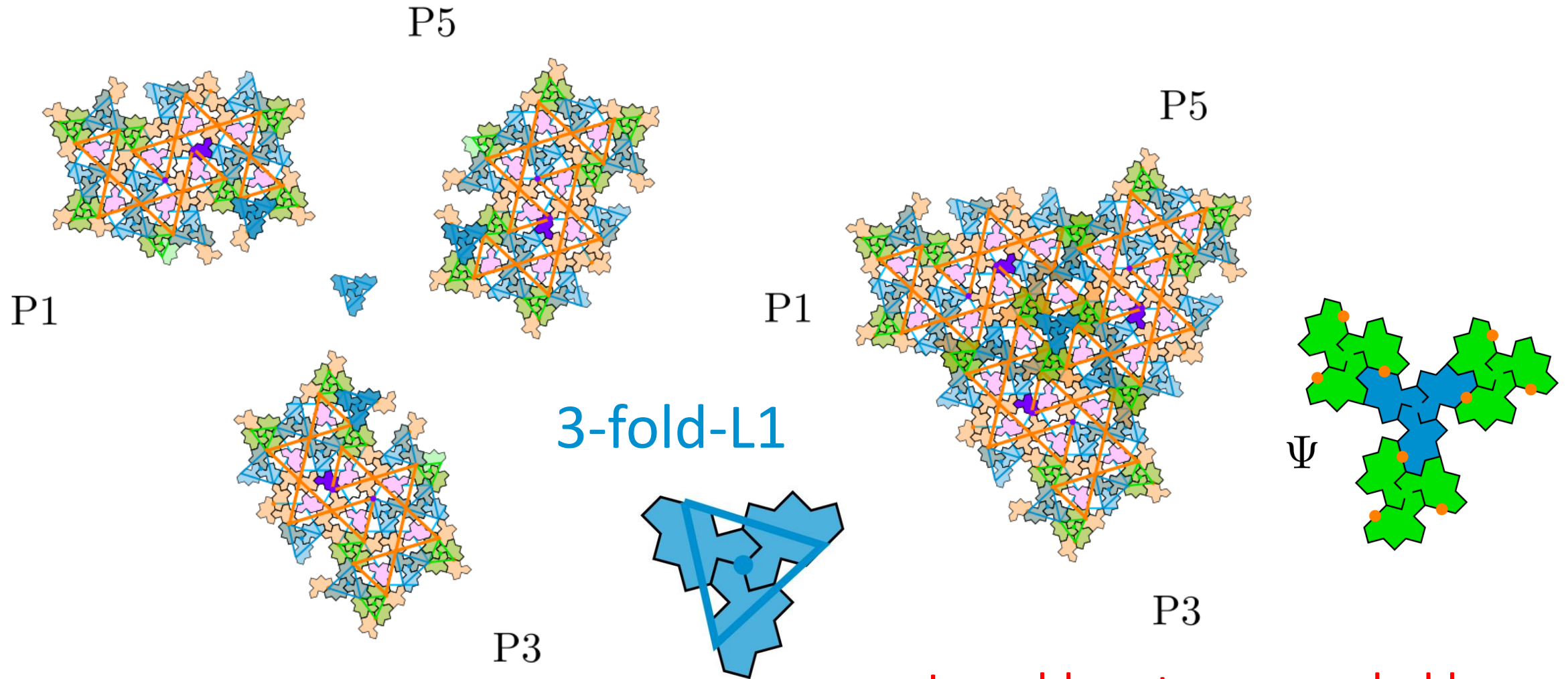


# Seed 1 with 3-fold symmetry



- Green star S1 at the center

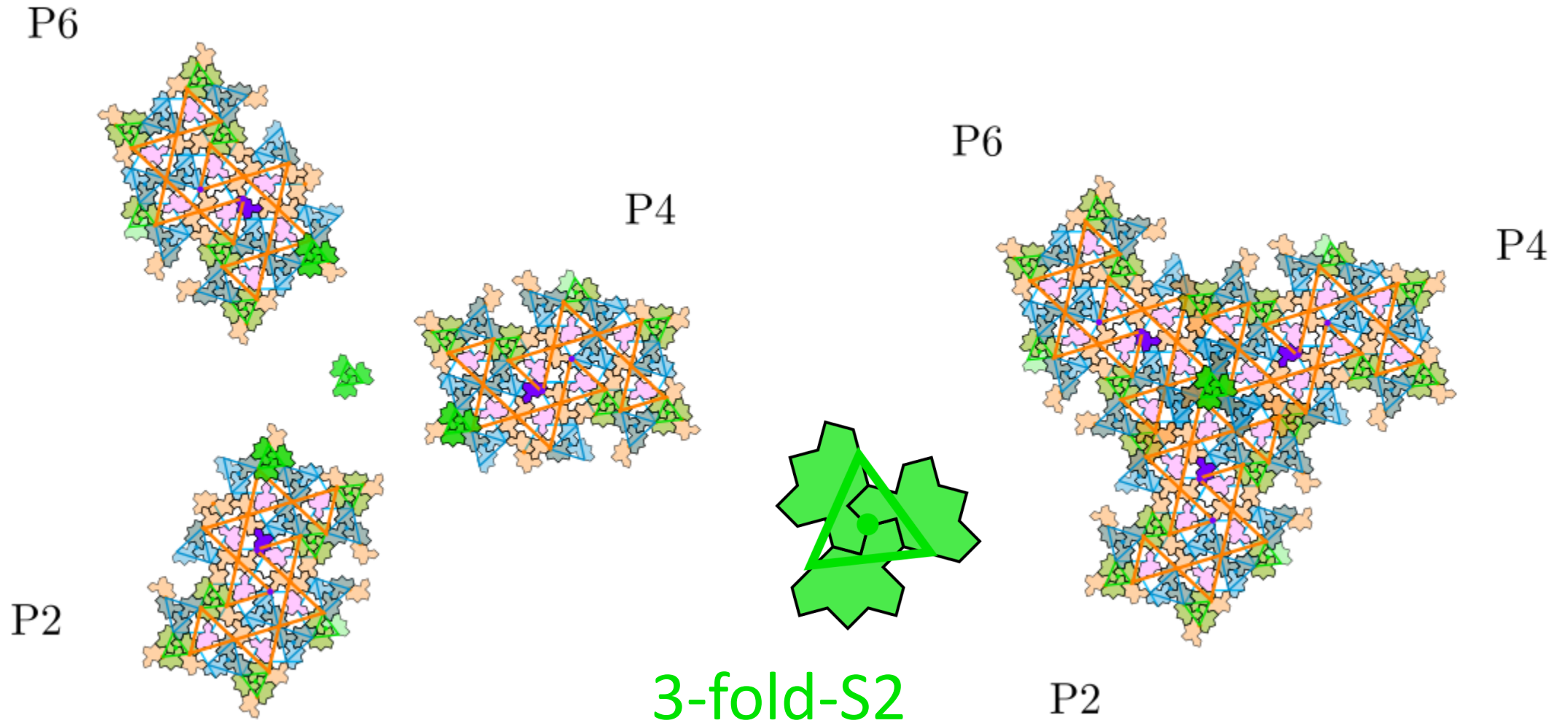
# Seed 2 with 3-fold symmetry



- blue star L1 at the center
- Lone blue star surrounded by three S2 green stars

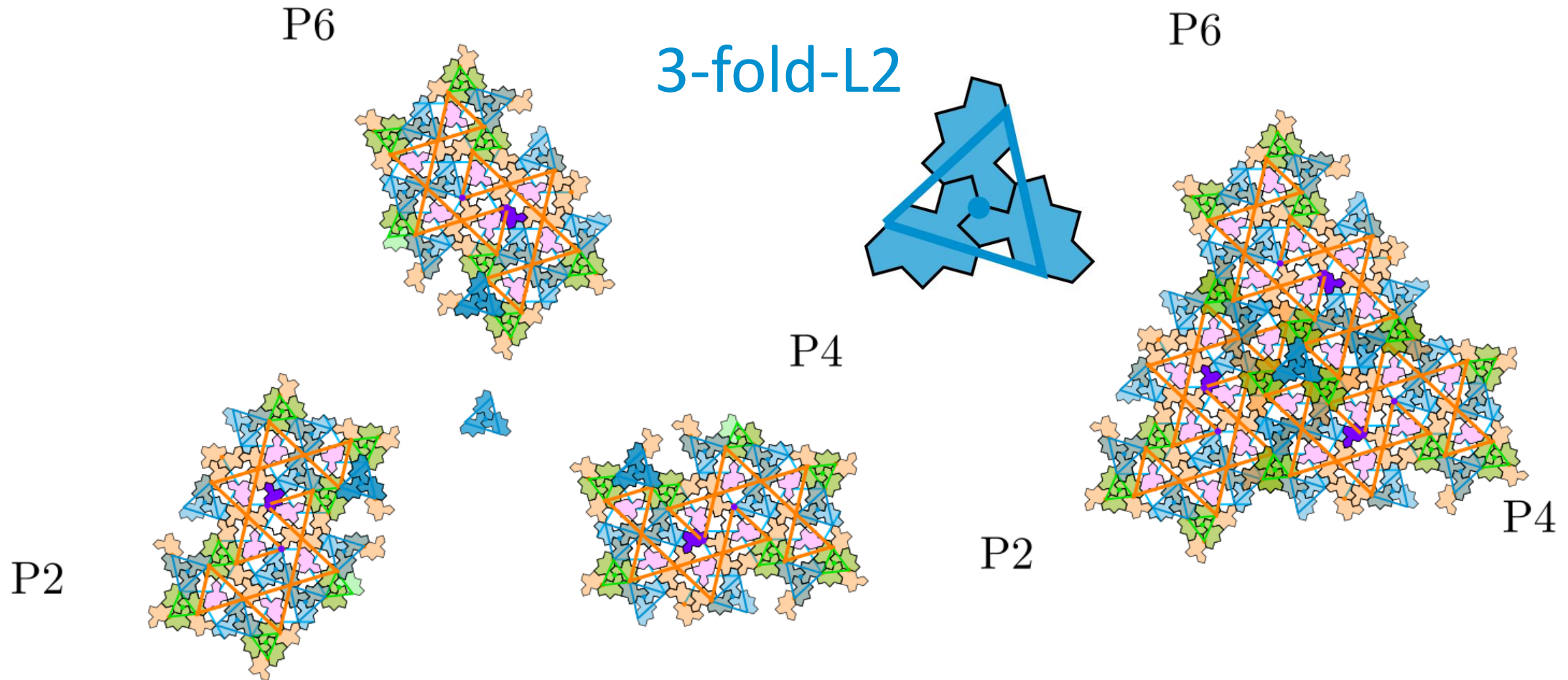


# Seed 3 with 3-fold symmetry



- Green star S2 at the center

# Seed 4 with 3-fold symmetry



- blue star L2 at the center

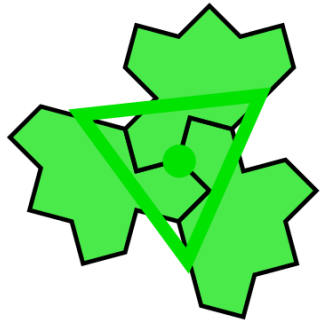
# Two possible seeds with global 3-fold symmetry

seed1

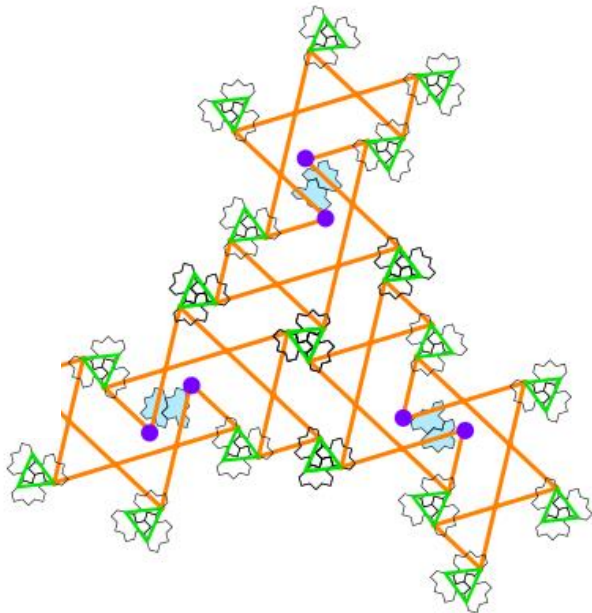
- Green star at the center

3-fold-S1

P5



P1

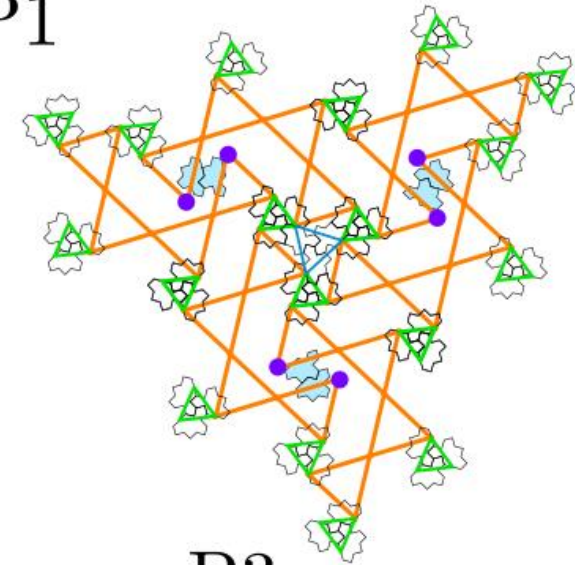


P3

seed2

- blue star at the center

P1



P3

P5

3-fold-L1



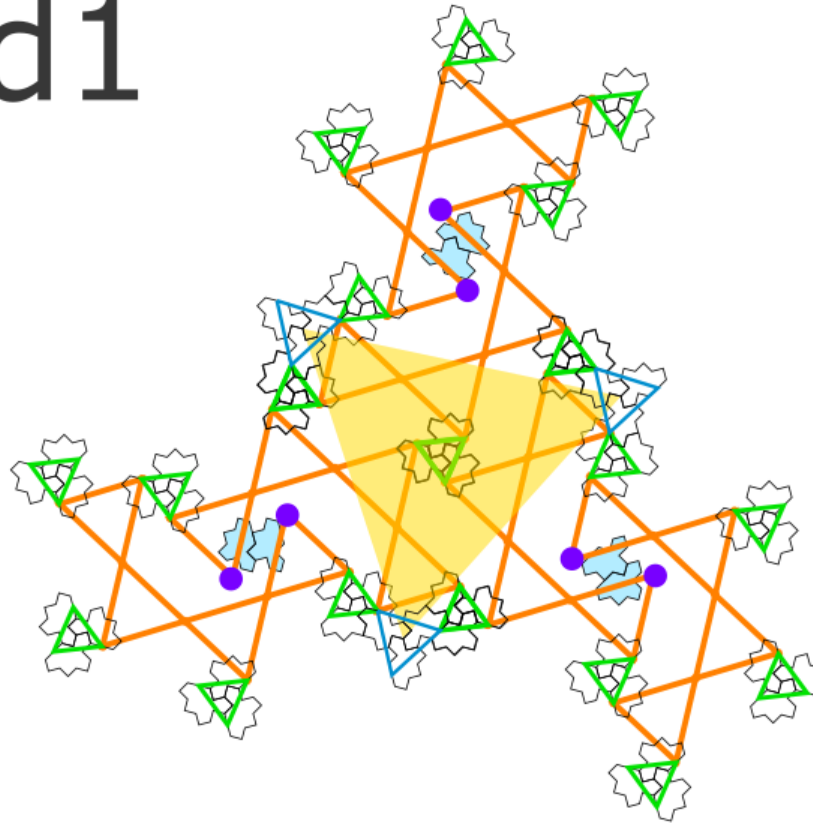
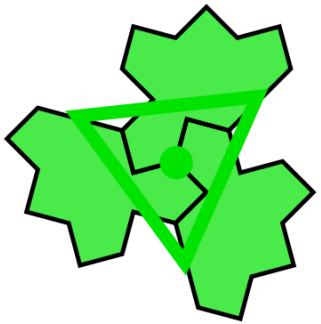
- Three 2-arms rotated from each other by  $2\pi/3$

Seed 1

P5

seed1

3-fold-S1

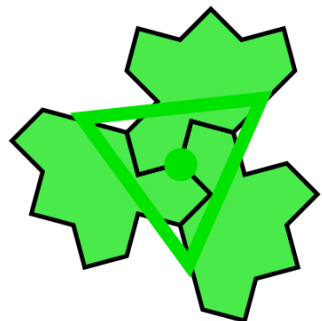


P1

P3

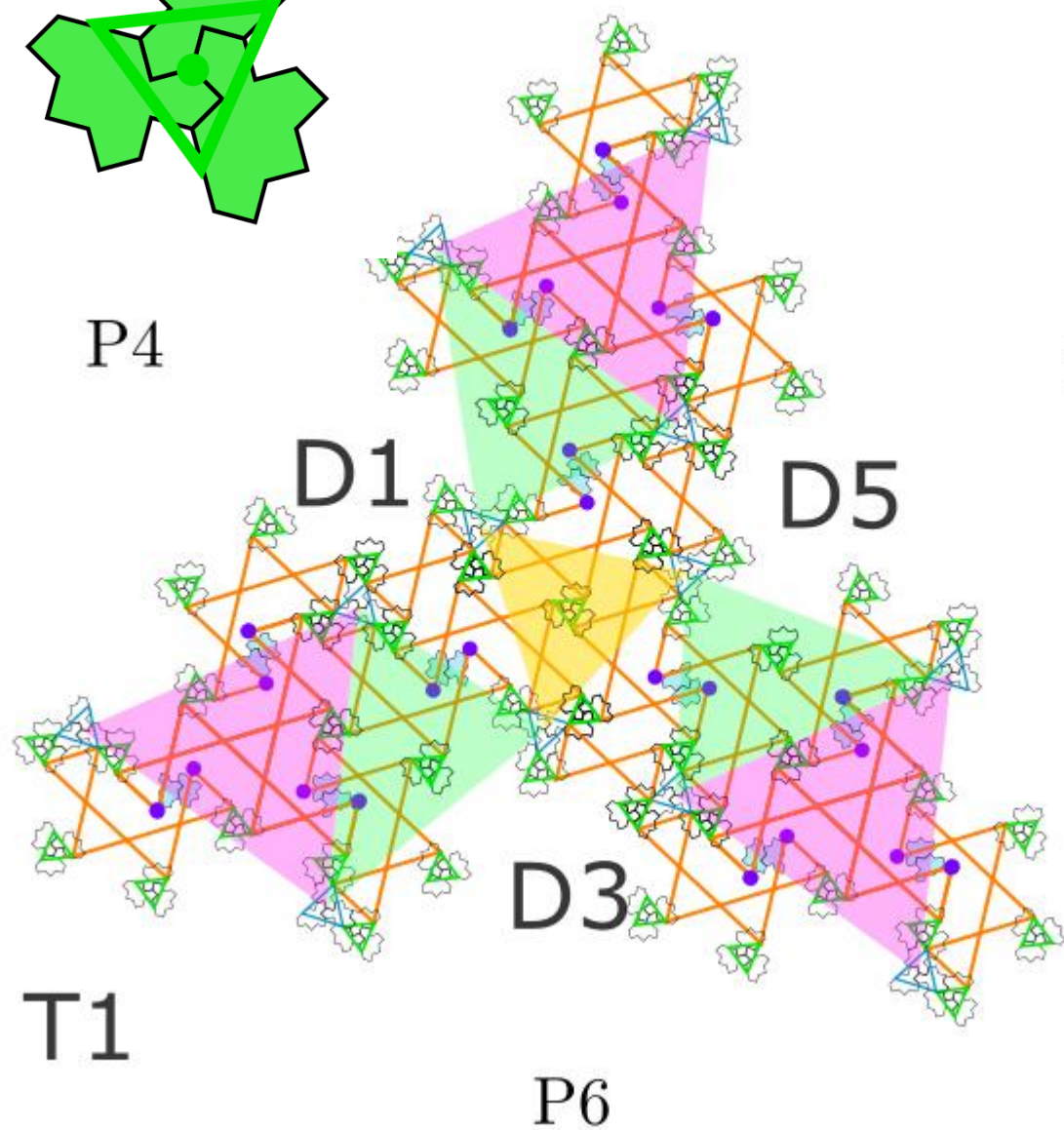


3-fold-S1



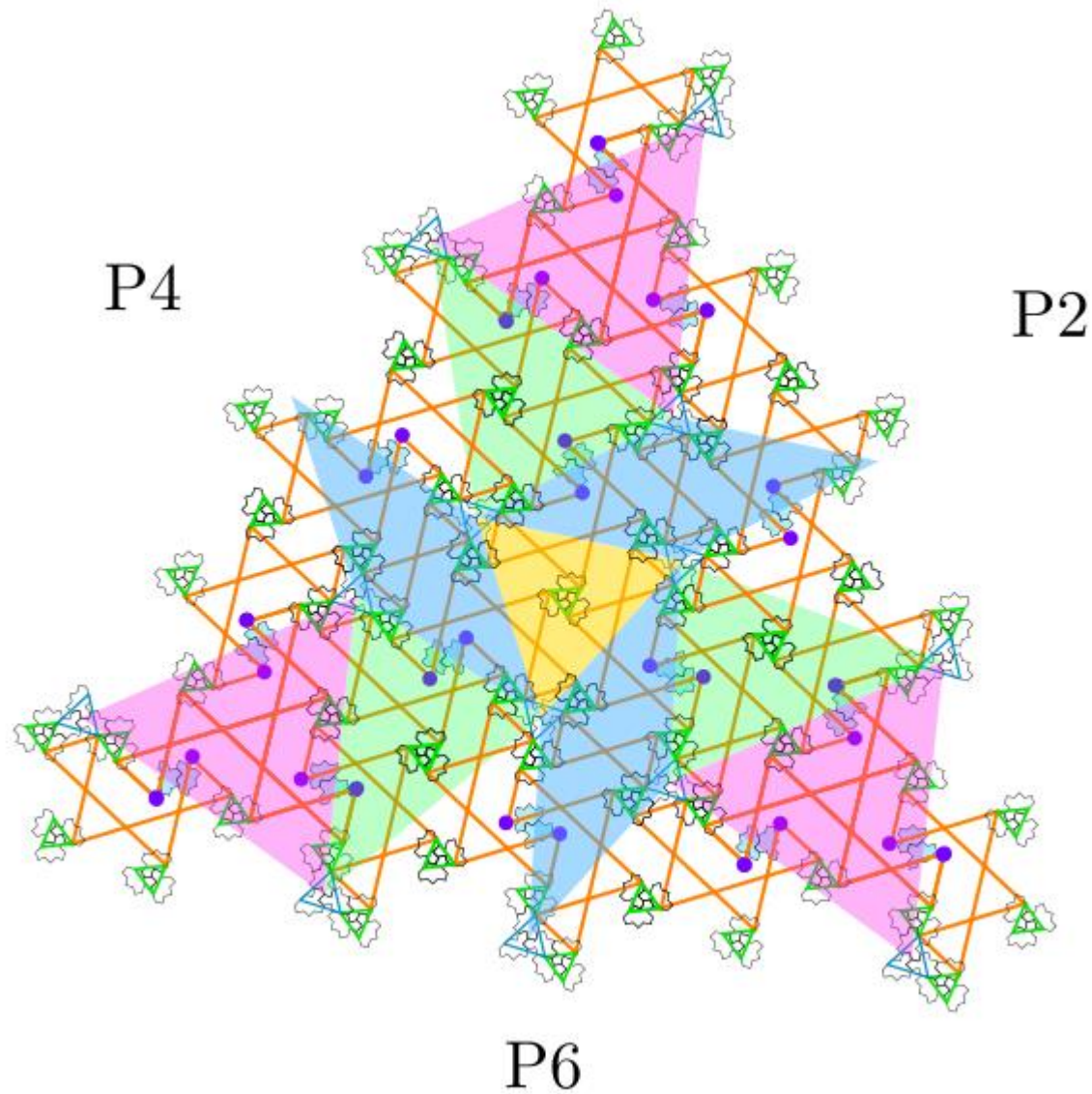
T5

Seed 1



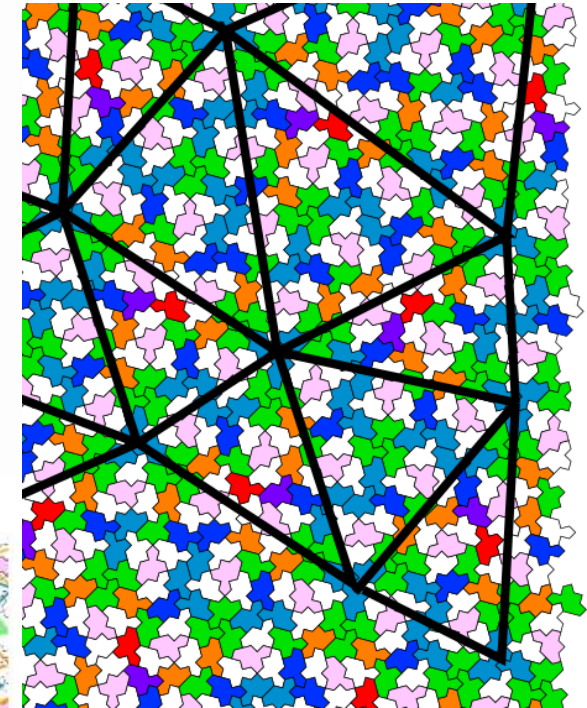
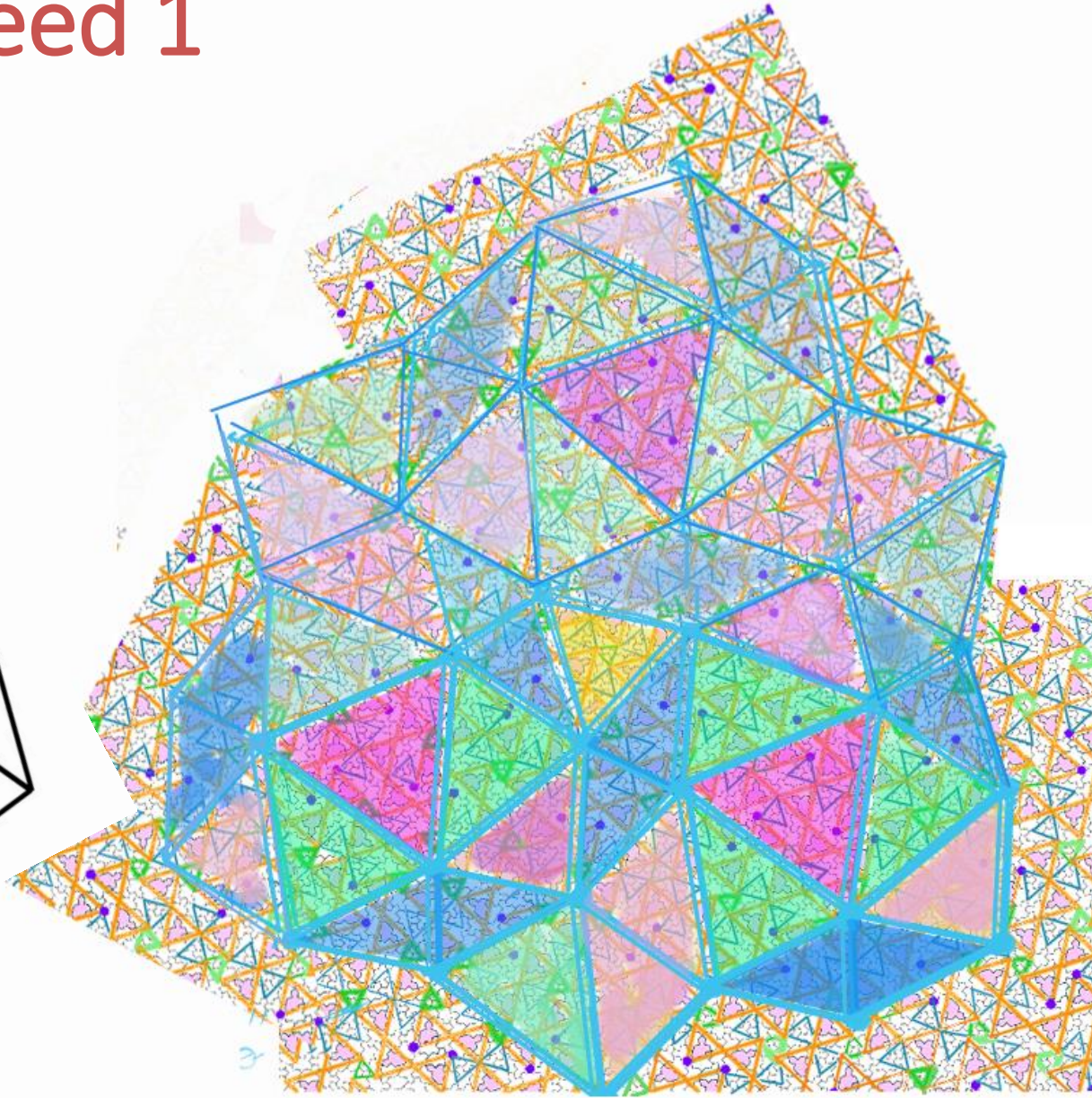
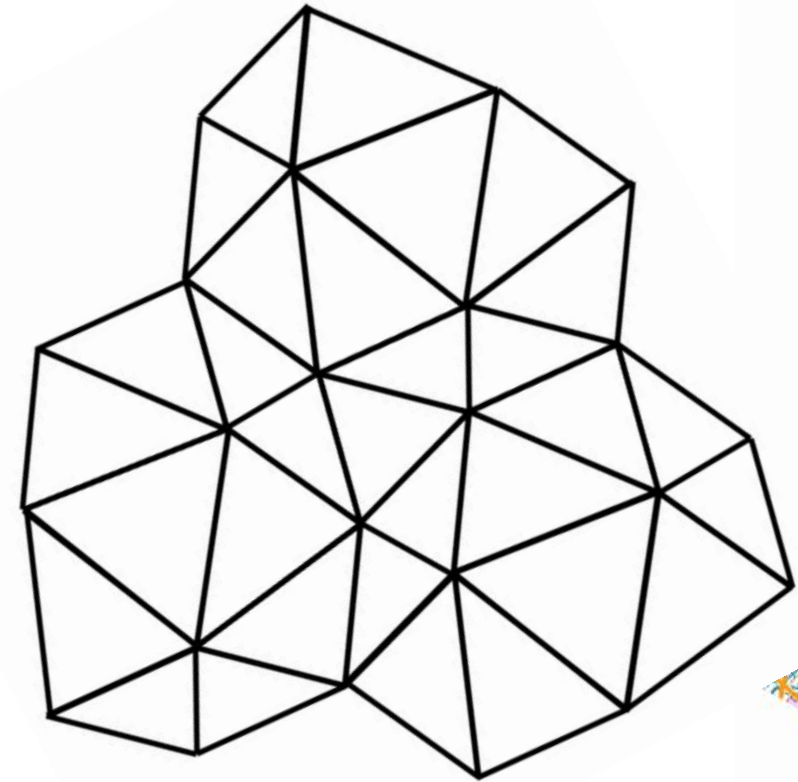
P2

T3





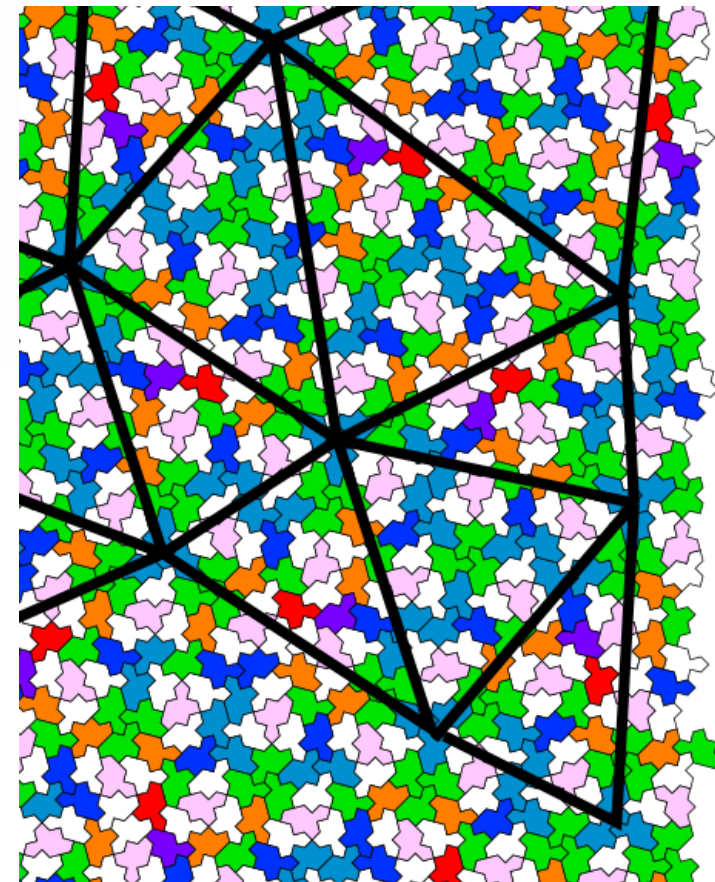
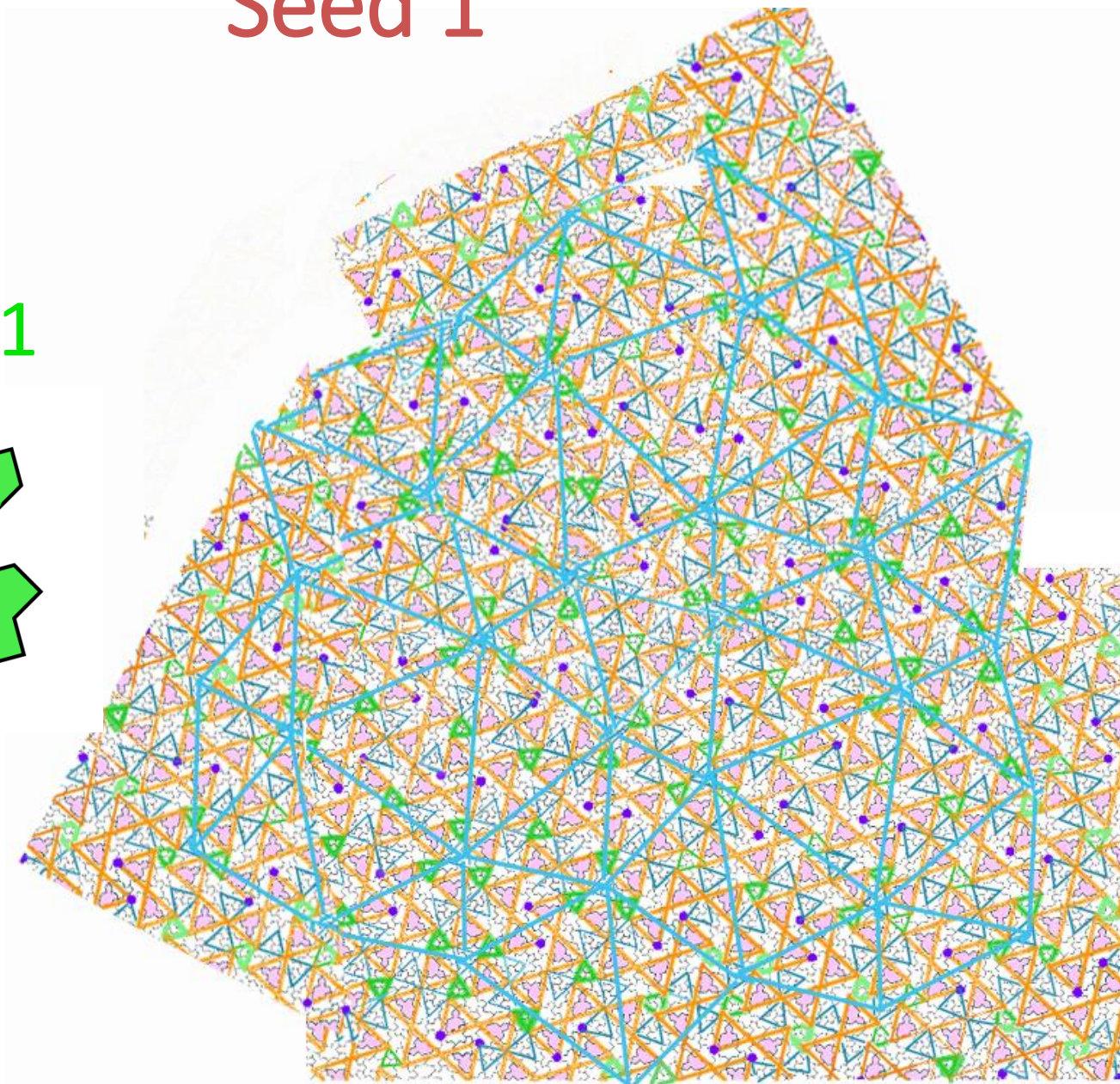
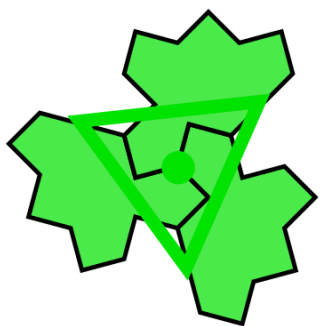
# Seed 1





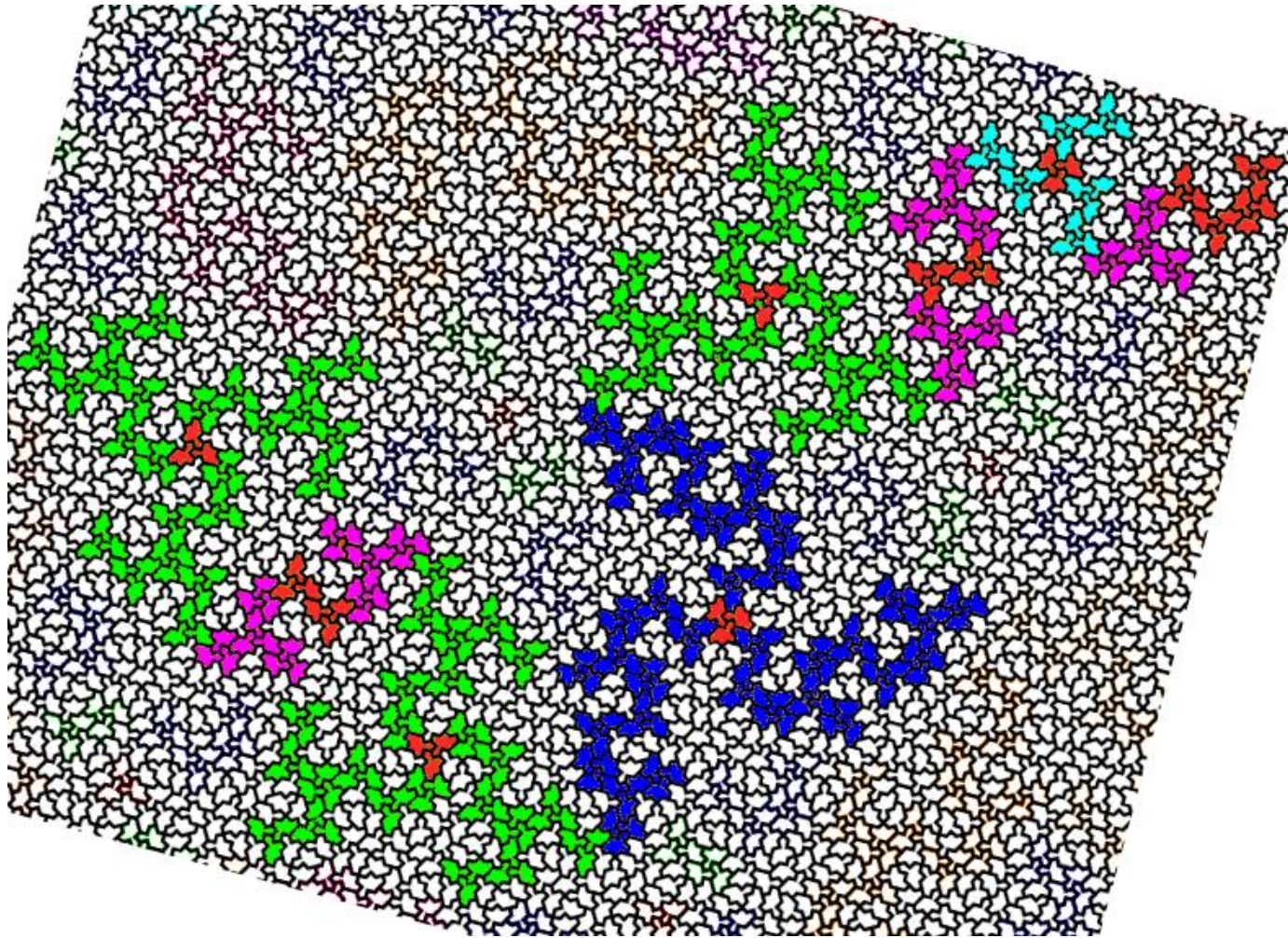
# Seed 1

## 3-fold-S1





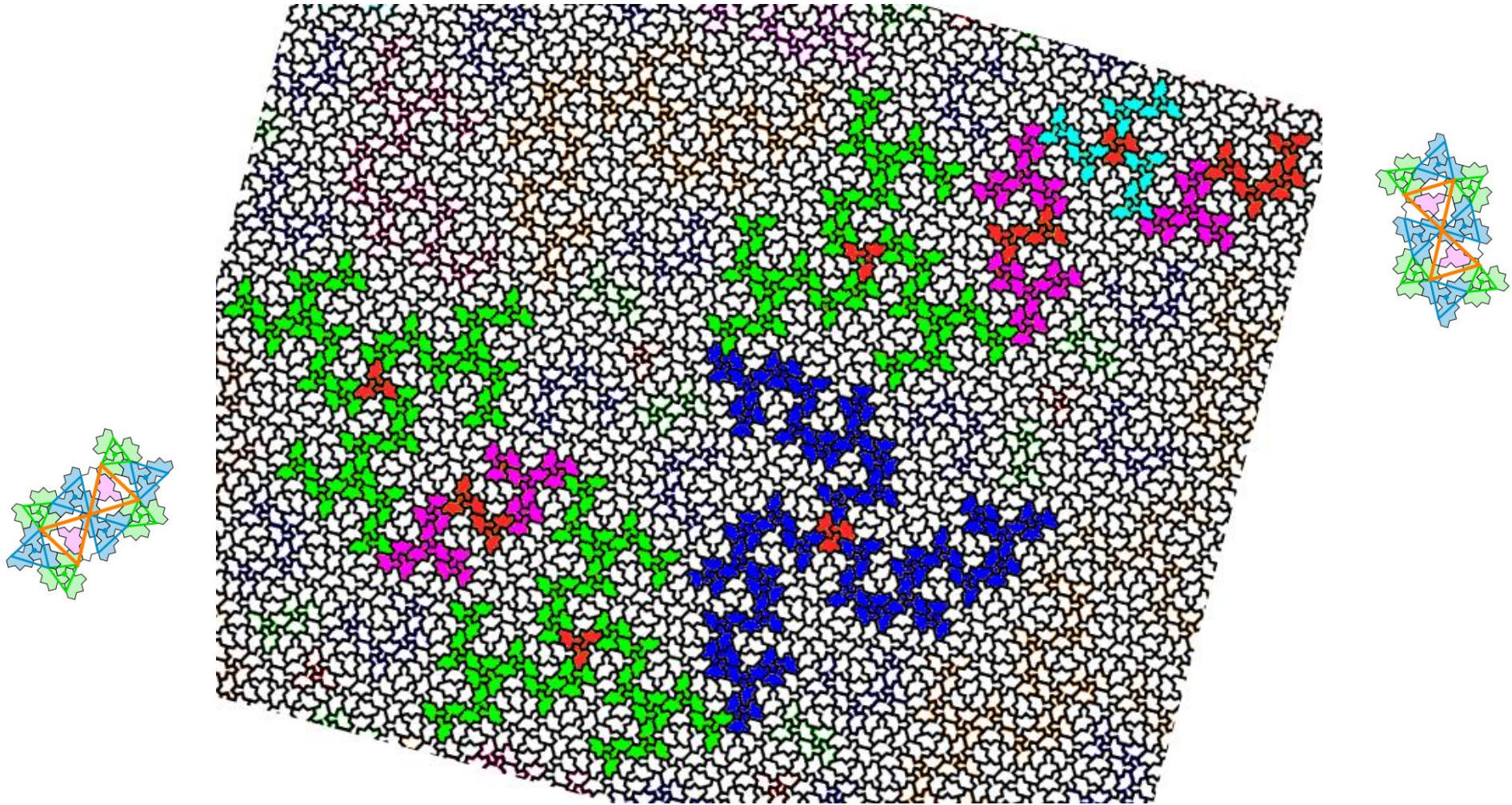
# Star clusters by Boris Horvat: example of cluster 147



- Mirror + rotation of  $\pi/12$  for comparison



## cluster 147: two diabolos



- -> to finish ... Identify the type of the local 3-fold symmetry centers

# Results so far

- Local 3-fold and 2-fold symmetry centers are identified
- Violet points -> different way to identify (chimera ...)
- Analysis of the size/amount of stars clusters depending on the level (Boris Horvat)
- Different patterns can be used to decompose the whole tiling:
  - 2-arms plus 3-arms+diabolo
  - full clusters
  - versions of 'anatomy' etc ...

Helpful to identify the global symmetry centers !

- Identification of 2-fold global symmetry center (center of diabolo figure)
- two possible 3-fold global symmetry centers: center of green triangle and center of blue triangle. They correspond to three 2-arms (P1/P3/P5) rotated by  $2\pi/3$  around the center)
- triangular array of blue stars surrounded by three green stars

# TO DO: inflation schemes around symmetry centers

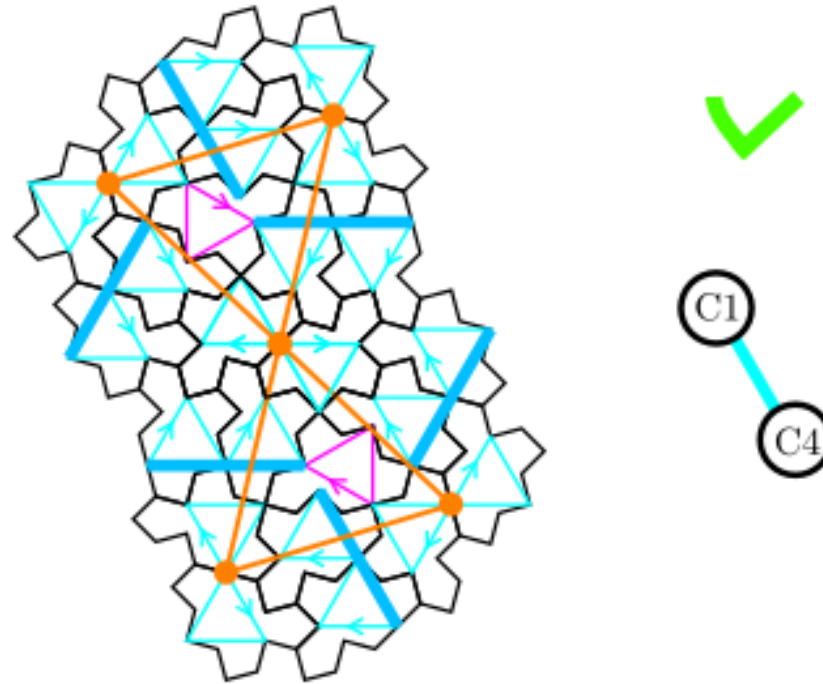
Different possible methods should give the same result

- H7/H8 (by S. Tathan) applied to a symmetric subset of clusters  
-> might be the most straightforward method for implementation as code is already running (but in RUST language) ?
- Full clusters with translations (Marianne ...)  
-> will need more work to do a complete python implementation
- Substitution rules for large colored triangles (JFS)  
-> will need more work to figure out the substitution rules
- generate only the star clusters based on 2-fold and 3-fold local rotations and some additional rules ???  
-> to test the idea first ...



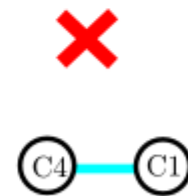
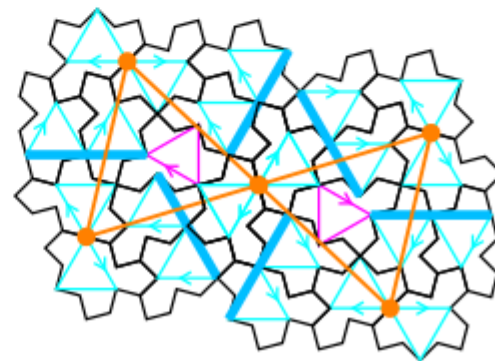
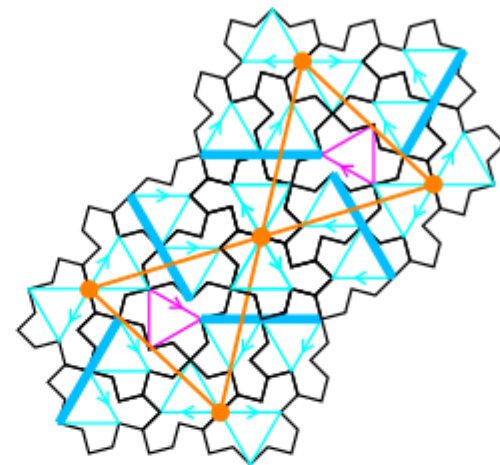
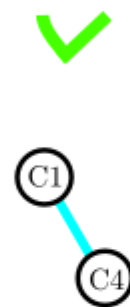
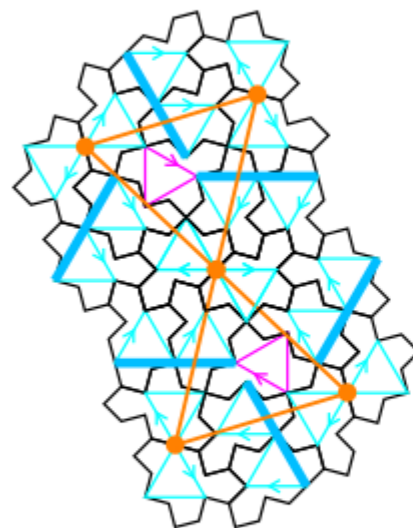
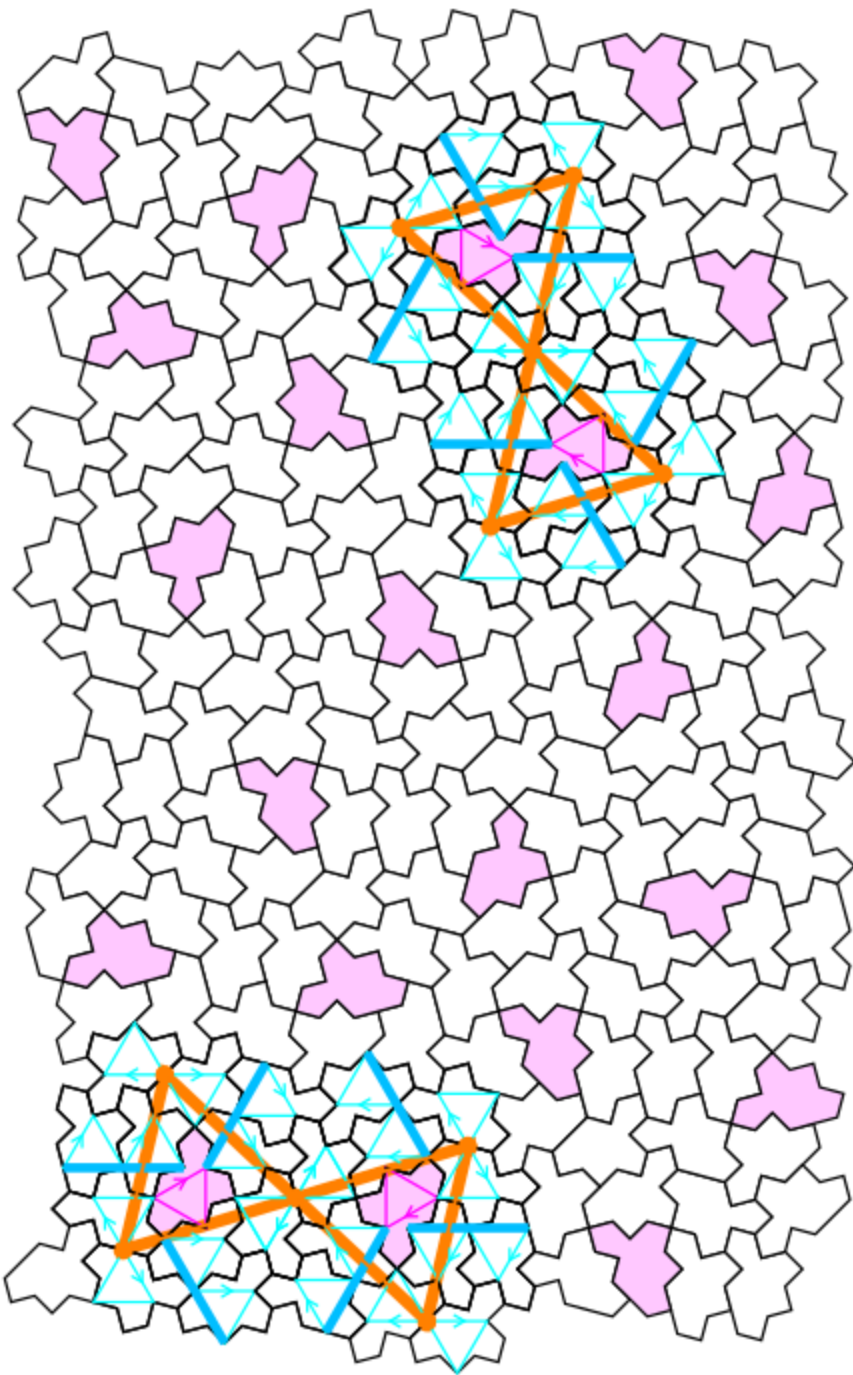
# Homochiral inflation: use H7/H8 by Simon Tathan ?

- For 2-fold symmetry, apply inflation to two clusters in C1-C4 relative orientations :



- Should keep the 2-folds symmetry at all inflation steps
- TO DO: test it manually and contact S. Tathan afterwards

# Diabolo pattern



# Identification of the greek letters for tiles

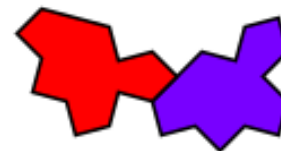
- ODD tiles  
Pink color

$\Gamma$

$\Gamma_2$



$\Theta$



$\Lambda$

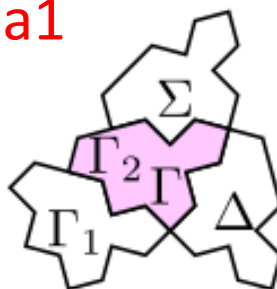
- $\backslash$ theta and  $\backslash$ lambda always in couple

- Small cluster  $\backslash$ sigma,  $\backslash$ delta,  $\backslash$ gamma1  
white color

$\Gamma_1$

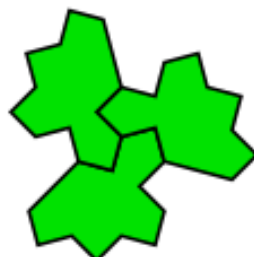
$\Sigma$

$\Delta$



$\Xi$

$\Phi$



$\Pi$

$\Psi$



$\Pi$

- All  $\backslash$ chi tiles are linked to a  $\backslash$ pi tile and consequently to a large star.

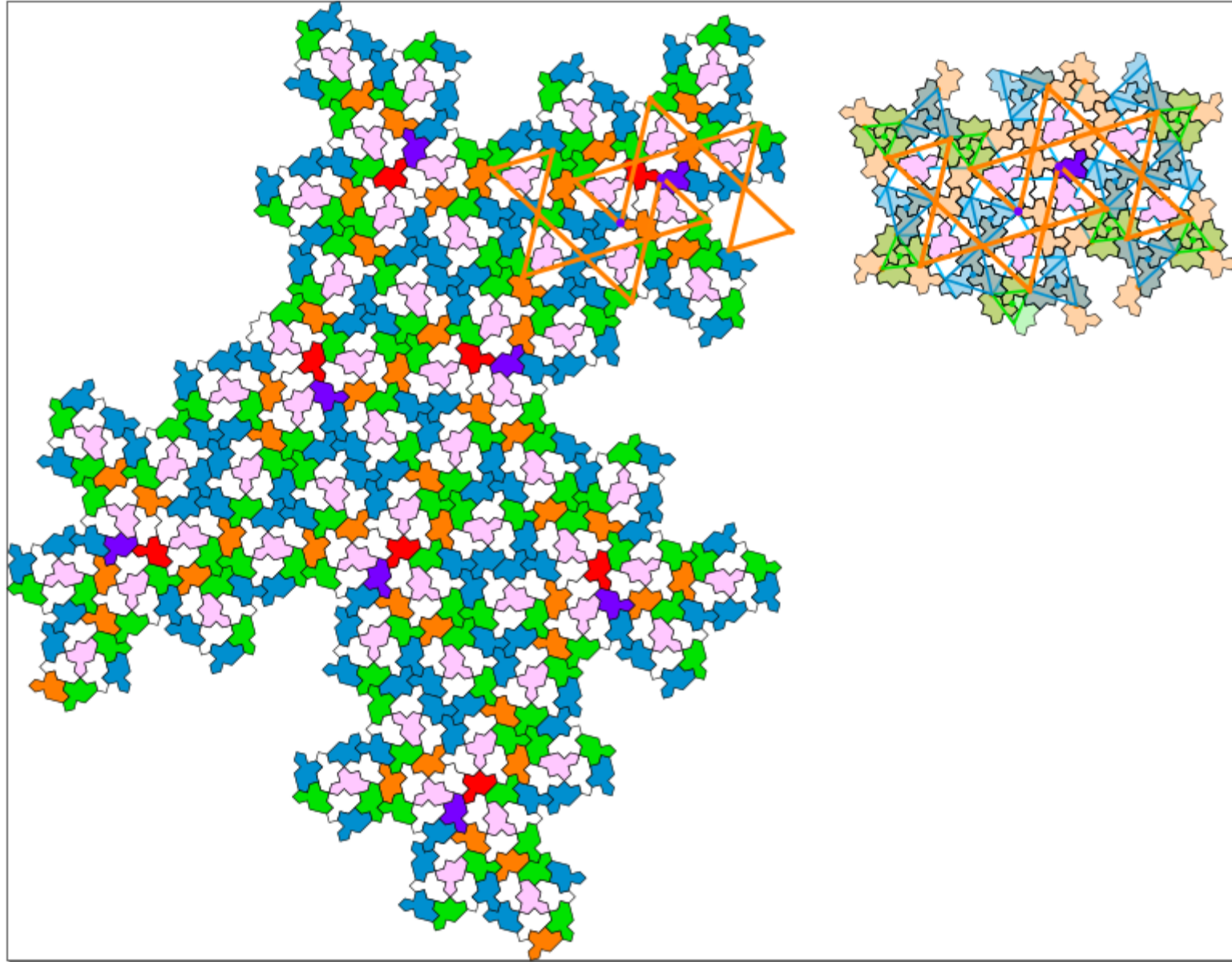
- $\backslash$ phi: Small green stars + some isolated EVEN tiles



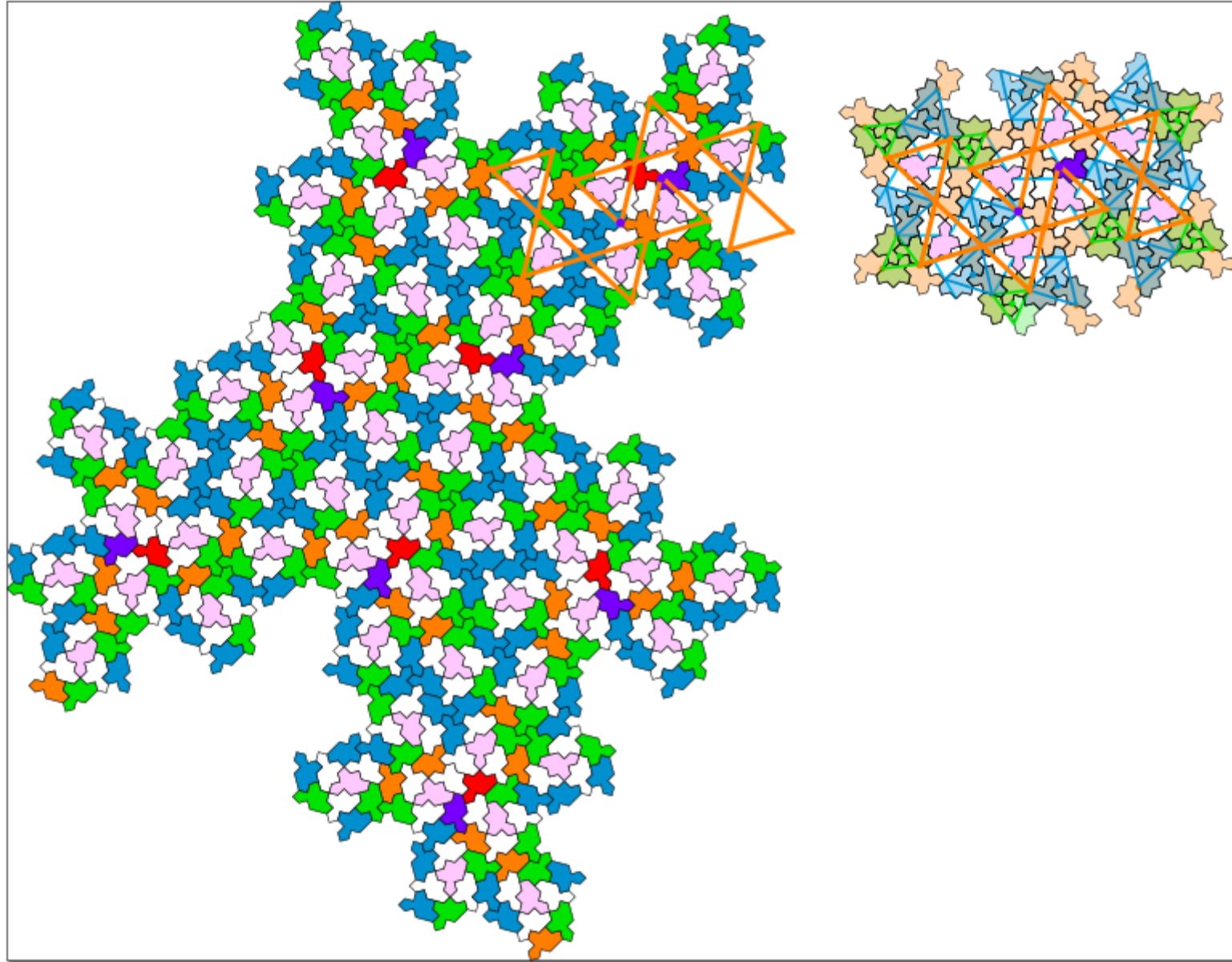
- Blue large stars correspond to all  $\backslash$ Pi and  $\backslash$ Psi tiles with different combinations



# Inflation using python code



# Inflation using python code



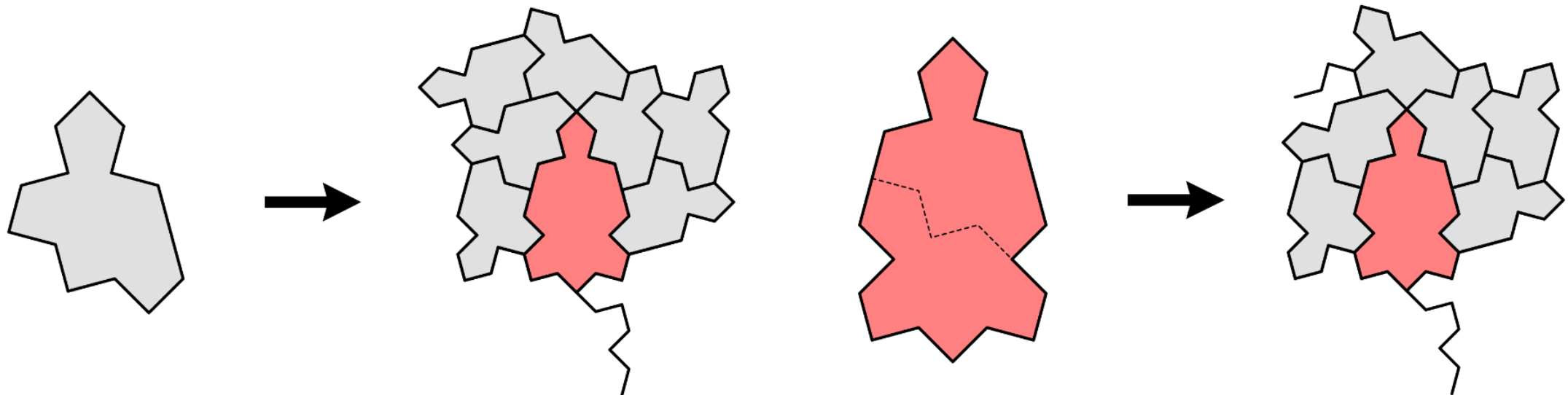
# H7/H8 by Simon Tathan

<https://www.chiark.greenend.org.uk/~sgtatham/quasiblog/aperiodic-refine/>

<https://arxiv.org/abs/2512.16595>

Code in RUST language:

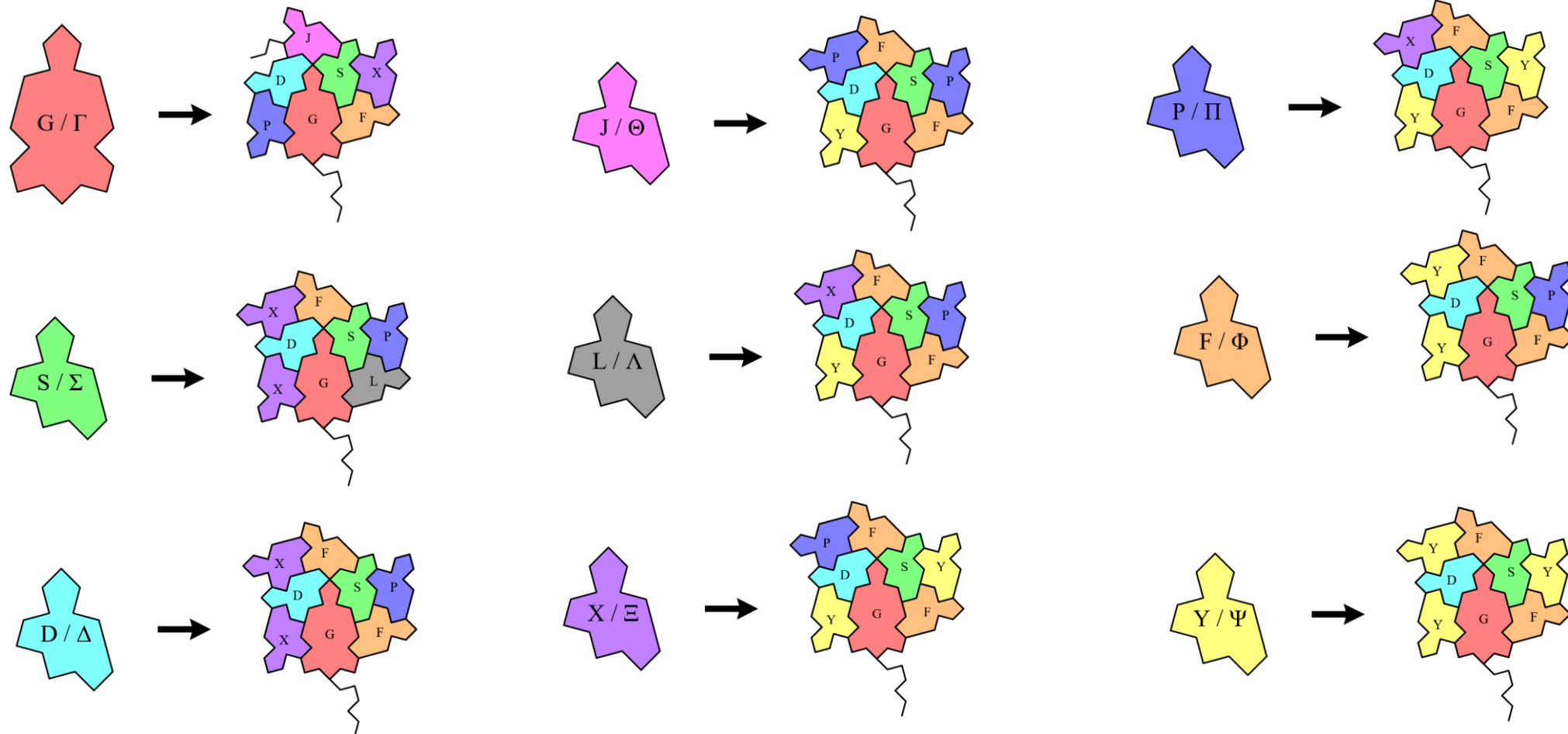
<https://crates.io/crates/substitution-tiling-transducer>





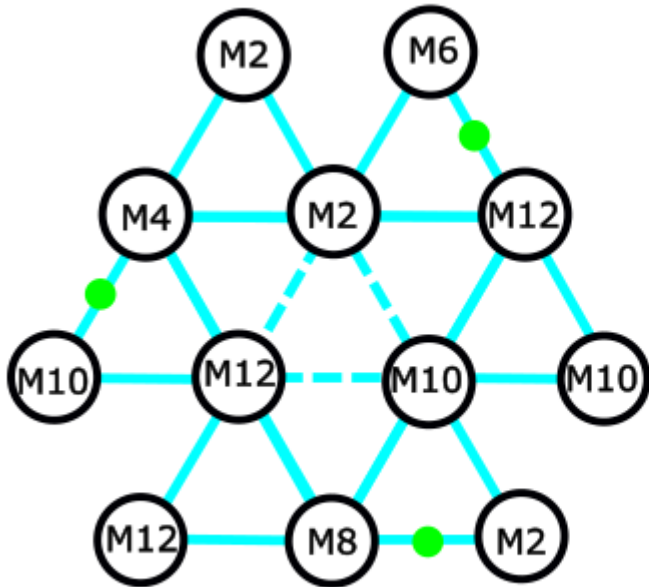
# The nine homochiral substitutions by Simon Tathan

<https://www.chiark.greenend.org.uk/~sgtatham/quasiblog/aperiodic-refine/>

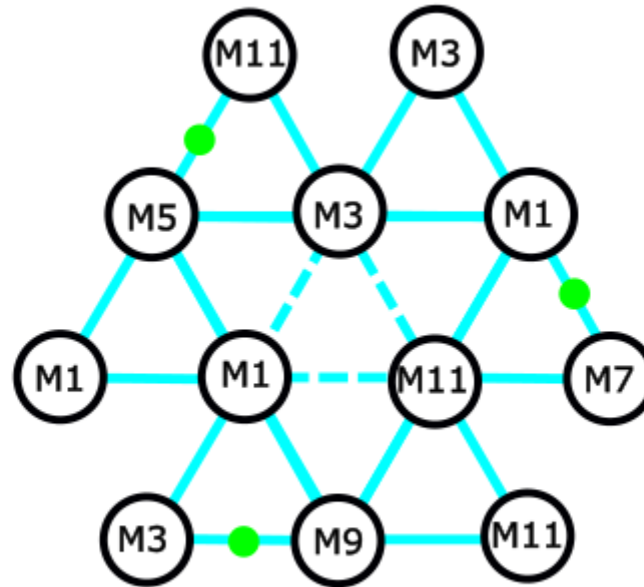


# Inflation step 1: substitution rule for C1

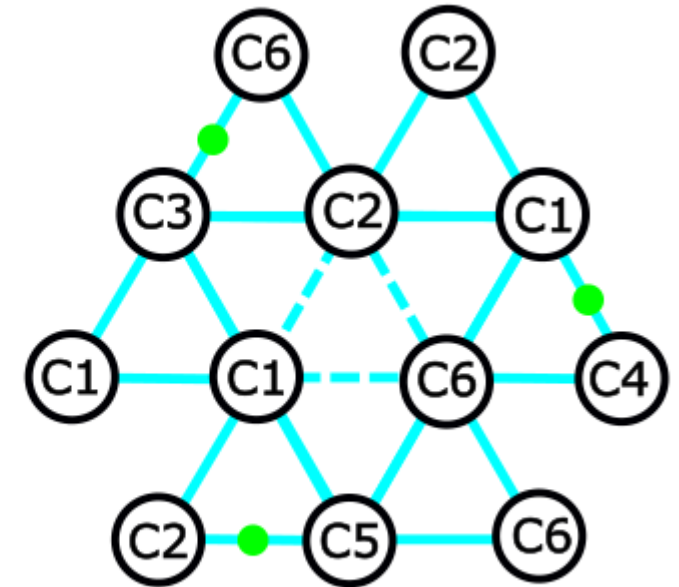
EVEN tiles in C1



ODD tiles for step 1

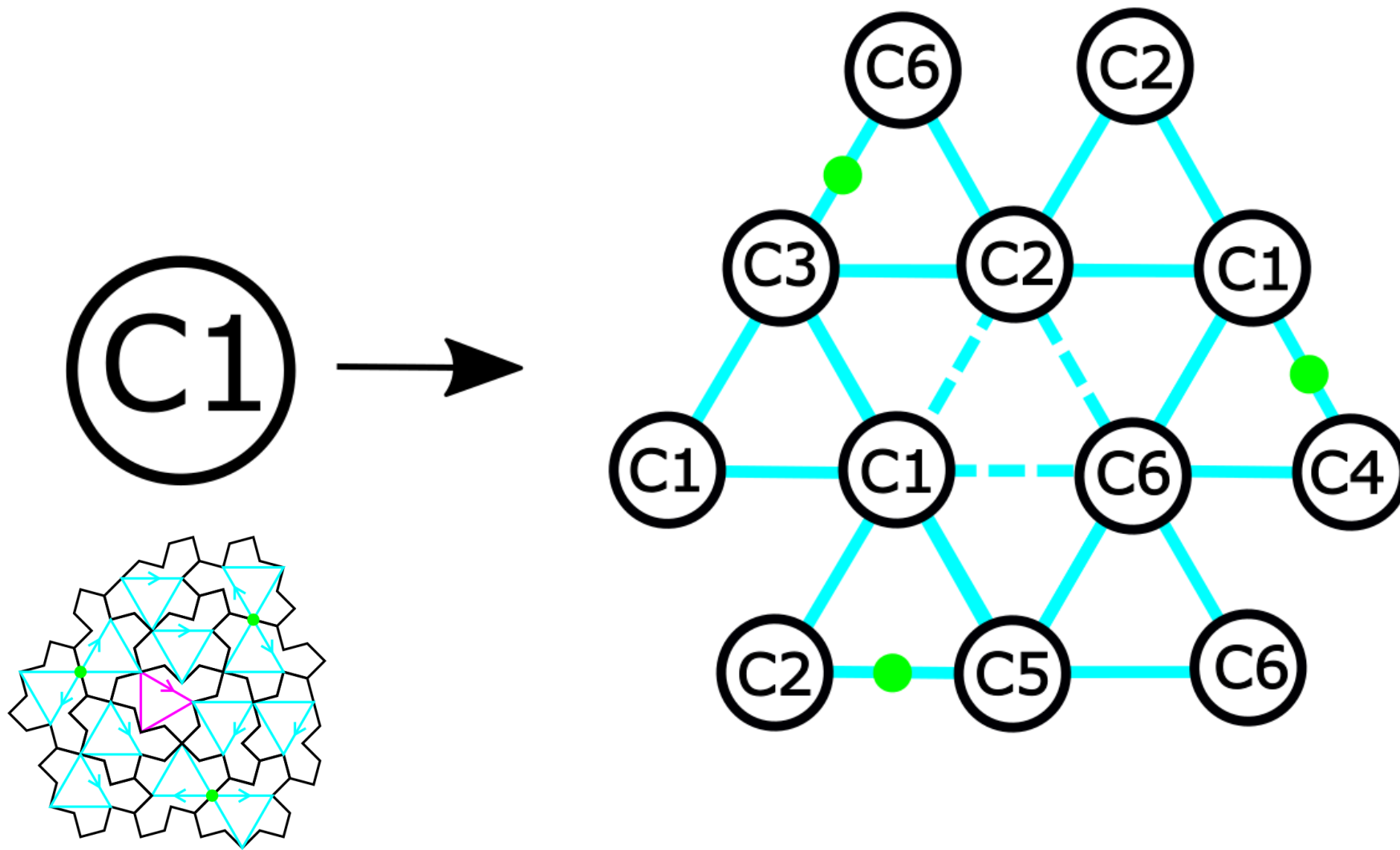


same as full clusters



- ODD tiles for step 1 are obtained by adding 1 (mod 12) to all EVEN tiles in the C cluster (6 tiles)
- Directions of translated and half-turned tiles are exchanged (**see the change of position for green points**). Then three tiles are translated (M1, M3, M11) and three are half-turned (M5→M11, M1→M7, M9→M3)
- ODD tile index is equivalent to the orientation of a full C cluster 😊

# Inflation step 1





# Homochiral python code on github

- <https://github.com/marimperorclerc/homochiral>

python code for drawing clusters with  $\text{Tile}(a,b)$  in svg graphics

## References

Code is based on the drawsvg package:

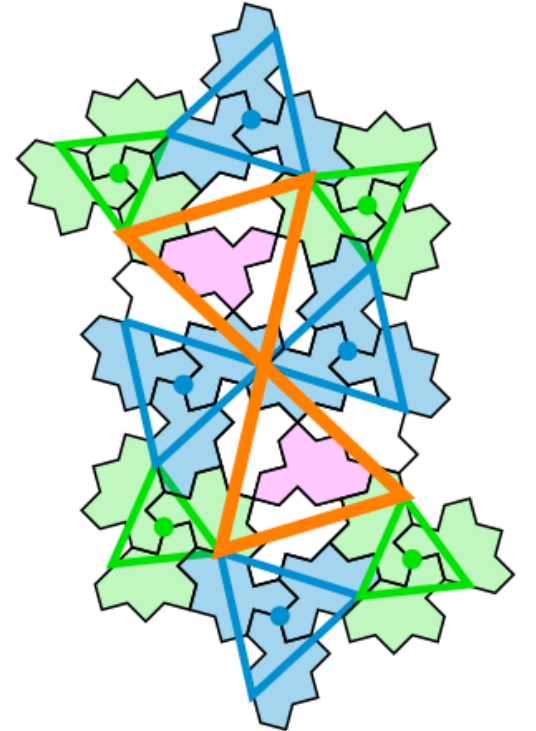
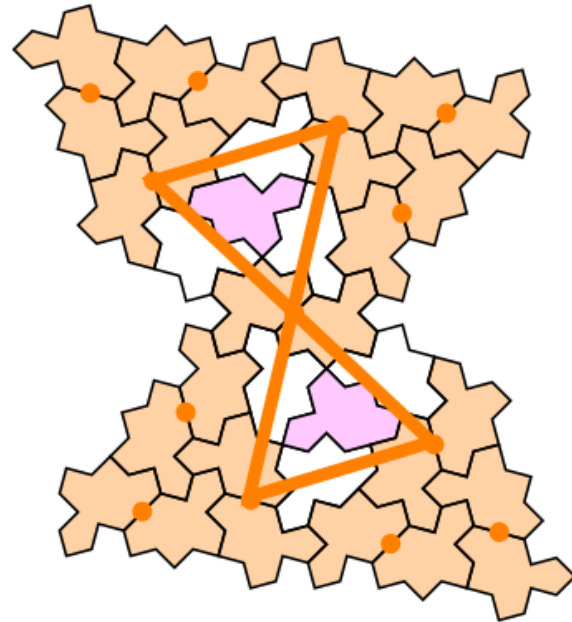
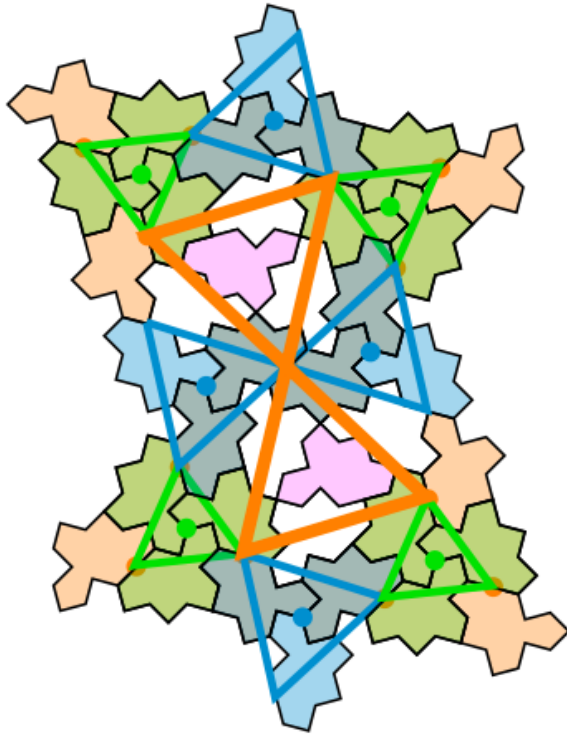
<https://pypi.org/project/drawsvg/>

And the python code with alternate chirality for inflation for spectre tiling:

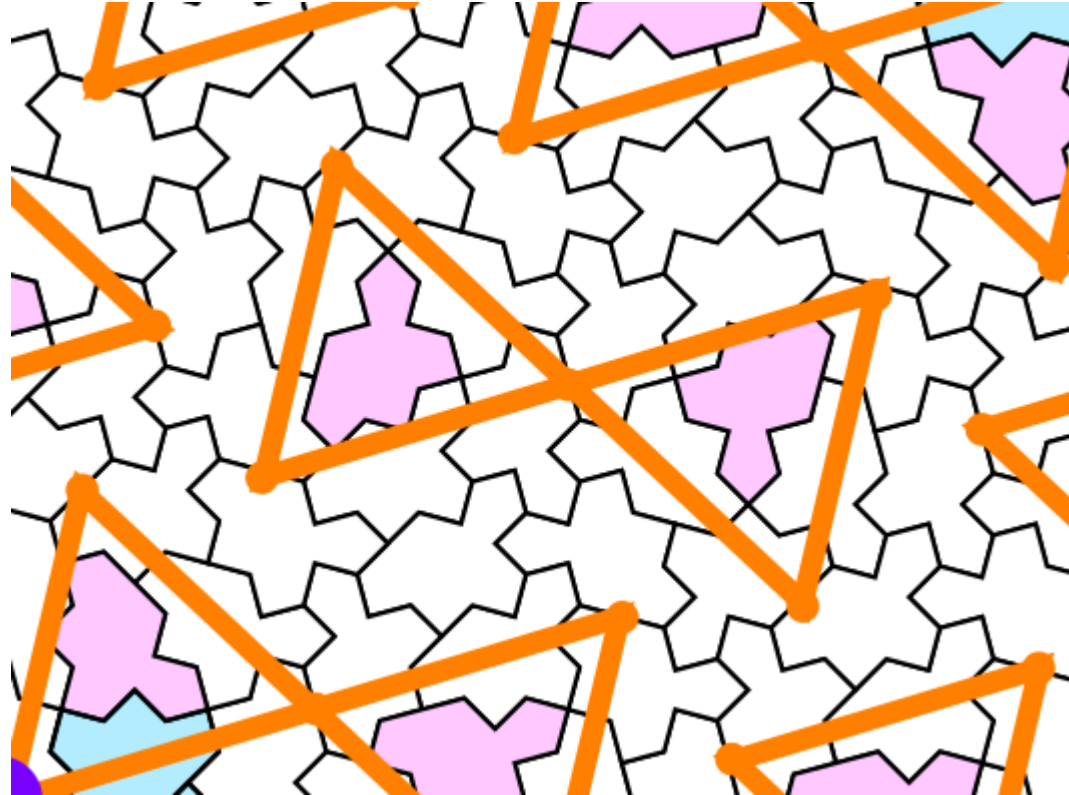
<https://github.com/shrx/spectre/blob/master/spectre.py>

- Adapted here (spectre-colored-tiles.ipynb) with different color choices
- See some .svg files in \svg-spectre folder

# DIABOLO

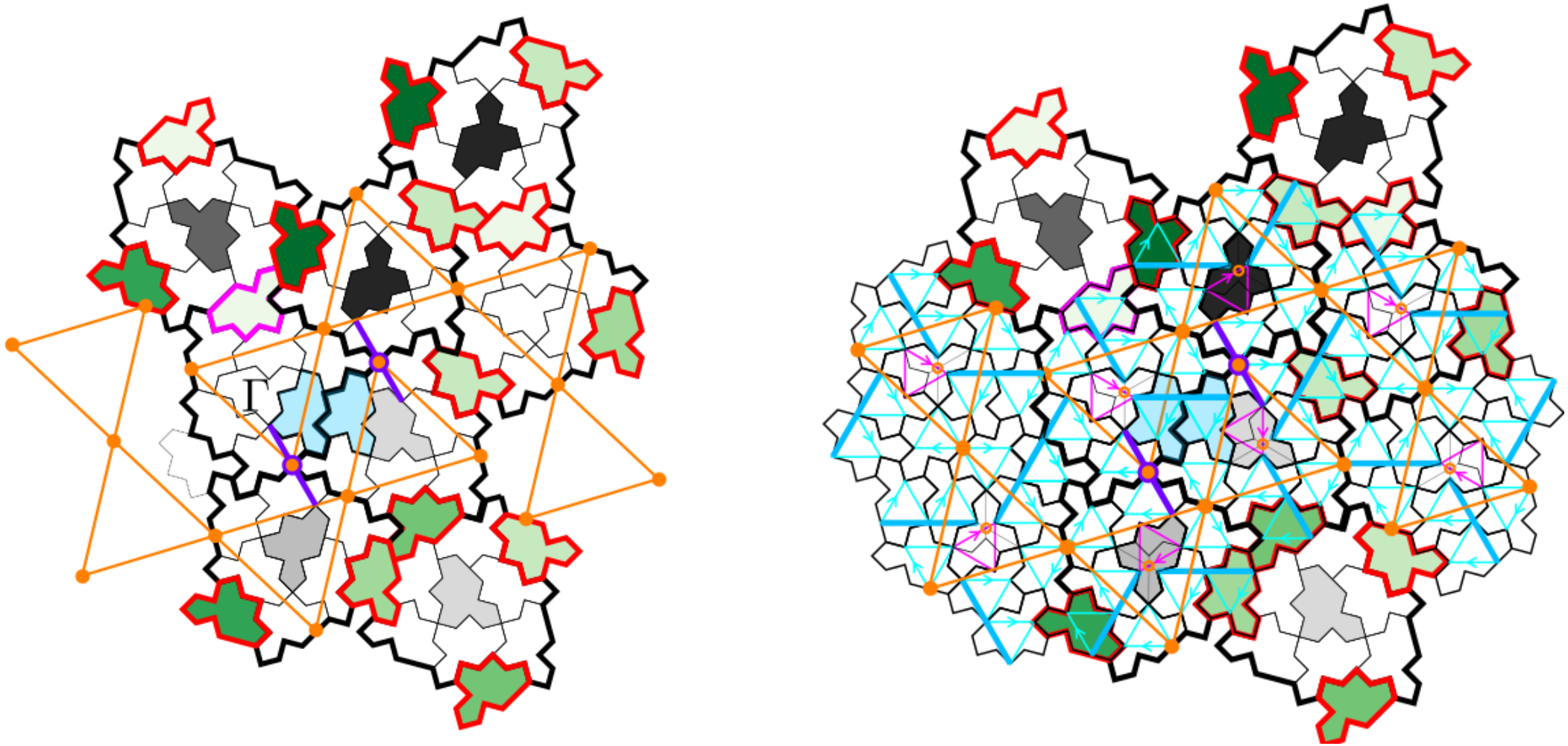


# Diabolo pattern



- Local 2-fold axis

# Link with triangular pattern

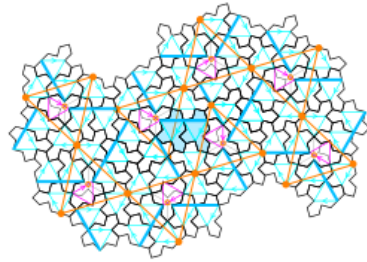


- The two other environnements with a violet lines are always coupled together

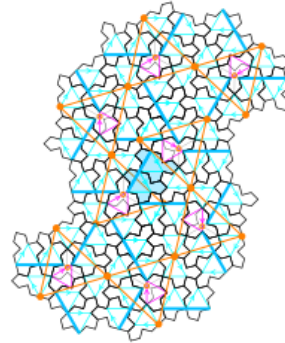


# Six orientations of the 2-arms

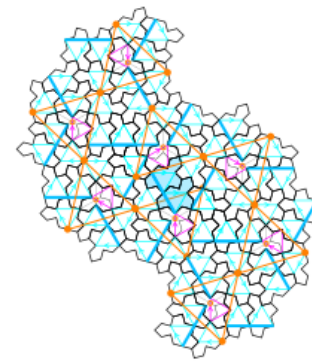
P1



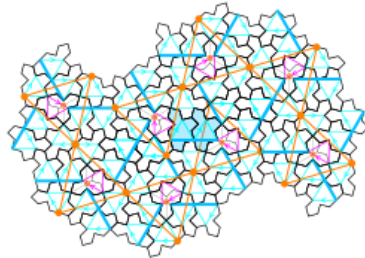
P2



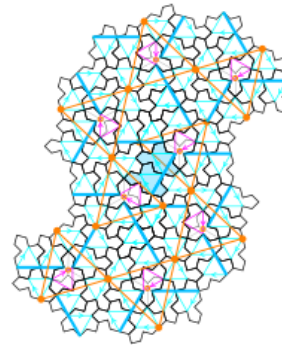
P3



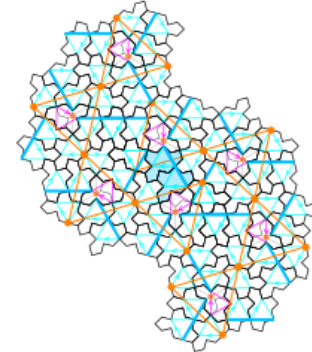
P4



P5

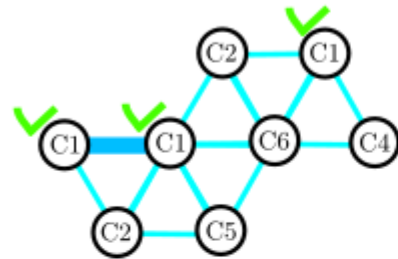


P6

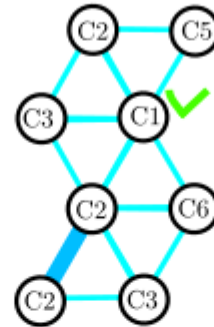


# Six orientations of the 2-arms

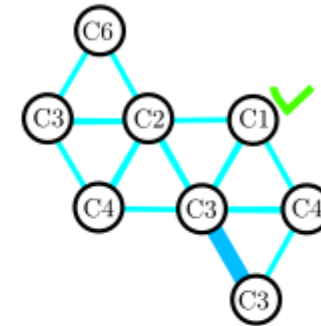
P1



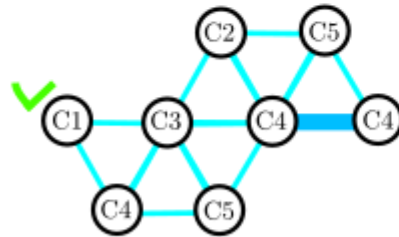
P2



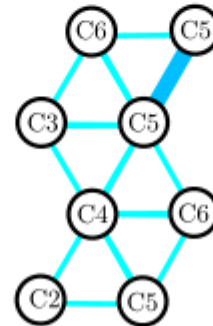
P3



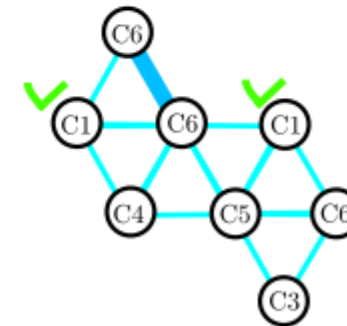
P4



P5



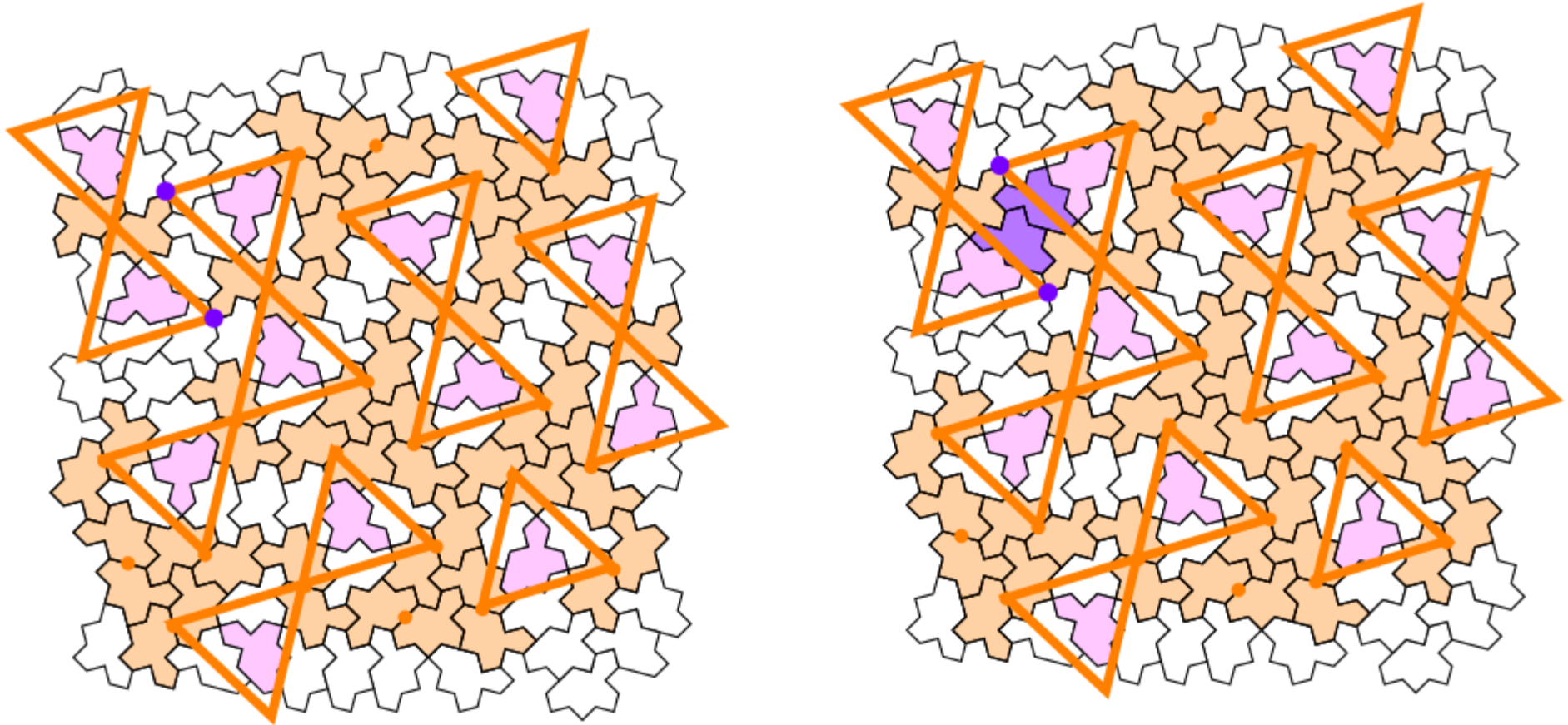
P6



- 8 combinations for a given cluster orientation (green crosses)

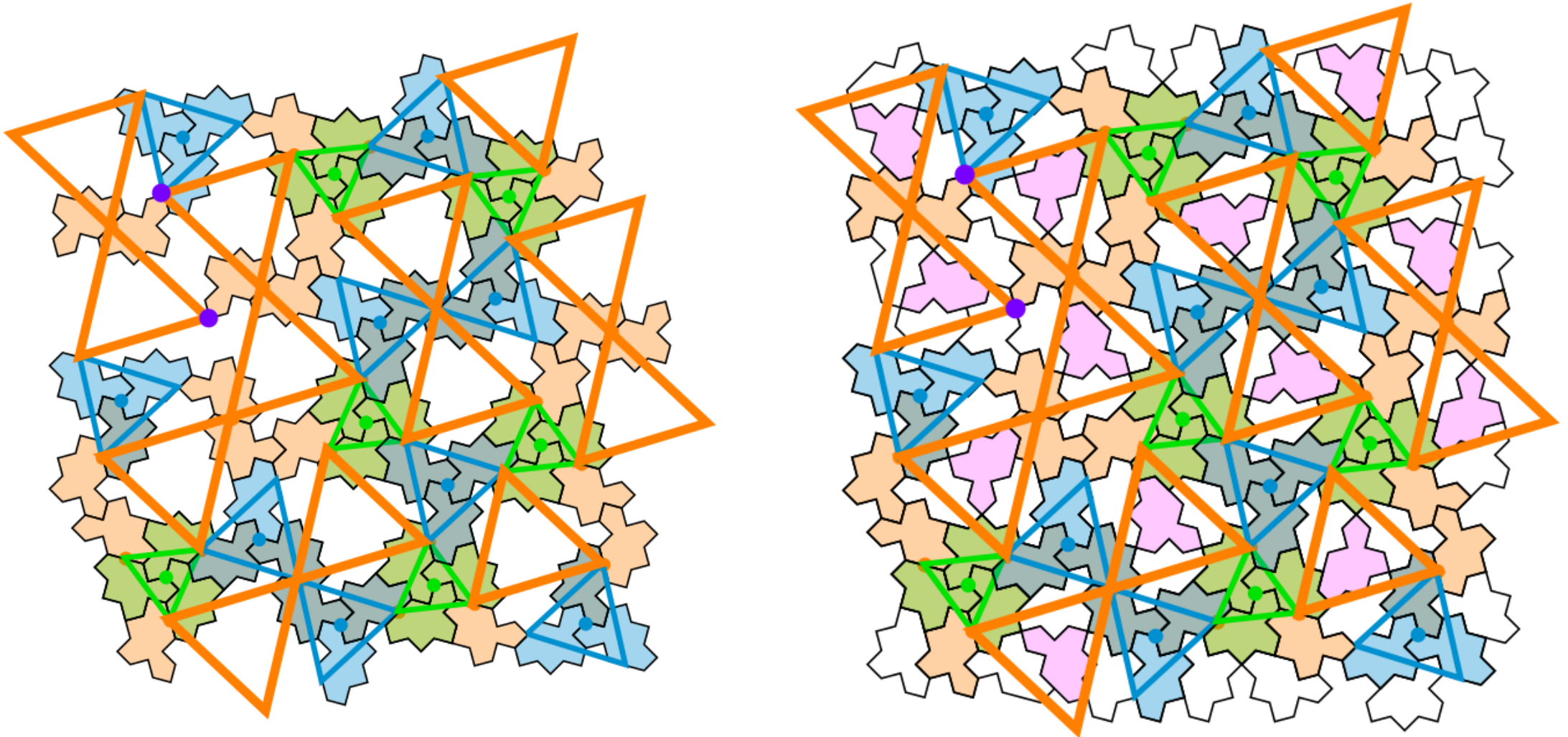
# Example: Set of all local 2-fold symmetry centers

2-fold



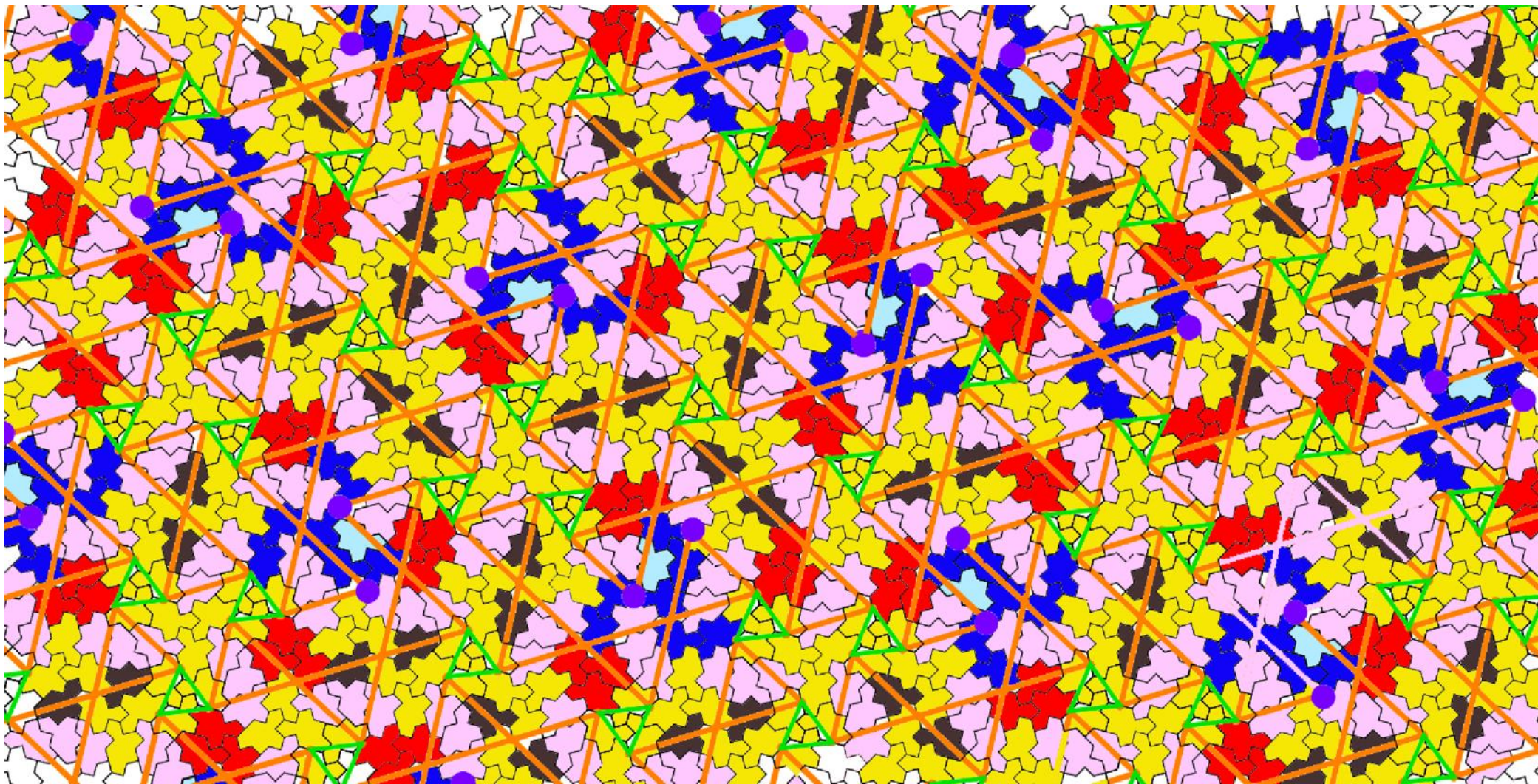
- around each ODD monotile (pink), three points (orange triangle)
- Violet points are not local 2-fold centers, but all other points are (orange)

# superposition



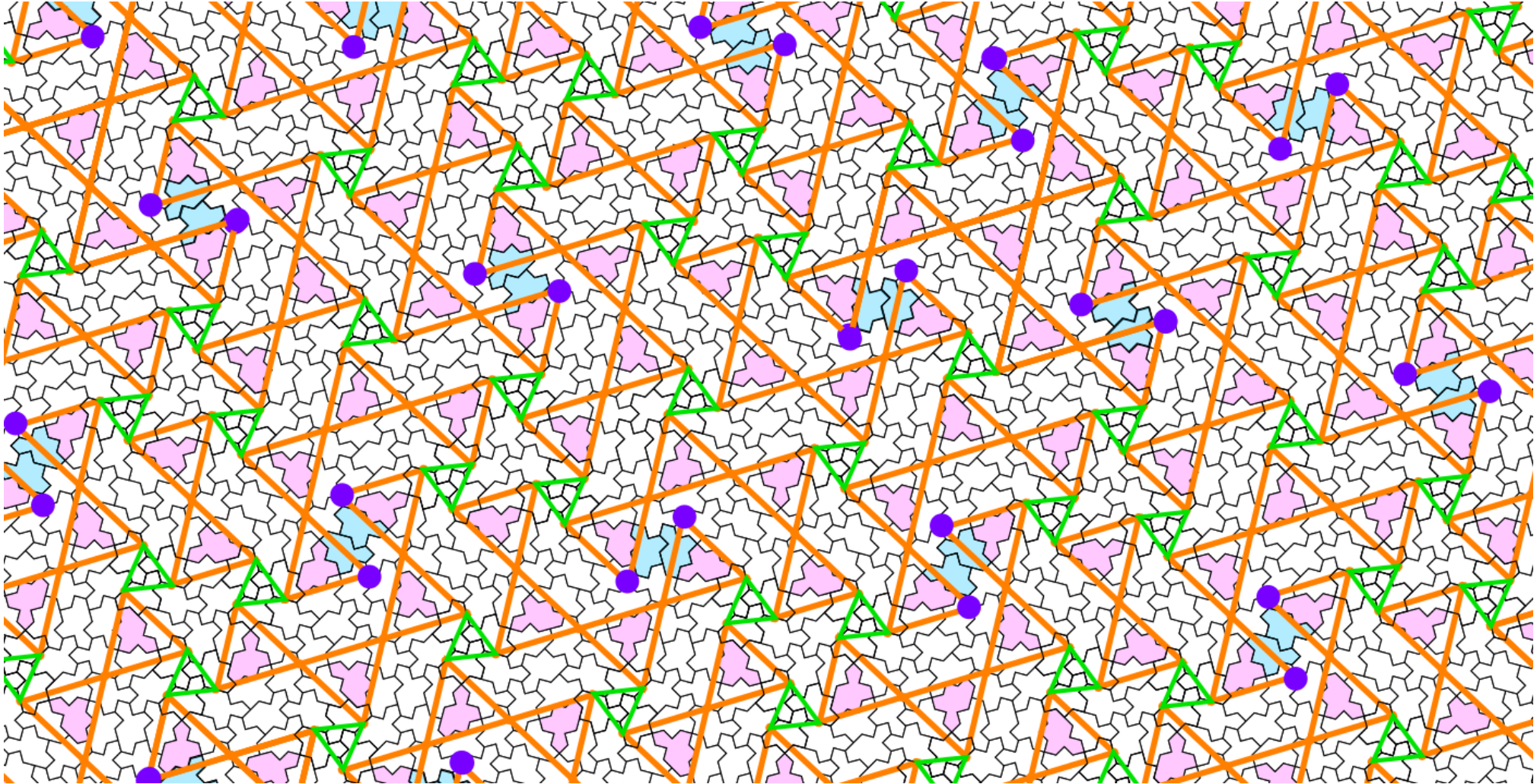


# Comparision





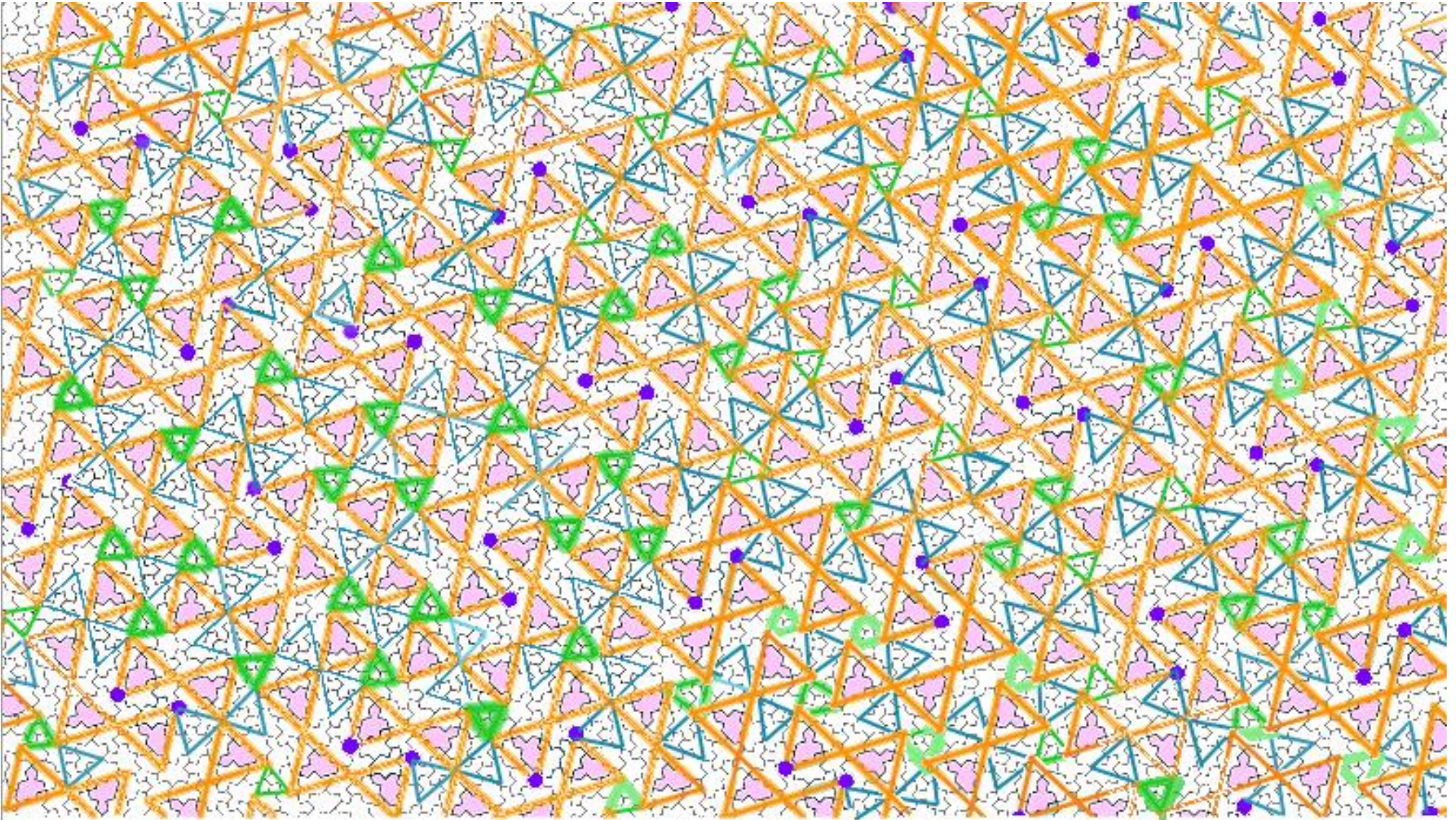
# Small stars connect the orange triangles



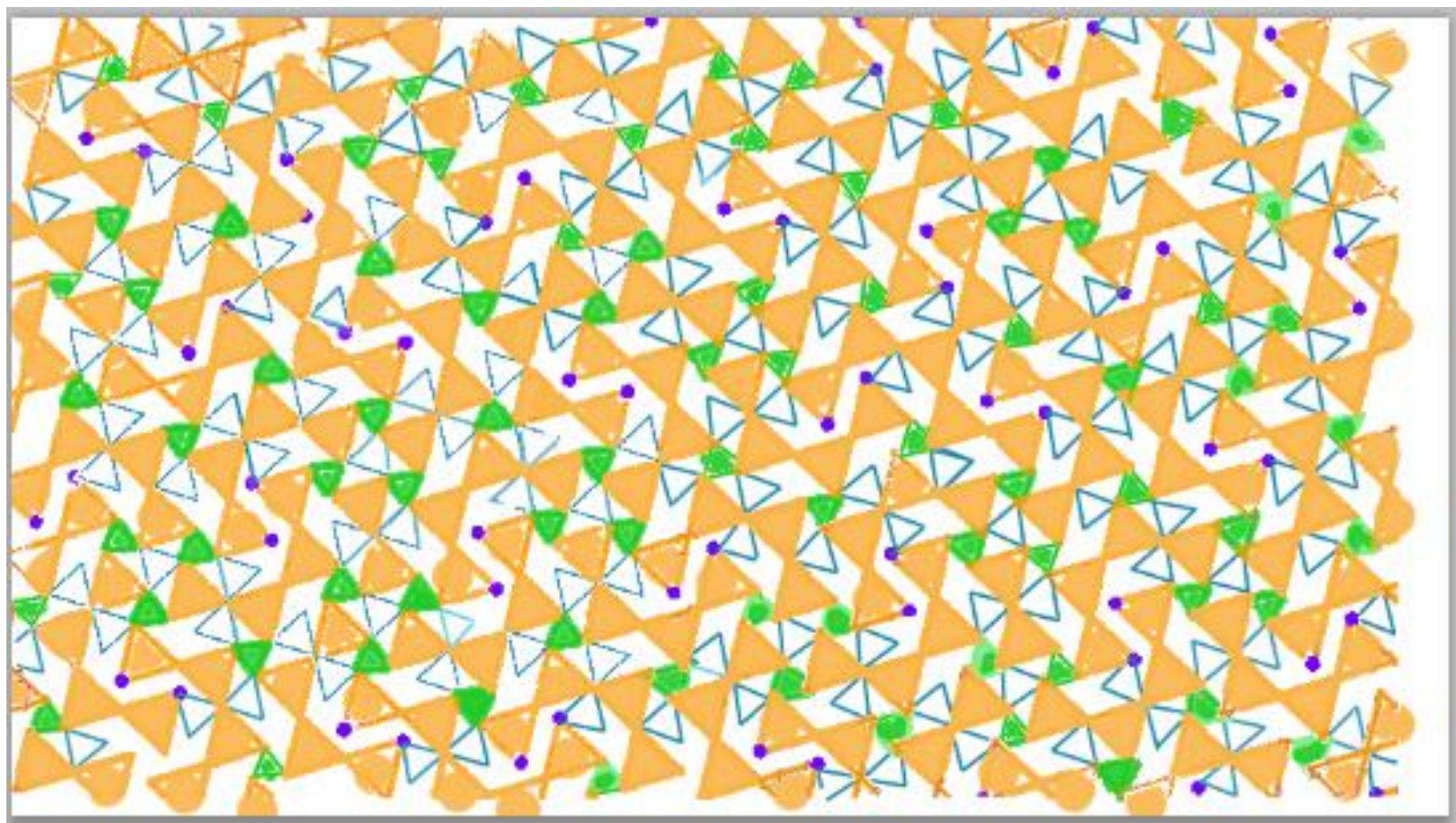
- Possibility to obtain the relative positions between diablo, 2-arms and 3-arms 77



# SMALL AND LARGE STARS

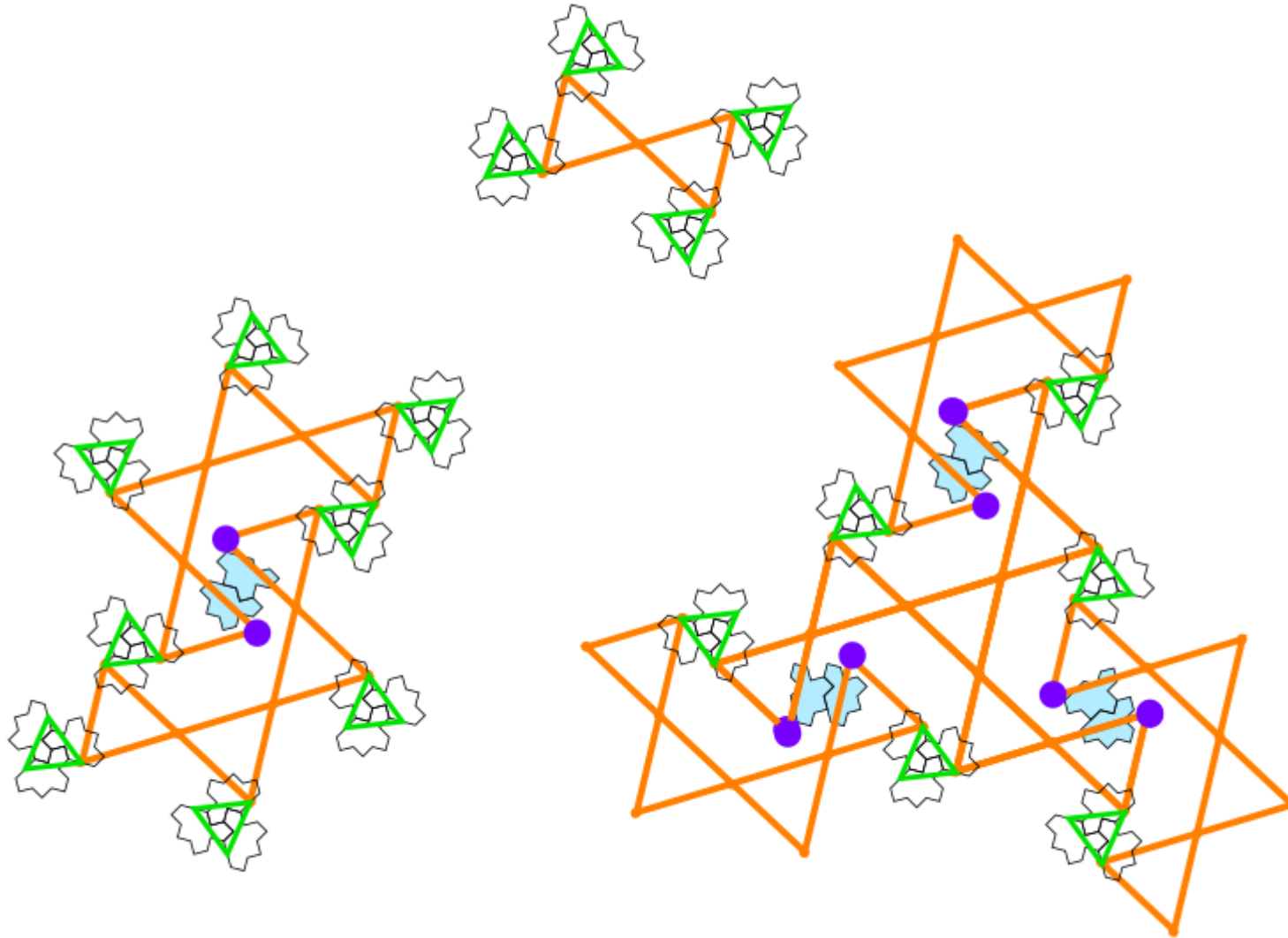


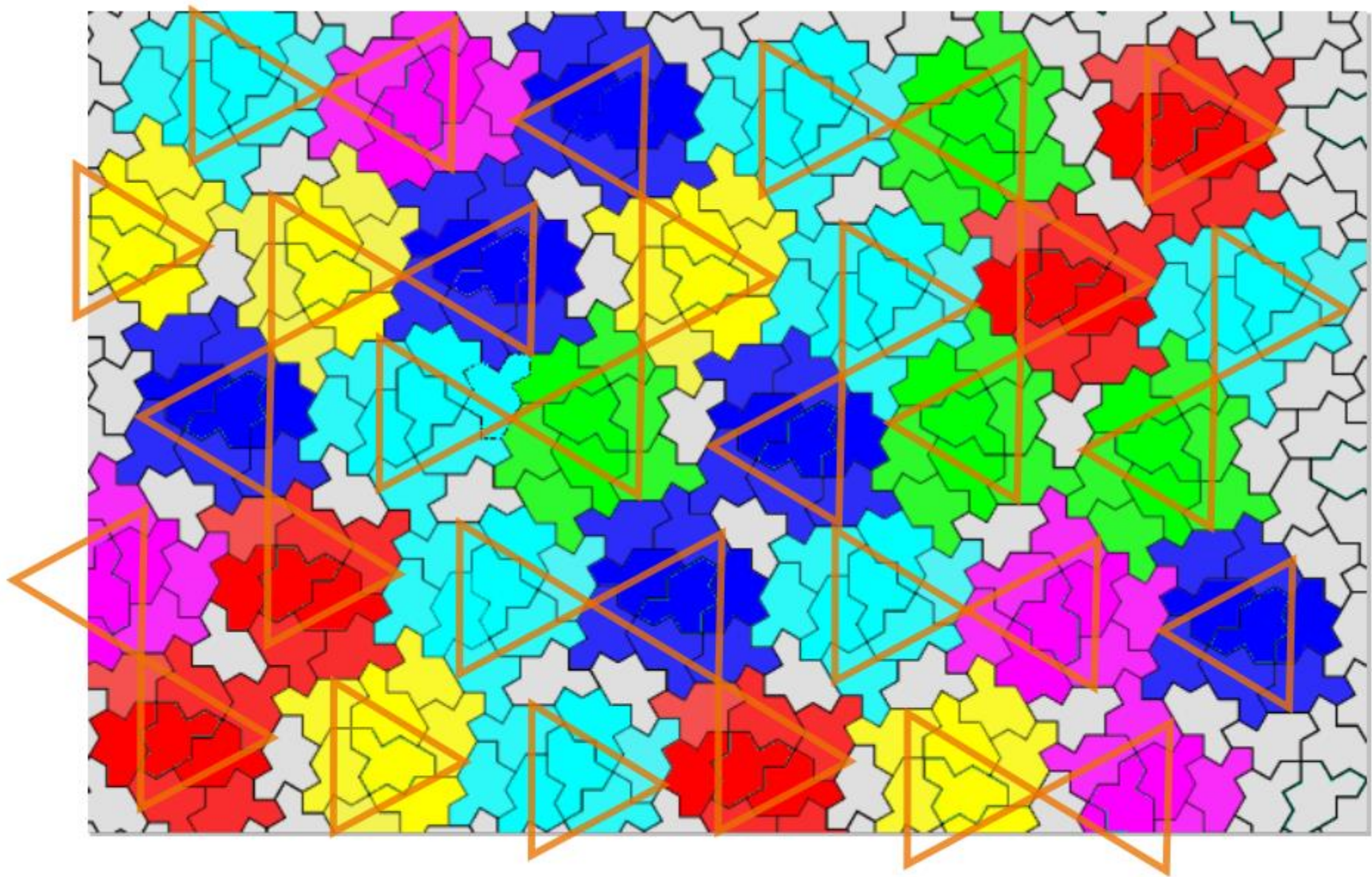




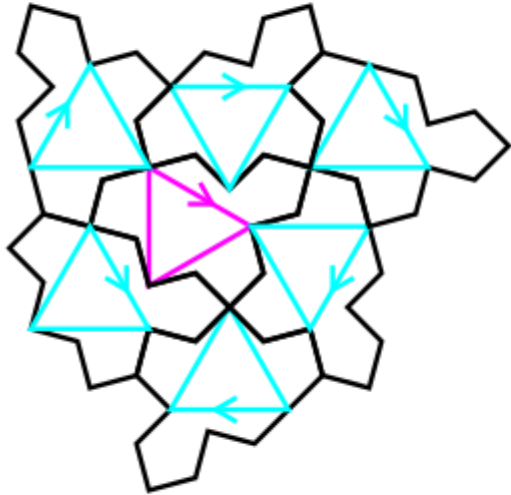


# New puzzle

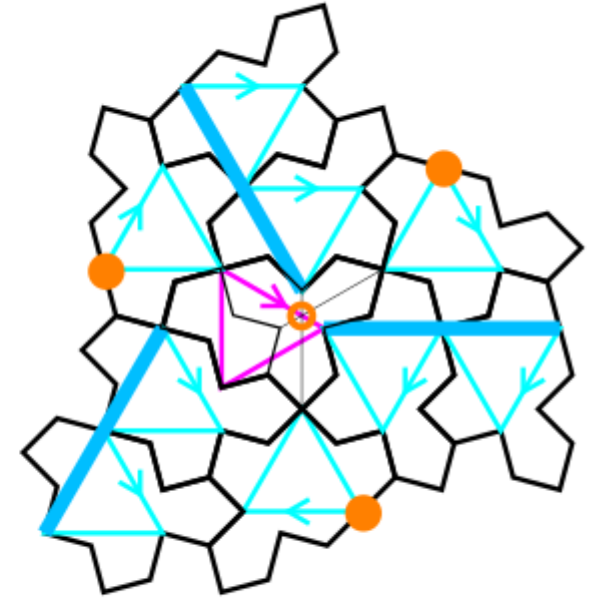
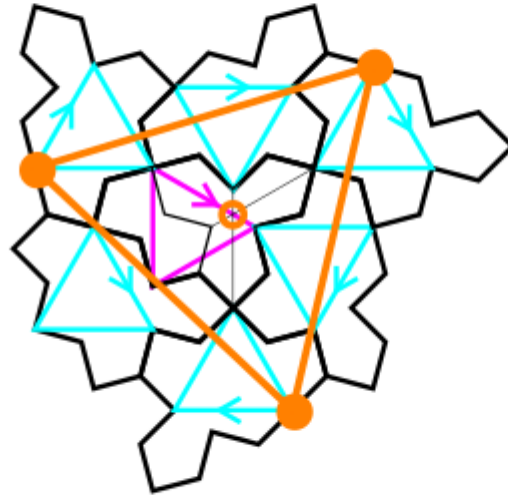




- Tile(1,1)



## full cluster



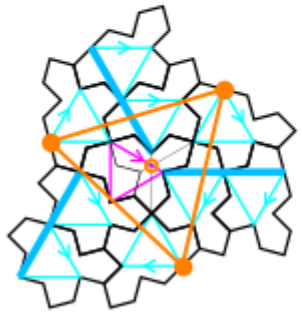
- C cluster: one ODD tile at the center + 6 EVEN tiles around
- Three glue points (orange color): regular triangle centered on the trifle shape of the ODD tile
- Full cluster: 3 more EVEN tiles obtained by translations (three dark blue lines)



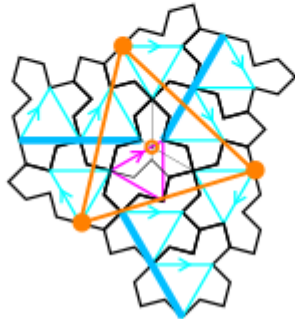
# full cluster with its six rotations: $n(2\pi/6)$

- Tile(1,1)

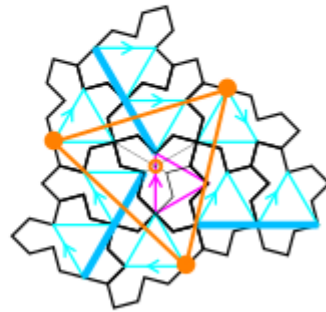
C1



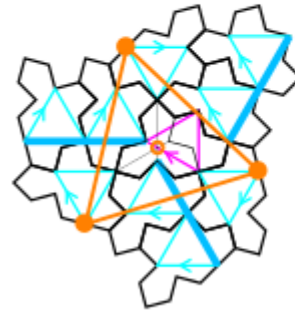
C2



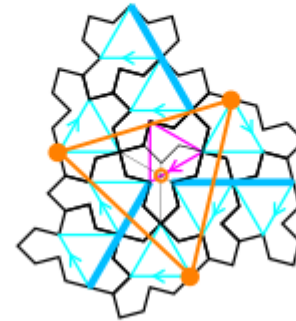
C3



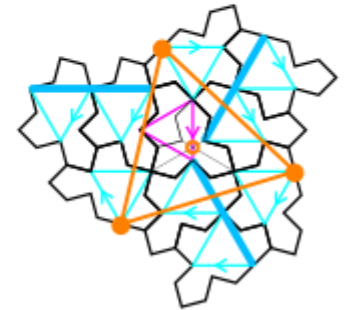
C4



C5

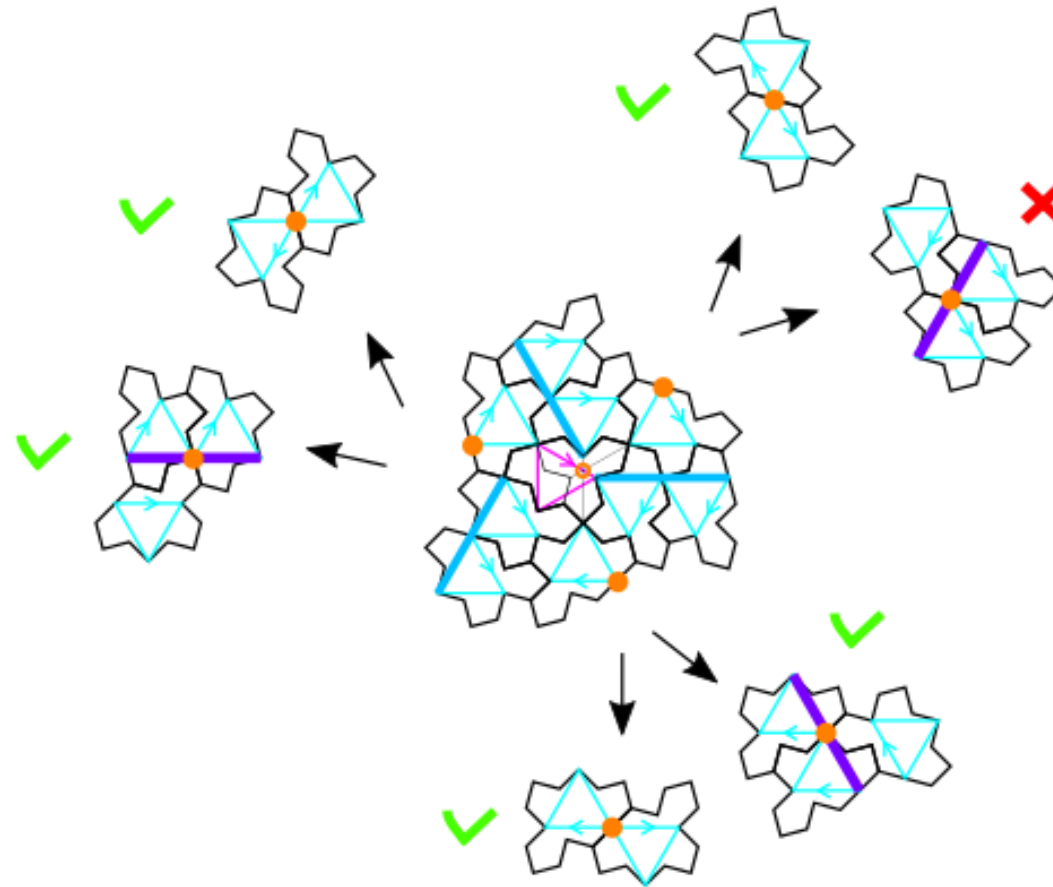


C6



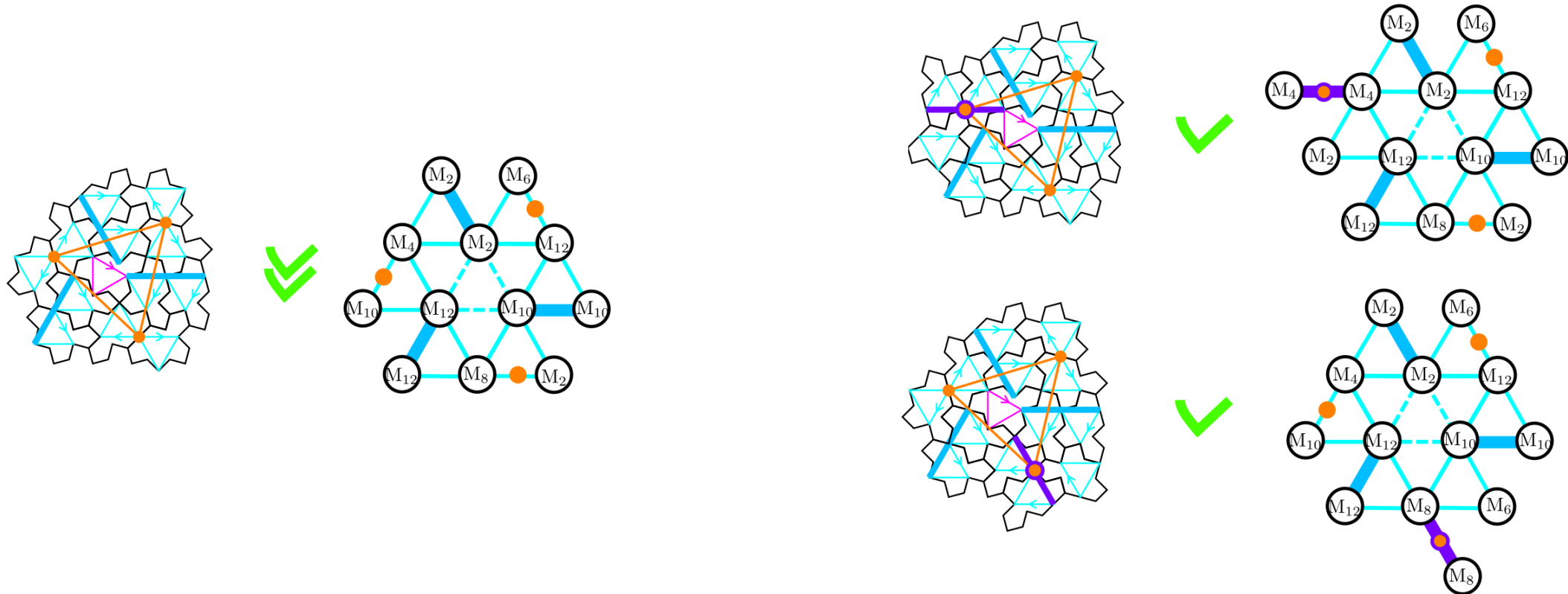
- 13 tiles in a full cluster: one ODD tile at the center + 9 EVEN tiles around
- Blue triangles (EVEN) have one horizontal edge when pink (ODD) have a vertical one

# Possible configurations at the glue points ?



- Half-turned or translated monotile (violet line)

# Three possible configurations at the glue points

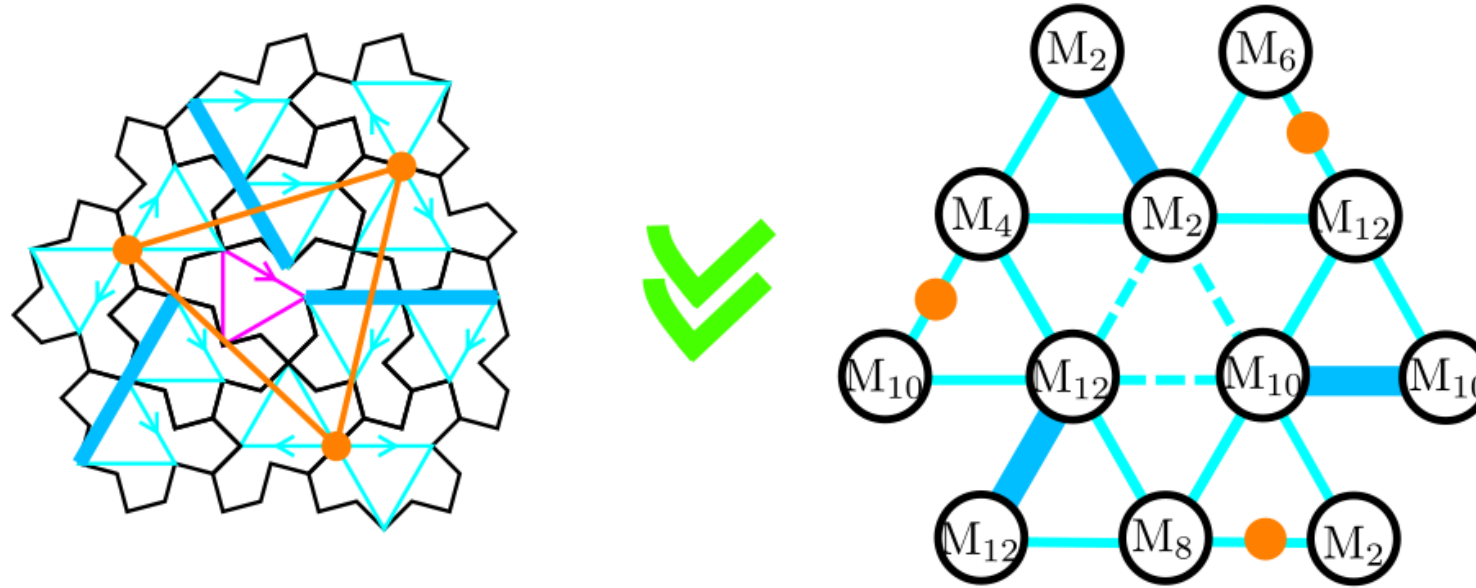


- configuration with three half-turned tiles is the most common
- The other two are always coupled together in the tiling:
- One violet line in a 2-arms and three violet lines in a 3-arms



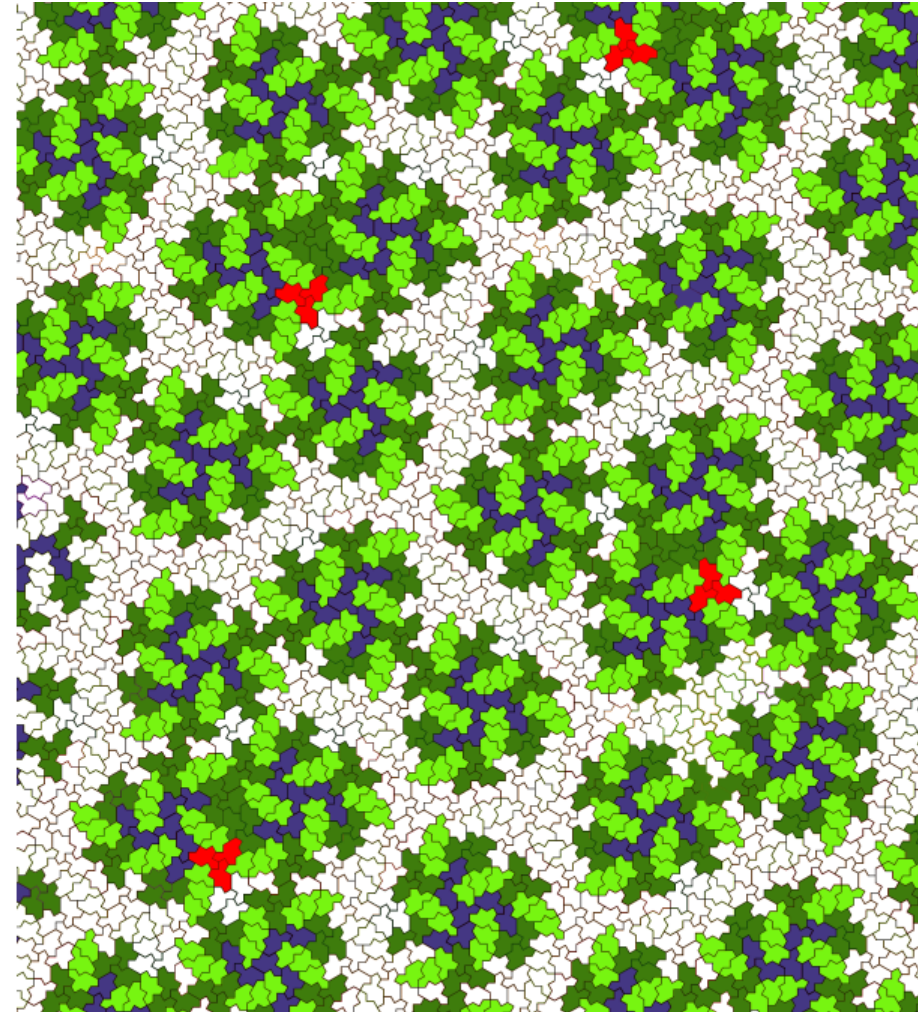
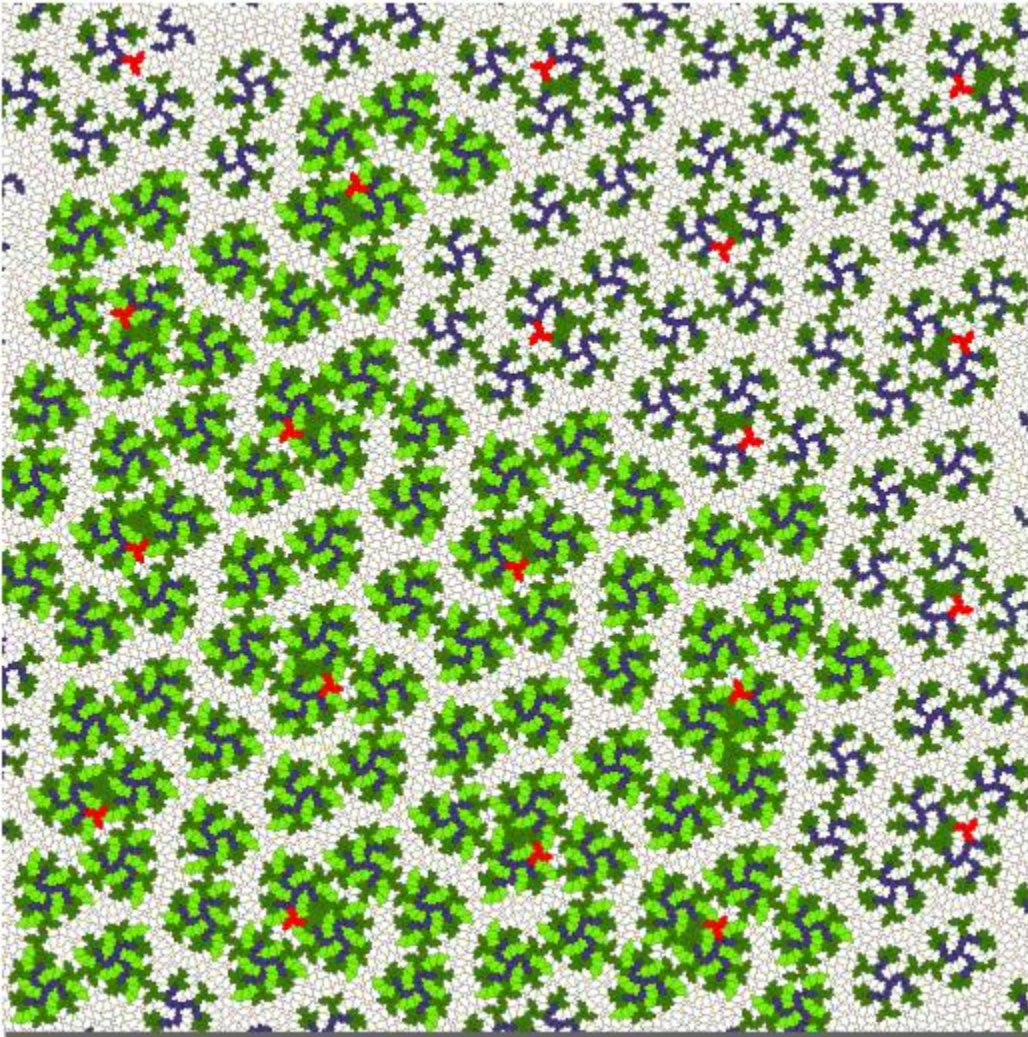
# Orientation of the EVEN tiles in the most frequent configuration

- Tile(1,1)



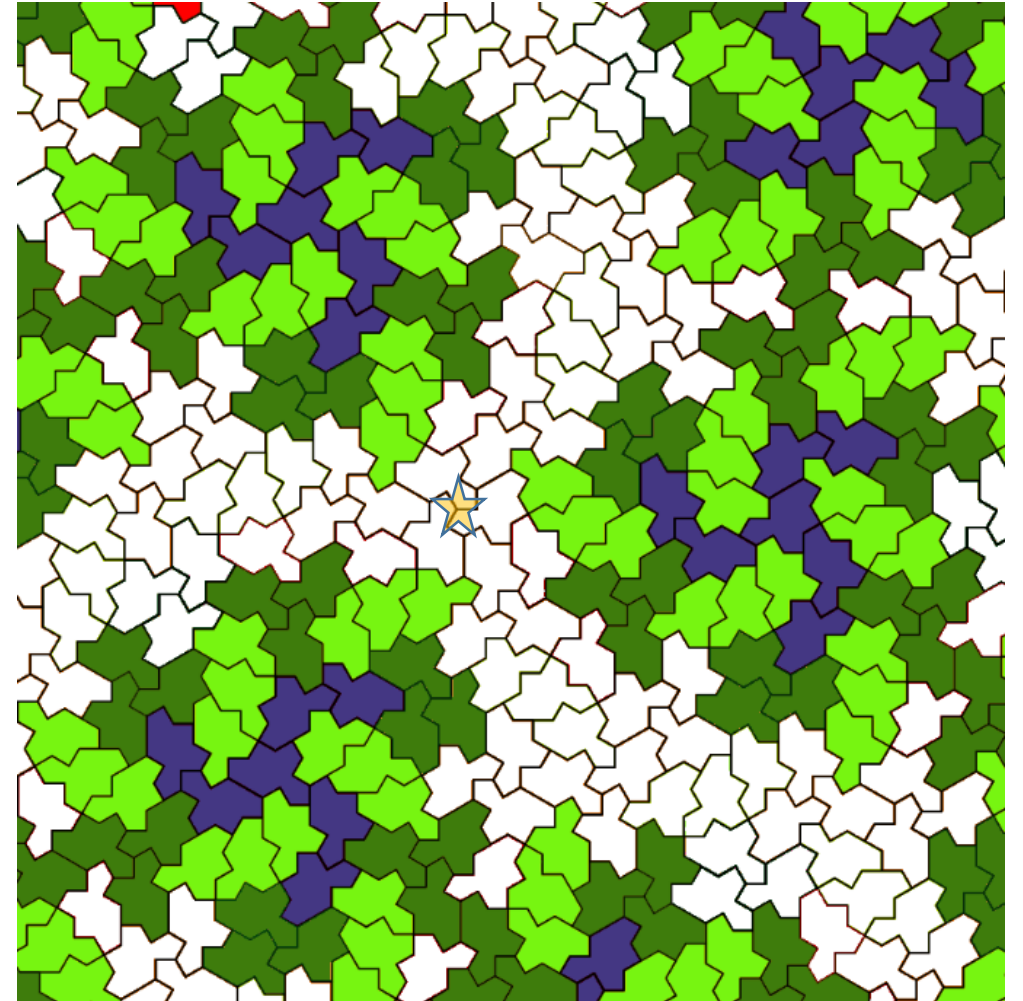
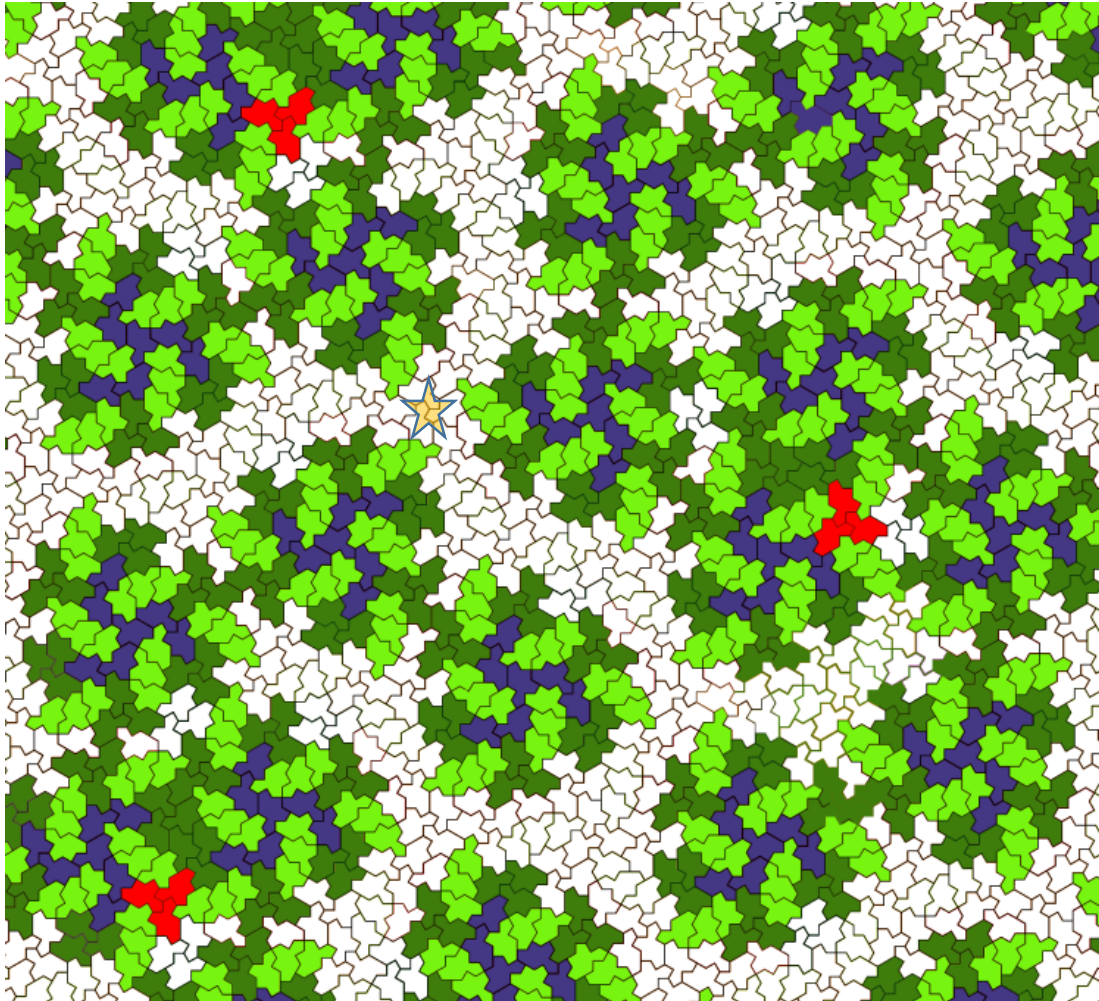
- 6 tiles in the C-cluster (corona made of M12, M4, M2, M12, M10, M8)
- 3 translated (M2, M10, M12) (along blue lines, the edges of blue triangles)
- 3 half-turned (M4 → M10, M12 → M6, M8 → M2)

# Diabolo/2-arms/3-arms at level 2 by Boris Horvat



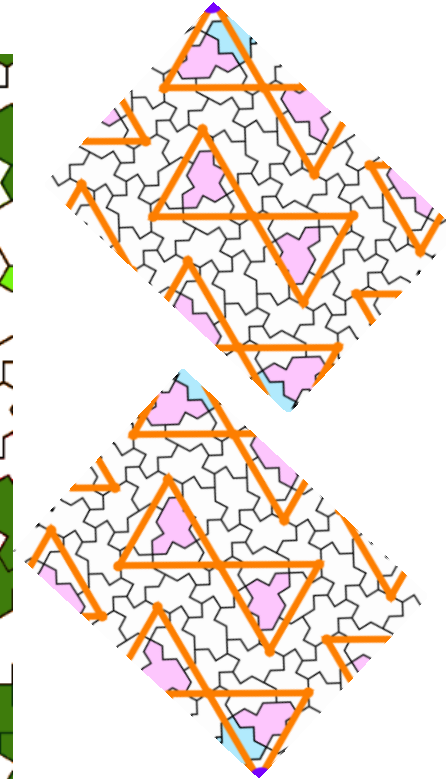
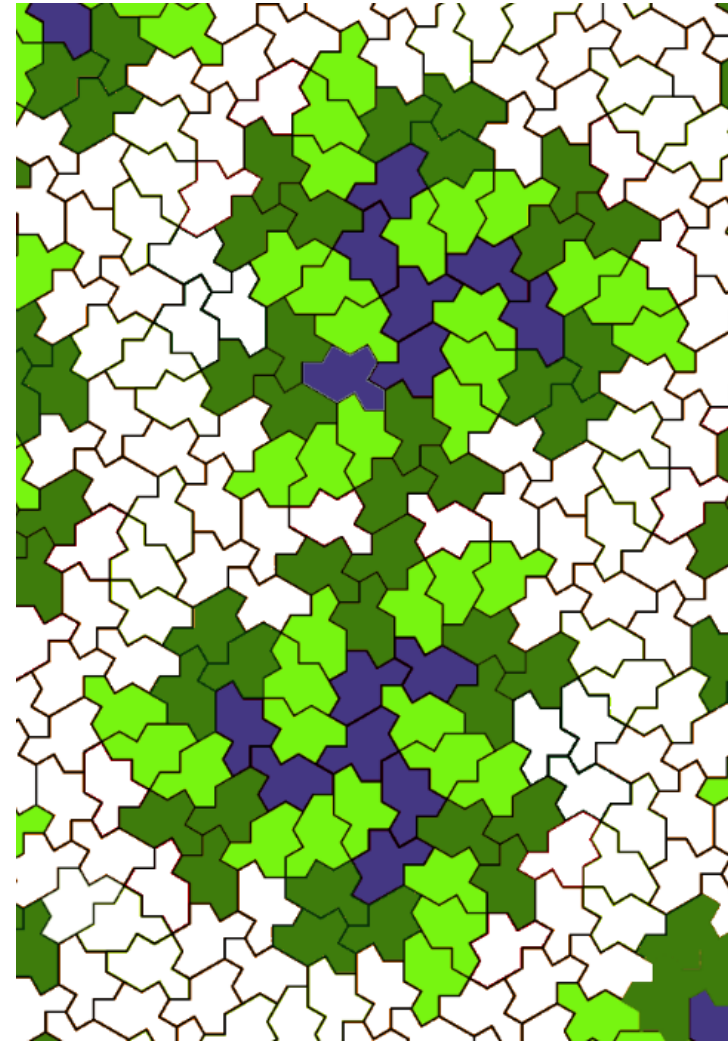
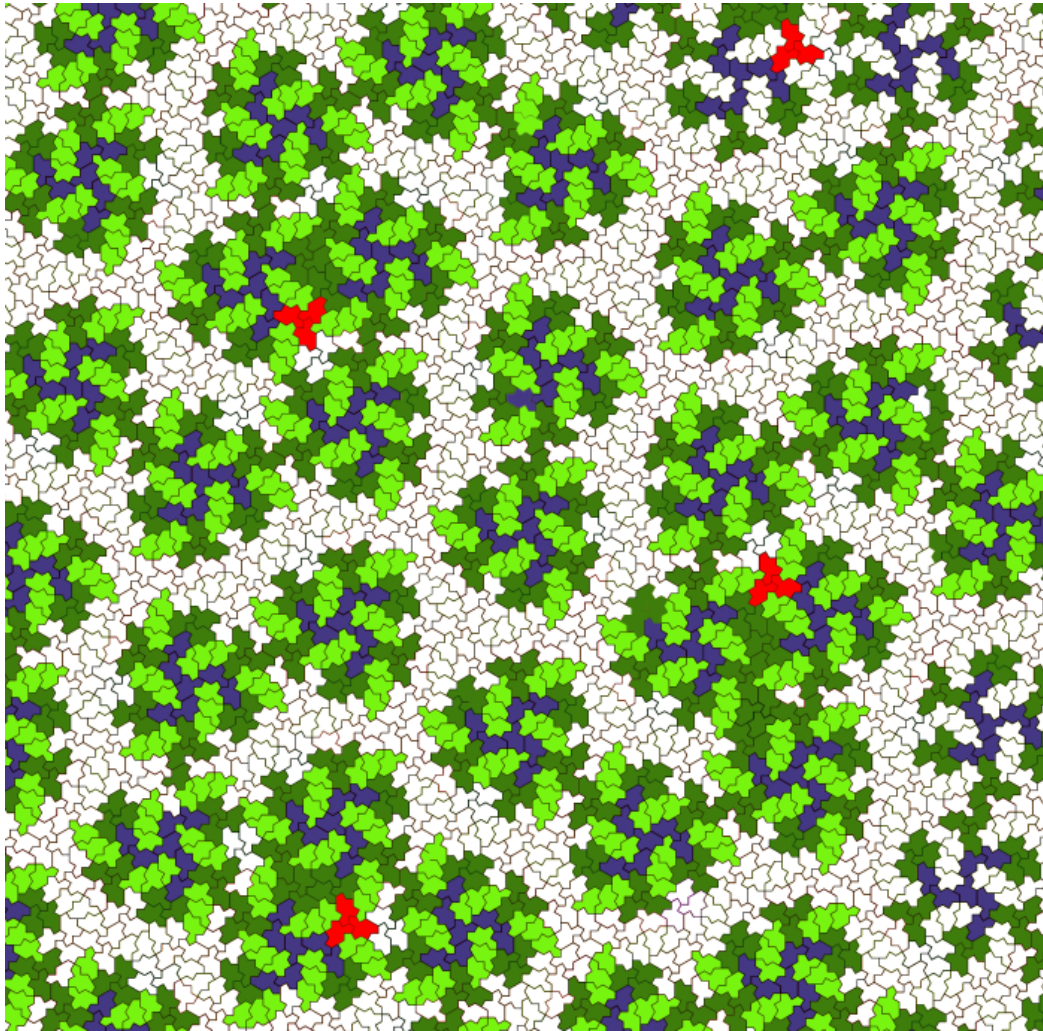


# Diabolo/2-arms/3-arms at level 2 by Boris Horvat

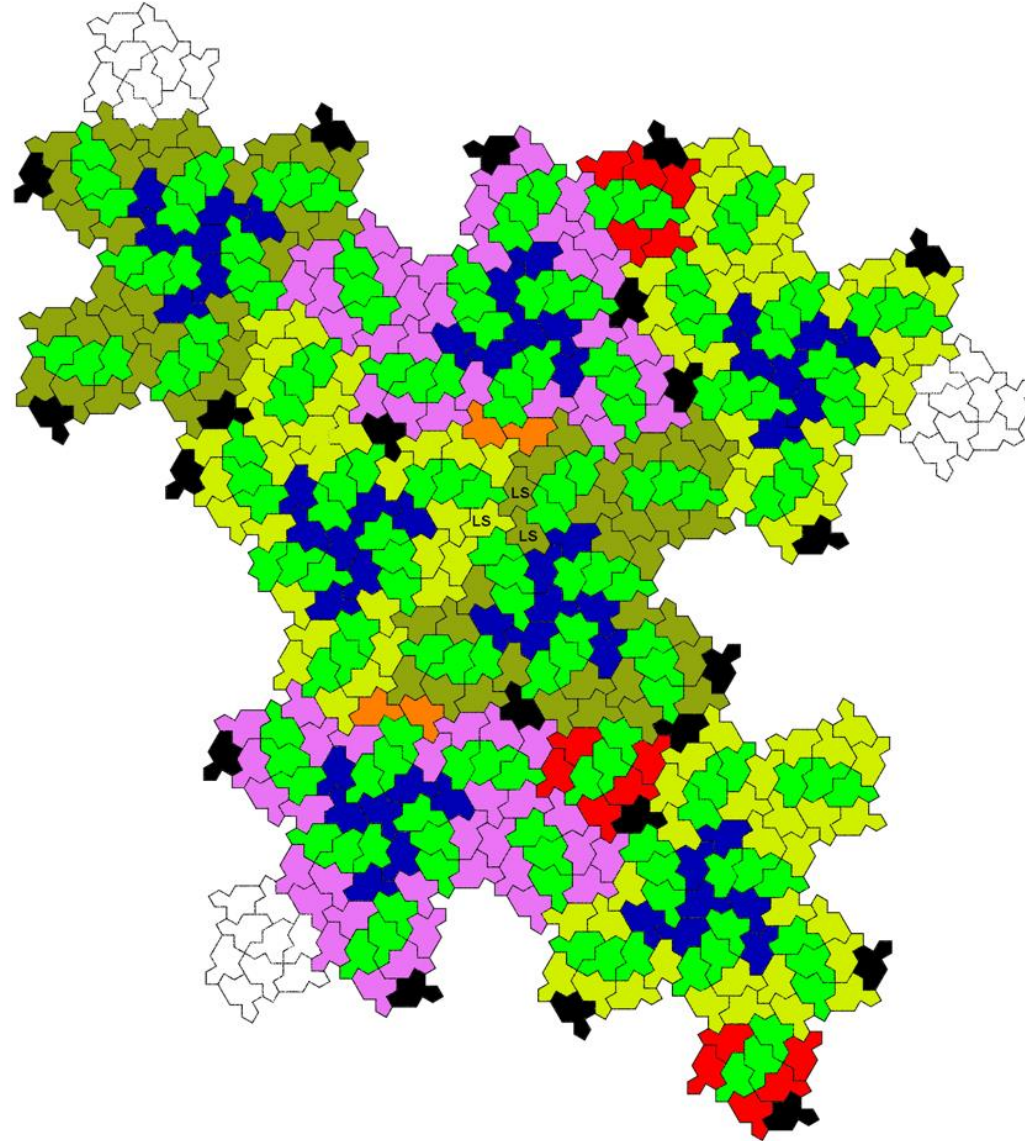




# Diabolo/2-arms/3-arms at level 2 by Boris Horvat

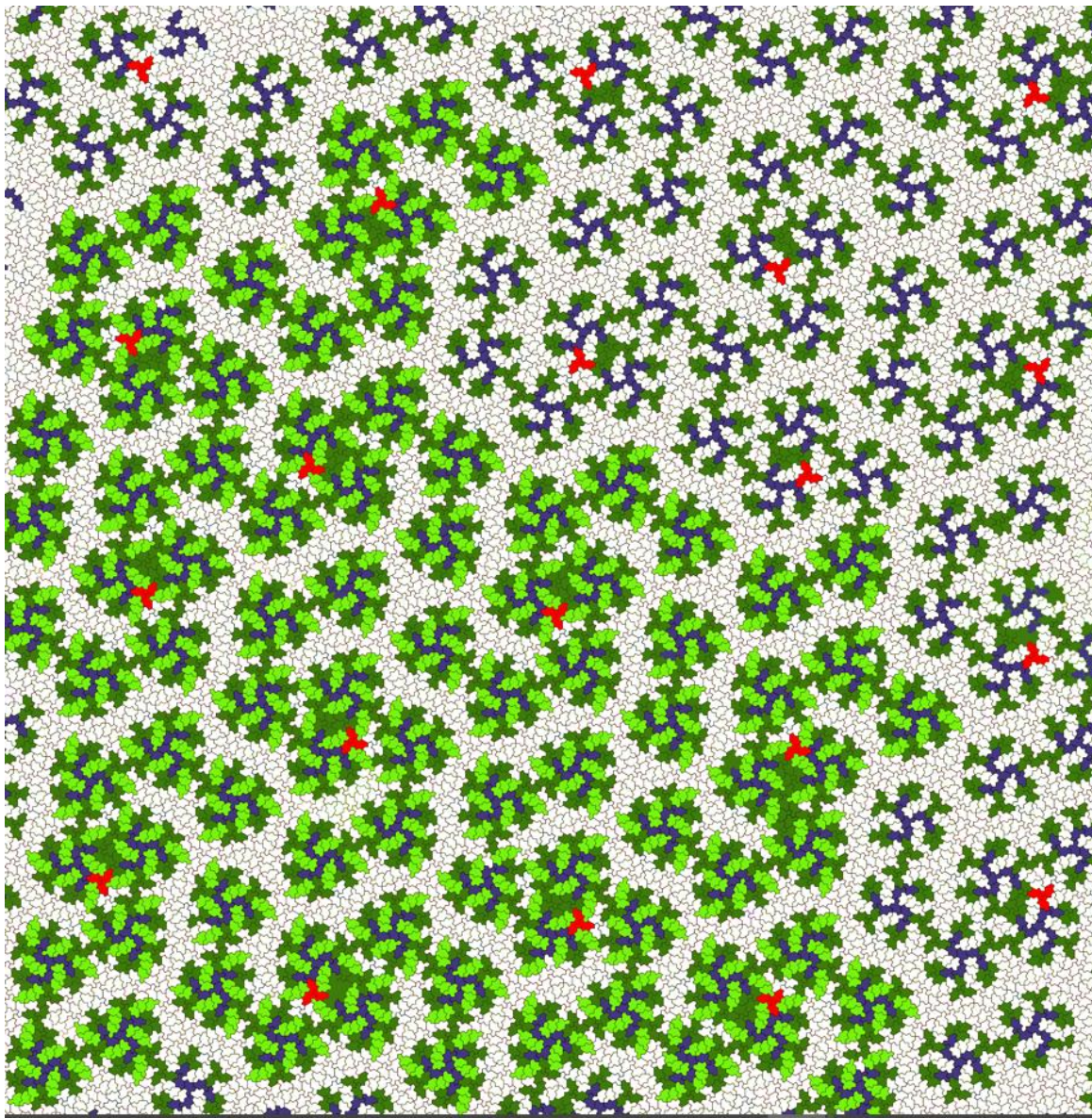
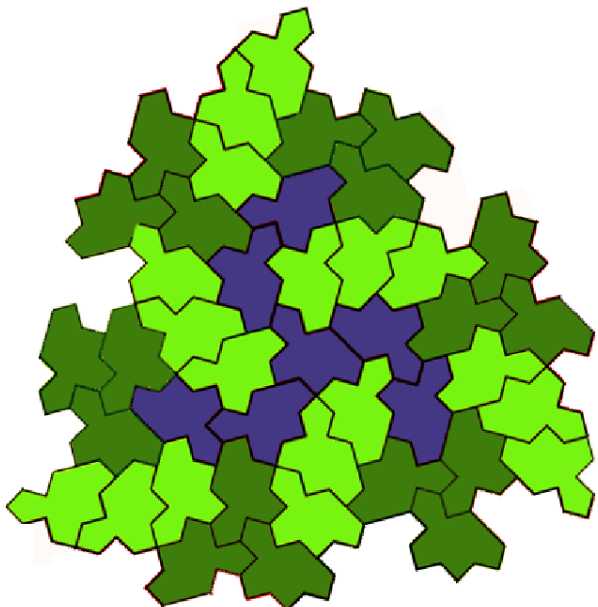


# Diabolo/2-arms/3-arms at level 2 by Boris Horvat





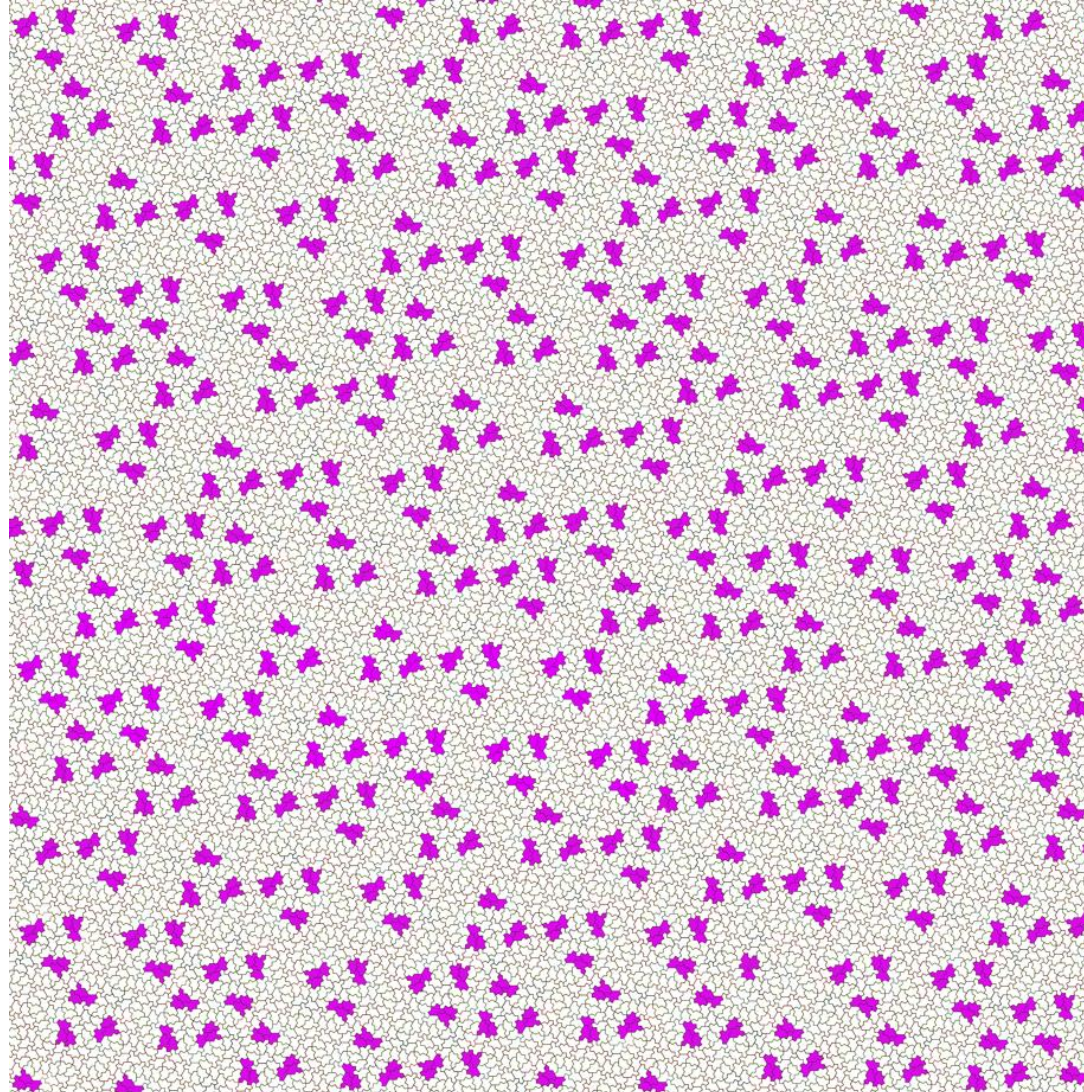
# Chimera level 2 by Boris Horvat





# Triplets Marianne by Boris Horvat

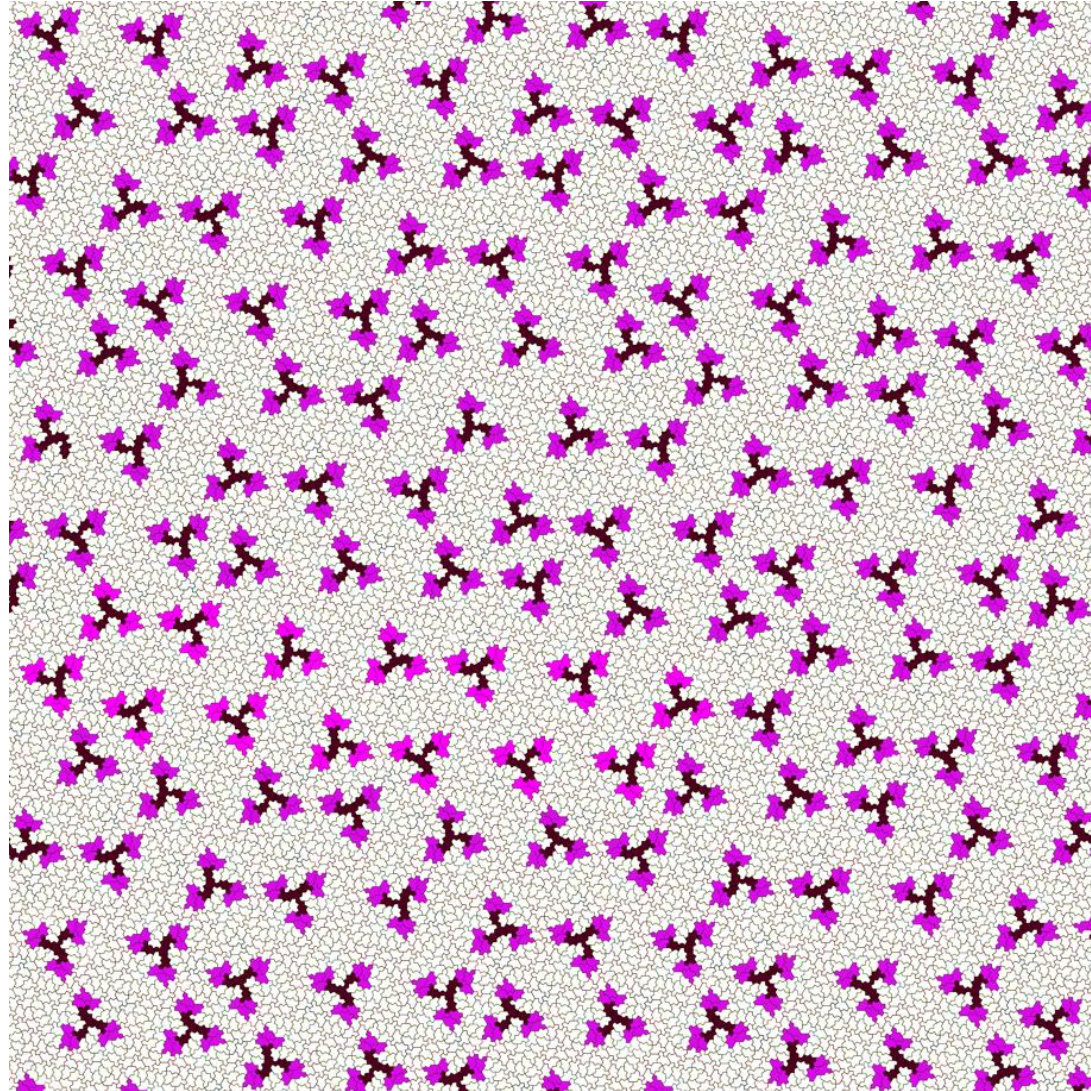
**Diabolo  
2-arms  
3-arms  
are visible !**





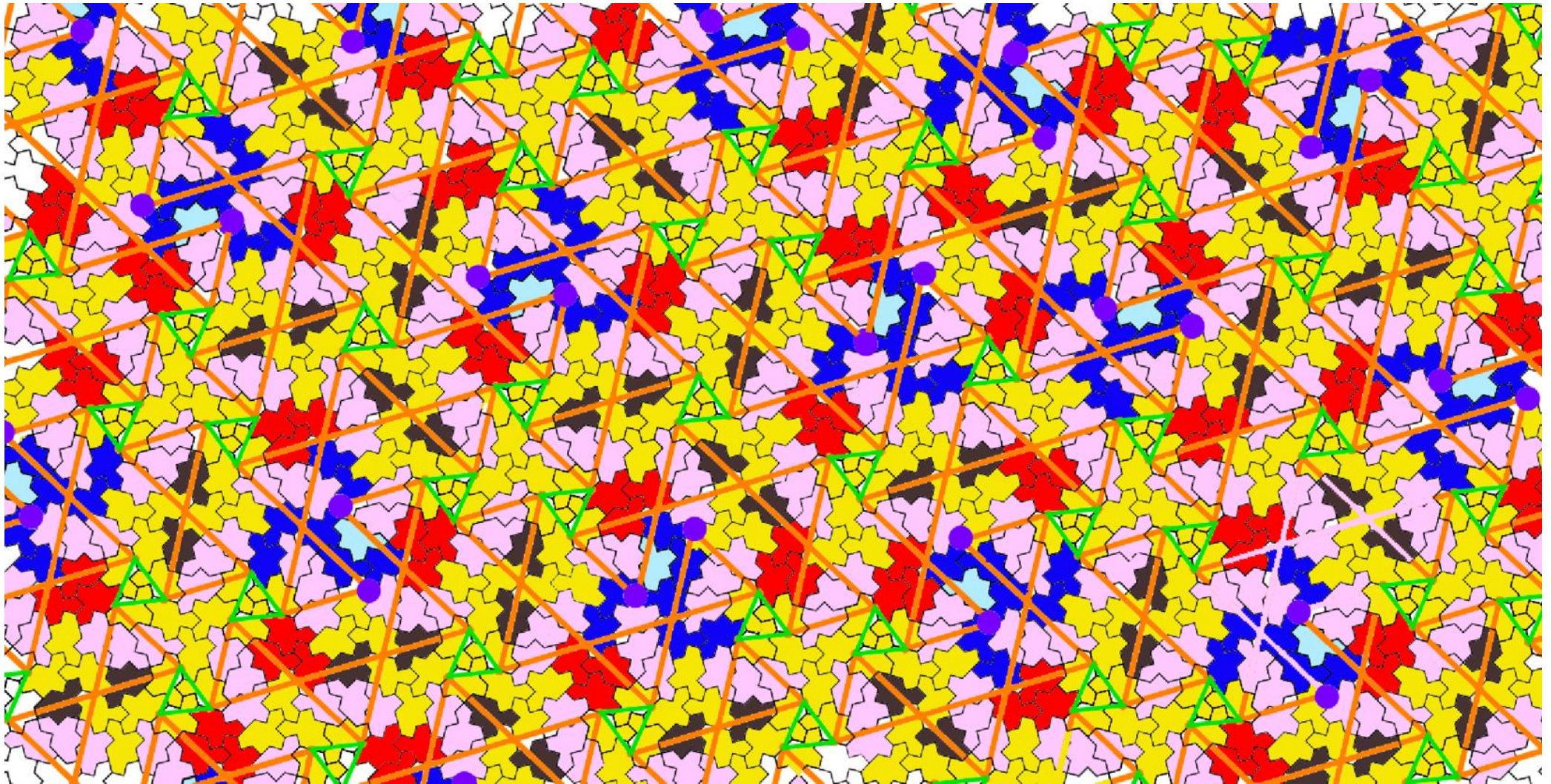
# Triplets and propellers by Boris Horvat

**Diabolo**  
**2-arms**  
**3-arms**  
**are visible !**





# Comparision





# Espaces entre triangles oranges

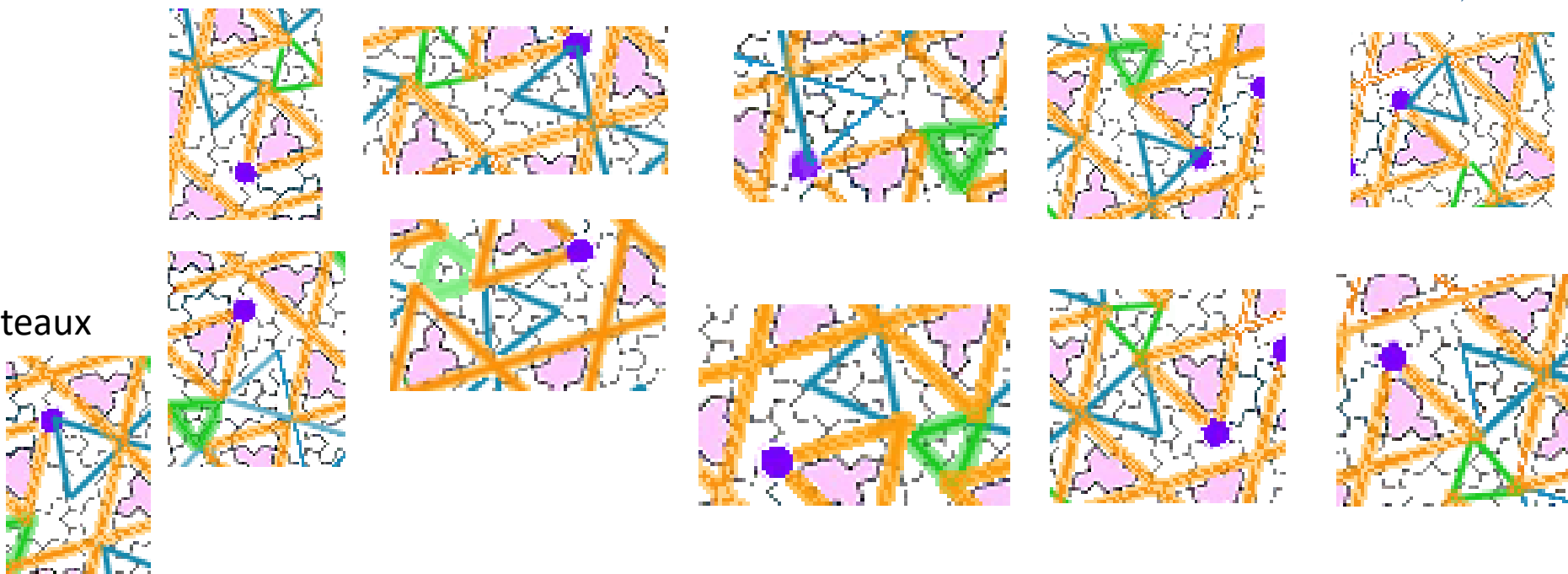
Parallélogrames



Triangles



Bateaux



Petits parallélogrames  
2 orientations de tuiles

